Computational Supplement:

Frame covariance of CMBR lensing in the exact $1 + 3$ covariant radiation hierarchy

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v1.3.0-dev — the derived recombination grid, the structural audit, the acoustic modes against the antecedent's own $1 + 3$ acoustic equations, and the localisation of the frame discrepancy (§11)
Previous release: v1.2.0, the full linear spectrum with the acoustic peaks in both frames. First public analytical reproducibility release: v1.0.0, 5 October 2026
20 October 2026

Abstract

This supplement is the analytical companion to *Frame covariance of CMBR lensing in the exact* $1 + 3$ *covariant radiation hierarchy* (Gebbie, Paper 1, `arXiv:astro-ph/9912072v4`; the suffix is load-bearing: `v4` is Paper 1, `v3` the correction, `v1`–`v2` the 1999 originals). It documents an independent Python reconstruction, from first $1 + 3$ covariant principles, of the almost-Friedmann–Lemaître results of Gebbie, Dunsby & Ellis, *Annals of Physics* **282**, 321–394 (2000), written bottom-up, not ported from a Boltzmann code.

It carries the derivations the bundle rests on — the exact hierarchy, the source closing at rank two, the frame law, the acoustic oscillator, free streaming, the matching and integral solution, and the angular spectrum — each tied to the gate that tests it, with the open questions and approximations collected in one place.

Four claims. The antecedent's equations, re-derived and evaluated independently, reproduce what the antecedent says they do, and where they do not it is located, named and measured: fifteen findings against the antecedent and forty-eight against this bundle. Multipoles of rank two and above are frame-invariant at first order, and both released frames come from one solve. The released integral solution (102) is not the exact integration by parts of its own integral, and the discrepancy is itemised in four rows. The first acoustic peak does not yet sit where a matched external spectrum puts it, and the residual has no established mechanism.

This is not a calibration, a fit, or a claim of new physics; the 1999 claim of one was withdrawn and nothing here reinstates it. Distance to CAMB and CLASS is measured and itemised by approximation, not minimised. One limit ships stated rather than discovered: the recombination grid sets the primary source at η_* and nothing derives it, worth 1.2% — larger than every tolerance claimed here, and invisible to the acceptance gate.

Contents

1 Scientific scope **2**

2 Theory-to-computation map **2**

3 The acceptance spine **9**

4 Conventions and their limits **10**

5 The exact hierarchy **17**
5.1 The expansion, and why it is PSTF . . . 17
5.2 The transport equation . . . 17
5.3 What "exact" claims, and what it does not . . . 17
5.4 Where the hierarchy is gated . . . 18

6 The generic source derivatives **18**

7 Frames and specialisation **20**

8 The acoustic oscillator **25**
8.1 The pair, and why (154) is not the object . . . 26
8.2 What drives it, and the one place the backgrounds differ . . . 26
8.3 Two corrections to the antecedent, both settled by a stated limit . . . 27
8.4 The phase is the sound horizon, and the start is derived from k . . . 27
8.5 Where the oscillator is gated, and the one thing no gate sees . . . 28

9 Free streaming, and where every $\ell \geq 3$ comes from **28**
9.1 What is left when the source and the scattering go . . . 28
9.2 The $\ell = 0$ row is the rank-zero source, seen from the other side . . . 29
9.3 The closed form, and why the kernel is a spherical Bessel function . . . 29
9.4 Three normalisations, none privileged, and one printed slip . . . 29
9.5 What gates this, and what each gate can and cannot see . . . 30

10 The transport assembly **30**
10.1 The by-parts, and the assembly that does not need it . . . 30
10.2 The $\ell = 0$ matching row . . . 33
10.3 The κ' ledger . . . 34
10.4 What this section establishes, and what it does not . . . 35

11 The frame discrepancy: localisation and triage **39**
11.1 The statement being tested . . . 39
11.2 Triage A — sites the discrepancy cannot come from . . . 39
11.3 The two measurements that closed the upstream end . . . 40
11.4 Triage B — the one frame-sensitive site that remains . . . 41
11.5 Triage C — frame-blind sites . . . 41
11.6 Triage by conformal time . . . 42
11.7 What would change this account . . . 43
11.8 The repair, attempted five times, and withdrawn . . . 43
11.9 What one term does to the integrand . . . 43
11.10 What this section does not establish . . . 44

12 The angular spectrum **44**
12.1 Two printed forms, and a cancellation that must be exact . . . 44
12.2 Which of the two is the quantity, and why every figure plots it . . . 45
12.3 C_ℓ is quadratic, and three readings have gone wrong on it . . . 45
12.4 Criterion 5b, and the two different "two routes" . . . 45
12.5 Where the spectrum is gated, and what none of it establishes . . . 46

13 Comparison and its range **46**
13.1 Criterion 5b's tolerance . . . 47

14 Evidence **48**

15 Release progression **48**
15.1 v1.0.0 — the reconstruction, and whether it is faithful 48
15.2 v1.1.0 — the recombination that makes peaks possible 58
15.3 v1.2.0 — the spectrum with structure, and the audit of it 58
15.4 The arc, in one sentence each 58
15.5 v1.3.0 — the structural audit, and the discrepancy localised 59

16 The scattering-strength expansion, re-derived **59**
16.1 The recursion is geometric 59
16.2 Two consequences the closed form makes immediate 60
16.3 What this costs the oscillator 60
16.4 What appears at second order that does not appear at first 60

17 The CDM-frame division of the integral solution, term by term **62**
17.1 Conventions, stated 62
17.2 What has to hold 62
17.3 The three shifts 62
17.4 The shift of (102), and why ΔC_0 has no partner 63
17.5 The k^2 is (90)'s, not a mixed convention 63
17.6 Measured 64
17.7 Why the antecedent never meets this 64
17.8 Three repairs tested, and one of them is untested 64
17.9 What is open 65

18 What is open **65**

19 The almost-EGS bounds **67**

20 Parameter and numerical evidence **67**
20.1 The three backgrounds, side by side 68

21 Numerical resolution and reproducibility **68**

22 Interpretive limits **69**

1 Scientific scope

This supplement records the analytical companion to *Frame covariance of CMBR lensing in the exact* $1+3$ *covariant radiation hierarchy*, Gebbie, Paper 1, University of Cape Town, 2026, `arXiv:astro-ph/9912072v4`, doi:10.48550/arXiv.astro-ph/9912072, submitted to *General Relativity and Gravitation* 2026-10-16, CC BY 4.0. **Cited versioned throughout**: the bare identifier has carried three documents — `v3` is the correction and `v1`/`v2` the legacy 1999 work, neither of them this paper. The bundle is an independent Python reconstruction, from first $1+3$ covariant principles, of the almost-Friedmann–Lemaître results of Gebbie, Dunsby & Ellis, *Annals of Physics* **282**, 321–394 (2000), `doi:10.1006/aphy.2000.6034`. It is archived at `doi:10.25375/uct.34069509`, live since v1.0.0. That is the **concept** DOI and always resolves to the latest version, so it is the one to cite.

It is written bottom-up: not a port of, and not a wrapper on, an existing Boltzmann code, and it inherits no other code's conventions. It reproduces; it does not argue. No new effect is claimed.

Criterion 5b compares against CAMB and CLASS. It does not port them. From v1.2.0 this bundle ships a `reference/` directory holding TT spectra generated by CAMB and CLASS, and that is the only form in which either code appears. **Nothing under `functions/` imports `camb` or `classy`, and nothing ever may**: a reconstruction that imported the codes it is checked against would not be an independent one, and a reader reproducing figure F1 should not need a Fortran toolchain. `scripts/make_reference_spectra.py` is the single caller, it is not a figure stage, and what enters the repository is **data with a stamp** — the spectra, the versions that produced them, every setting, and the derived parameters — never vendored source. Where a published result is carried on this side so that a comparison is like for like, it is taken from the literature and named, not read out of either code. **The external codes are a check on this reconstruction, never an input to it.**

The background expansion is *derived* from the $1+3$ covariant energy constraint rather than transcribed from a printed expansion law, so that the antecedent's own Eqs. (185) and (G.3) are checks on this bundle and not its sources. That ordering is what makes a reconstruction an audit, and it is load bearing: a port would have inherited the misprint recorded below.[1]

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2 Theory-to-computation map

Table 1 links each published equation the numerics implement to its provenance and to the `.py` file and object implementing it, together with the test that checks it. Table 2 gives the notation and the names the quantities carry in the code.

Both are *generated* from `config/implementation-register.toml` by `scripts/make_tables.py`, and `tests/test_register.py` resolves every module, object and test named in them. They cannot drift from the code: a rename breaks the build rather than quietly falsifying a table printed here.

The harmonic normalisation is $Q_{A_\ell} = (-k_{\rm phys})^{-\ell}\mathrm{D}_{\langle A_\ell\rangle}Q$. Both candidate conventions are run end to end; only this one reproduces the real, opposite-sign form of the external hierarchies.[2]

Nomenclature is kept strictly separate: τ_ℓ are **mode** coefficients and τ_{A_ℓ} are **multipole** coefficients. *Annals II* notes that the distinction is not made in Bardeen-variable treatments. Every released figure

[1] The Gauss constraint at (G.3) prints $+\frac{2}{3}\Theta^2$ where the companion thesis and the Cargèse lectures Eq. (55) carry $-\frac{2}{3}\Theta^2$. In the Friedmann limit the printed sign gives $H^2 = K/a^2 - (\mu+\Lambda)/3$, which for a flat section is $-(\mu+\Lambda)/3$ and so has no solution at positive energy density. A misprint, recorded; the derived background already carries the correct sign, and a test now pins it so that a later edit towards the printed form fails rather than passes.

[2] *Annals I* prints $(-1)^\ell$ at the corresponding step where its own definition gives i^ℓ. The covariant multipole τ_{A_ℓ} is invariant under the choice, so the misprint never produced a wrong result; the mode coefficients are not invariant, and the two statements are distinct. Both are asserted as tests. A misprint, recorded in `provenance/ANNALS-II-ANTECEDENT-AND-CORRECTIONS-v1.md`.

plots a multipole mean-square, which holds for general geometries, rather than a mode mean-square, which holds only for almost-Robertson–Walker ones.

Table 1: Equations implemented in the numerics, with their provenance and the code that implements them. Equation numbers are those of the *published* papers. Rows with no code are not yet implemented.

Quantity	**Source**	**Equation**	**File**	**Object**	**Checked by**
$\beta_\ell = O^{A_\ell} O_{A_\ell} = (\ell!)^2 2^\ell/(2\ell)!$	Annals I	(24)	`functions/harmonics/weights.py`	`beta`	`test_beta_module_matches_symbolic_exactly`
$\beta_\ell = \frac{\ell}{2\ell-1}\beta_{\ell-1}$, $\beta_{\ell+1} = \frac{\ell+1}{2\ell+1}\beta_\ell$	Annals I	(25)	`functions/harmonics/weights.py`	`beta`	`test_beta_recurrence_descending_general_ell`
$\Delta_\ell = 4\pi\beta_\ell/(2\ell+1)$	Annals I	(119)	`functions/harmonics/weights.py`	`delta_over_four_pi`	—
α_ℓ, defined by $\beta_\ell = \alpha_\ell^{-1}(2\ell+1)$	Annals II	App. F, p. 380	`functions/harmonics/weights.py`	`alpha`	`test_f4_is_direct_substitution_into_f3`
free-streaming weight $w_\ell = (\ell+1)^2/[(2\ell+3)(2\ell+1)]$	Annals II	(F.1), p. 379	`functions/harmonics/weights.py`	`free_streaming_weight`	`test_f1_weight_is_the_mode_recursion_weight_shifted`
covariant recursion $-\dot{\tau}_\ell = (k_{\rm com}/a)[w_\ell\tau_{\ell+1} - \tau_{\ell-1}]$	Annals II	(F.1), p. 379	`functions/harmonics/free_streaming.py`	`free_stream`	`test_the_three_normalisations_are_one_solution`
β-normalised recursion, conformal time	Annals II	(F.3), p. 380	`functions/harmonics/free_streaming.py`	`free_stream`	`test_the_three_normalisations_are_one_solution`
α-normalised recursion	Annals II	(F.4), p. 380	`functions/harmonics/free_streaming.py`	`free_stream`	`test_f4_variable_is_exactly_the_spherical_bessel_function`
free-streaming solution $\alpha_\ell^{-1}\tau_\ell = j_\ell(k_{\rm com}\Delta\eta)$	Annals II	App. F, p. 380	`functions/harmonics/free_streaming.py`	`analytic_projection`	`test_f4_variable_is_exactly_the_spherical_bessel_function`
$\dot{\Theta}_\ell = k[\frac{\ell}{2\ell-1}\Theta_{\ell-1} - \frac{\ell+1}{2\ell+3}\Theta_{\ell+1}]$	Hu & Sugiyama	(6), p. 2601	`functions/harmonics/external.py`	`WEIGHTS`	`test_external_hierarchies_match_the_bundle_at_machine_precision`
$\dot{F}_{\nu\ell} = \frac{k}{2\ell+1}[\ell F_{\nu(\ell-1)} - (\ell+1)F_{\nu(\ell+1)}]$	Ma & Bertschinger	(49), (50)	`functions/harmonics/external.py`	`WEIGHTS`	`test_external_hierarchies_match_the_bundle_at_machine_precision`
$\dot{\Delta}_{T\ell} = \frac{k}{2\ell+1}[\ell\Delta_{T(\ell-1)} - (\ell+1)\Delta_{T(\ell+1)}]$	Seljak & Zaldarriaga	(3d)	`functions/harmonics/external.py`	`WEIGHTS`	`test_external_hierarchies_match_the_bundle_at_machine_precision`

Quantity	Source	Equation	File	Object	Checked by
relative-sign control: the $c^2 = -1$ normalisation, run as a wiring check	Annals II	(F.1), p. 379	functions/harmonics/free_streaming.py	free_stream	test_sign_control_fails_in_a_known_shape
$\int_0^\infty z^{-m} j_\ell^2(z)\, dz$, corrected to carry $[(m/2)!]^2$	Annals II	(199), p. 372	scripts/derive_ell_range.py	bessel_integral_corrected	test_corrected_bessel_identity_matches_quadrature
$\ell_A = \pi \Delta\eta_* / r_s$, bounding criterion 5's range	Annals II	§8.3.2, p. 372	scripts/derive_ell_range.py	acoustic_scale	—
$(a'/a)^2 = H_0^2[\Omega_r a^{-2} + \Omega_m a^{-1} + \Omega_k + \Omega_\Lambda a^2]$	this work	the $1+3$ energy constraint	functions/background/flrw.py	Background	test_the_gauss_constraint_sign_is_the_corrected_one_and_the_printed_one_has_no_solution
$\mathcal{P}(k) = \mathcal{A}(k\eta_0)^{n-1}$	Annals II	(J.1)	functions/spectra/transfer.py	power_spectrum_j1	test_j1_is_constant_at_n_equals_one_and_eta0_drops_out
$\mathcal{P}(k) = \mathcal{B} k^n T^2(k)$	Annals II	(J.3)	functions/spectra/transfer.py	power_spectrum_j3	test_j1_and_j3_differ_by_one_power_of_k
$T(k) = [1 + (ak + (bk)^{3/2} + (ck)^2)^\nu]^{-1/\nu}$	Annals II	(J.5)	functions/spectra/transfer.py	transfer_function	test_b10_the_printed_transfer_function_diverges_where_it_must_vanish
$\dot\delta_\ell = -n_e \sigma_T \delta_\ell - ikT\{\frac{\ell}{2\ell-1}\delta_{\ell-1} + \frac{\ell+1}{2\ell+3}[1 - \frac{\ell(\ell+2)K}{k^2}]\delta_{\ell+1}\}$	Wilson	(8), p. 4	functions/harmonics/external.py	integrate_wilson	test_wilson_reproduces_the_bundle_and_tests_the_phase_convention
$x_e(z)$, equilibrium Saha	this work	standard; not in the antecedent	functions/spectra/decoupling.py	saha_ionisation	test_saha_is_monotone_and_spans_the_full_range
$\mathcal{V}(k,\eta) \simeq \kappa' e^{-\kappa}$	Annals II	§5.2	functions/spectra/decoupling.py	RecombinationHistory	test_the_visibility_is_a_normalised_probability_density
$\mathcal{D}(\eta_0, k) = \int d\eta\, \mathcal{V} e^{-(k/k_D)^2} \simeq e^{-(k/k_D)^2}$	Annals II	(101), §5.2	functions/spectra/decoupling.py	RecombinationHistory	test_the_visibility_is_a_normalised_probability_density
the integral solution with $e^{-\kappa}$ on gravitational and $\kappa' e^{-\kappa}$ on scattering sources	Annals II	(98), §5	functions/spectra/decoupling.py	Weighting	test_the_two_weightings_are_distinct_objects
the same solution with both weights replaced by $\mathcal{V}$	Annals II	(102), §5	functions/spectra/decoupling.py	Weighting	test_the_visibility_weighting_cannot_carry_a_late_isw

Quantity	Source	Equation	File	Object	Checked by
$\gamma \simeq \frac{k^2}{6\kappa'} \frac{R^2+\frac{4}{5}(1+R)}{(1+R)^2} = k^2/k_D^2$	Annals II	(168), §6	functions/spectra/decoupling.py	diffusion_scale	test_damping_suppresses_small_scales_and_leaves_large_ones
$-C_0 = \frac{5}{2}a\mathcal{B}_2$, $-C_1 = \frac{1}{k}(a\mathcal{B}_1+\kappa' v_B)$, $-C_2 = \frac{1}{k^2}a\mathcal{B}_2$	Annals II	(90), §5	—	—	—
$\tilde{\tau}_1' = -\frac{R'}{1+R}\tilde{\tau}_1 - \frac{k}{1+R}\delta\tilde{T} - k\Phi_A$; $\delta\tilde{T}' = -aH\Phi_A - \Phi_H' + \frac{k}{3}\tilde{\tau}_1$	Annals II	(152), (153), §7.1	functions/spectra/acoustic.py	solve_acoustic	test_the_pair_reduces_to_the_printed_oscillator_exactly
$\delta\tilde{T}'' + \frac{R'}{1+R}\delta\tilde{T}' + k^2 c_s^2 \delta\tilde{T} = -\Phi_H'' - \frac{R'}{1+R}\Phi_H' - \frac{k^2}{3}\Phi_A$	Annals II	(154), §7.1	functions/spectra/acoustic.py	solve_acoustic	test_the_pair_reduces_to_the_printed_oscillator_exactly
$c_s^2 = 1/(3(1+R))$, $R = 3\rho_M/4\rho_R$	Annals II	§6	functions/spectra/acoustic.py	sound_speed_squared	test_sound_speed_limits
$\delta\tilde{T} + \Phi_A \approx [\delta\tilde{T}(0) + (1+R)\Phi_A]\cos(kr_s) - R\Phi_A$	Annals II	(180), §7.1.2	functions/spectra/acoustic.py	sachs_wolfe_at_last_scattering	test_the_integration_converges_to_the_closed_form_180_as_r_is_held_constant
$[\delta\tilde{T} + \Phi_A](\eta_*) \approx \frac{1}{3}\Phi_A(0)(1+3R_*)\cos(kr_s^*) - R_*\Phi_A(0)$	Annals II	(181), §7.1.2	functions/spectra/acoustic.py	sachs_wolfe_at_last_scattering	test_b8_equation_181_sign_is_fixed_by_its_own_limit
$\frac{\beta_\ell \tau_\ell(\eta_0)}{2\ell+1} = S_P j_\ell(k\Delta\eta_*) + \int d\eta\,(S_{\rm DISW} + S_{\rm ISW}) j_\ell(k\Delta\eta)$	Annals II	(176), §7.1.1	functions/spectra/sources.py	temperature_transfer	test_176_with_an_impulsive_source_is_the_free_streaming_projection
$S_P = D(\eta_0, k)\,[(\tilde{\delta}_T + \Phi_A) + \kappa' v_B]$	Annals II	(177), §7.1.1	functions/spectra/sources.py	source_primary	test_176_with_an_impulsive_source_is_the_free_streaming_projection
$S_{\rm DISW} = V e^{-(k/k_D)^2}[\frac{1}{3}k\tilde{\tau}_1 + (\kappa'\tilde{v}_B' + \kappa''\tilde{v}_B)]$	Annals II	(178), §7.1.1	functions/spectra/sources.py	source_doppler	test_the_printed_doppler_bracket_equals_the_contracted_form
$S_{\rm ISW} = V e^{-(k/k_D)^2}[(\Phi_A' - \Phi_H') - aH(\tilde{\delta}_T + 3\Phi_A)]$	Annals II	(179), §7.1.1	functions/spectra/sources.py	source_integrated	test_b6_sign_alternative_is_not_a_no_op
$C_\ell = \frac{2}{\pi}\frac{\beta_\ell^2}{(2\ell+1)^2}\int \frac{dk}{k} k^3 \lvert\tau_\ell\rvert^2$	Annals II	(186), p. 368	functions/spectra/angular.py	cl_mode_route	test_the_harmonic_normalisation_cancels_out_of_186_exactly
$\langle \tau_{A_\ell} \tau^{A_\ell} \rangle = \frac{\beta_\ell}{2\pi^2}\int k^2 dk \lvert\tau_\ell\rvert^2$	Annals II	(187), p. 368	functions/spectra/angular.py	multipole_mean_square	test_188_is_delta_over_2l_plus_1_times_187

Quantity	Source	Equation	File	Object	Checked by
$C_\ell = \Delta_\ell (2\ell+1)^{-1} \langle \tau_{A_\ell} \tau^{A_\ell} \rangle$	Annals II	(188), p. 368	functions/spectra/angular.py	cl_covariant_route	test_the_two_routes_agree_to_machine_precision_on_real_input
$C_\ell^{\mathrm{SW}} = \frac{1}{2\pi} H_0^4 \Omega_0^{1.54} \int \frac{dk}{k^2} P(k) j_\ell^2(k\Delta\eta_*)$	Annals II	(198), p. 371	functions/spectra/sachs_wolfe.py	cl_sachs_wolfe_quadrature	test_the_analytic_reduction_matches_the_quadrature
large-scale closed form, $n = 1$	Annals II	(200), (201), p. 372	functions/spectra/sachs_wolfe.py	cl_large_scale_reduced	test_b5_printed_201_is_short_by_the_comoving_distance
closed-form k-integral kernel at complex m, $\int z^{-m} j_\ell^2(z) dz$	Annals II	(199), p. 372, corrected	functions/spectra/kernels.py	bessel_square_kernel	test_complex_kernel_reduces_to_the_real_identity
primary-term k-integral by power-law decomposition	Annals II	(176) primary term, (198)	functions/spectra/kernels.py	fftlog_primary_integral	test_fftlog_matches_quadrature_on_a_non_power_law_integrand
integrated terms, $\int d\eta\, S(k,\eta)\, j_\ell(k\Delta\eta)$	Annals II	(176), p. 365	functions/spectra/bessel_table.py	line_of_sight	test_line_of_sight_matches_direct_evaluation
truncation condition: four consecutive vanishing harmonics force $\sigma_{ab} = 0$	Annals II	App. E, (E.1), p. 378	—	—	—
C_0, C_1, C_2 from the rank-by-rank match of the integral solution	Annals II	(89)	functions/spectra/coefficients.py	match_coefficients	test_all_three_of_90_are_reproduced_by_the_match
$-C_0 \simeq +\frac{5}{2} a\mathcal{B}_2$, $-C_1 \simeq \frac{1}{k}(a\mathcal{B}_1 + \kappa' v_B)$, $-C_2 \simeq \frac{1}{k^2} a\mathcal{B}_2$	Annals II	(90)	functions/spectra/coefficients.py	c0_from_printed_89	test_b11_is_a_sign_in_89_not_in_90
$-\tilde{C}_1 \approx \kappa' \tilde{v}_B + (\delta\tilde{T} + \Phi_A)$	Annals II	(109)	functions/spectra/coefficients.py	newtonian_coefficients	test_the_newtonian_specialisation_is_equation_109
the slow-decoupling integral solution carried with a generic u^a: two boundary terms and one integrated term	Annals II	(102)	functions/spectra/integral_solution.py	integral_solution	test_the_specialisation_recovers_what_the_pipeline_computes
$(a\tilde{\mathcal{B}}_1)' \approx -a^2 H \tilde{\mathcal{B}}_1 + k(\Phi_A' - \Phi_H') - 2Hak\Phi_A + \frac{1}{3}k^2\tilde{\tau}_1$	Annals II	(106)	functions/spectra/coefficients.py	newtonian_c1_prime	test_t23_splining_the_amplitude_is_wrong_and_the_test_can_tell
$\dot{\sigma}_{\langle ab\rangle} + 2H\sigma_{\langle ab\rangle} - \mathcal{D}_{\langle a} A_{b\rangle} + E_{\langle ab\rangle} \simeq \frac{1}{2}\pi_{\langle ab\rangle}$, and its second derivative	Annals II	(G.12)	functions/spectra/coefficients.py	generic_shear_derivatives	test_the_rank_two_control_passes_symbolically

Quantity	Source	Equation	File	Object	Checked by
C_2, C_2' and C_2'' from $\mathcal{B}_2 = \sigma_{ab}$	Annals II	(90)	`functions/spectra/coefficients.py`	`c2_derivatives`	`test_no_numerical_derivative_survives_in_the_c2_chain`
$\sigma_{ab} = \mathcal{D}_{\langle a} v_{b\rangle}$ in the CDM frame, with $\dot{\sigma}$ and $\ddot{\sigma}$	Annals II	(31) + (G.12)	`functions/spectra/coefficients.py`	`cdm_frame_shear`	`test_B13_the_printed_n_t1_does_not_close_and_the_general_rule_does`
$C_0, C_1, C_2, C_1', C_2''$ in the CDM frame, $A_a = 0$	this work	(89) + (31)-(37) + (106) + (152)	`functions/spectra/coefficients.py`	`cdm_set`	`test_the_frame_shift_of_every_coefficient_is_the_closed_form_one`
$\tilde{\tau}_a = \tau_a - v_a$, as $\tilde{\tau}_1 \to \tilde{\tau}_1 - v/k$	Annals II	(37)	`functions/spectra/frame_transform.py`	`FrameShift`	`test_T27_the_dipole_carries_the_same_inverse_k_as_the_monopole`
$(\rho, p, \mathcal{H}')$ from the Ω's, consistent with $\dot{H} = -H^2 - (\rho + 3p)/6$	Annals II	Appendix G	`functions/spectra/coefficients.py`	`background_fluid`	`test_the_frame_shift_of_every_coefficient_is_the_closed_form_one`

Table 2: Notation, and the names the quantities carry in the code. The mode/multipole distinction is the one that matters most: *Annals II* notes that it is not made in Bardeen-variable treatments.

Symbol	In code	Meaning	Note
Φ_A, Φ_H	`phi_a, phi_h`	the two gauge-invariant potentials entering (177) and (179)	—
$\tilde{\delta}_T$	`delta_T`	covariant temperature perturbation	frame-dependent; $\tilde{\delta}_T + 3\Phi_A$ is the combination in dispute (B-6)
$\tilde{\tau}_1$	`tau_1`	the dipole, which supplies the Doppler term of (178)	—
$\tilde{v}_B$	`v_b`	baryon velocity	—
κ', κ''	`kappa_prime`	differential optical depth and its conformal-time derivative	supplied, never manufactured inside the source terms
V	`visibility`	visibility function, weighting the integrated sources	—
k_D	`damping_k`	diffusion damping scale, entering (178) and (179) as $e^{-(k/k_D)^2}$	—
aH	`conformal_hubble`	a'/a, the background combination (179) carries	derived from the $1+3$ constraint, not from a printed expansion law
S_P, $S_{\rm DISW}$, $S_{\rm ISW}$	`source_primary, source_doppler, source_integrated`	the primary, Doppler and integrated temperature sources of (177)–(179)	individually frame-dependent
$\tau(x, e)$	—	temperature anisotropy as a function of position and direction	—
τ_{A_ℓ}	—	**multipole** coefficients — the covariant objects	valid for general geometries; what every released figure plots
τ_ℓ	`tau, values`	**mode** coefficients	almost-Robertson–Walker only. Conflating these two is how a sign is lost
O^{A_ℓ}	—	$e^{\langle A_\ell \rangle}$, the PSTF basis on the sphere	Annals I (15)
Q_{A_ℓ}	—	harmonic basis tensor, $(-k_{\rm phys})^{-\ell} \mathrm{D}_{\langle A_\ell \rangle} Q$	the adopted normalisation; see criterion 3
β_ℓ	`beta`	$O^{A_\ell} O_{A_\ell}$	Hu & Sugiyama's normalisation
α_ℓ	`alpha`	$(2\ell+1)/\beta_\ell$	Ma & Bertschinger's and Seljak & Zaldarriaga's normalisation
$k_{\rm com}$	`k_com`	comoving wavenumber	**never a bare `k` in this codebase**
$k_{\rm phys}$	`k_phys`	physical wavenumber, $k_{\rm com}/a$	Appendix F's k is comoving; the conventions sheet's is physical
η	`eta`	conformal time, $dt = a\, d\eta$	—
$\Delta\eta_*$	`d_eta`	$\eta_0 - \eta_*$	—
j_ℓ	`spherical_jn`	spherical Bessel function	the free-streaming solution
u^a	—	the frame 4-velocity	carried generic; frames are specialisations made at the end
σ_{ab}	—	shear	σ is the shear and nothing else
A_a	—	acceleration, $\dot{u}_a$	—
Θ_ℓ	—	Hu & Sugiyama's multipole variable	β-normalised

3 The acceptance spine

Appendix F of the antecedent, pp. 379–380, Eqs. (F.1)–(F.4), links the covariant mode recursion to external treatments written in real coefficients. Reproducing that chain tests the harmonic normalisation

without introducing a representation, and it is where this bundle's independence lies.[3]

Five external treatments are read and **four** are drawn — Hu & Sugiyama Eq. (6), Wilson (1983) Eq. (8), Ma & Bertschinger Eqs. (49) and (50), and Seljak & Zaldarriaga Eq. (3d) — each transcribed from its own paper.[4] The fifth is Wilson & Silk (1981) Eq. (7), the earliest of them and likewise read from its own page: at $K = 0$ it *is* Wilson (1983) Eq. (8), so it is credited and not drawn, four panels carrying four independent numerical checks. Hu & Sugiyama and Wilson write the recursion in the β normalisation — the normalisation their Eq. (2) fixes — and the other two in the α normalisation, which is why Appendix F prints both (F.3) and (F.4). A basis phase leaking into the couplings would break at least one of the two matches.

Wilson is the strongest of the four, and not because of his weights, which are Hu & Sugiyama's. He writes the recursion in the *imaginary* convention, with both bracket terms positive under an overall $-ikT$, where the others carry opposite signs under a real k. The two are related by $\Theta_\ell = i^\ell \delta_\ell$, which turns one form into the other exactly. He is integrated complex and as printed, and the phase removed only afterwards; transforming the equation first would assume the identity under test. That the transformed solution is *real* to 10^{-10} is a separate assertion from that it agrees. Wilson therefore tests the phase convention — the subject of §2's footnote and of acceptance criterion 3 — from a source published before any of the others.[5]

4 Conventions and their limits

The same hierarchy is written four different ways in the literature this bundle checks against, and the differences are not cosmetic. Table 3 is the translation, generated from `config/implementation-register.toml` so that it cannot drift from the code, with the test that checks each row.

"RW", "FL" and "FLRW" are three different claims, and this bundle makes the middle one. The names are used interchangeably often enough that it is worth separating them once.

- **Robertson–Walker (RW)** is a statement about *geometry*: the spacetime is spatially homogeneous and isotropic. It says nothing about what drives the expansion, and it follows from symmetry alone — no field equation has been used.
- **Friedmann–Lemaître (FL)** is that geometry *with its dynamics*: the Einstein equations imposed on it, giving the Friedmann and Raychaudhuri equations and hence $a(\eta)$ for a given matter content. FL is RW *plus* the field equations.
- **FLRW** is the two together, and is the usual compound name.

Annals II and this bundle say **almost-Friedmann–Lemaître**, and the choice is not stylistic. The background here is integrated from the $1 + 3$ energy constraint rather than assumed, so the dynamical

[3] The comparison applies the $(2\ell + 1)^{-1}$ division that the text preceding (F.4) specifies and the display omits; (F.4) is otherwise correct as printed. A misprint, recorded.

[4] Ma & Bertschinger print the same recursion twice. Eqs. (49) and (50) are the *massless neutrino* hierarchy, synchronous and conformal Newtonian, and are collisionless; their Eqs. (63) and (64) are the photon hierarchy, carrying the identical recursion plus a Thomson term $-an_e\sigma_{\rm T}F_{\gamma\ell}$. Appendix F's (F.4) is a free-streaming recursion, so (49) and (50) are the comparison and (63) and (64) are not. Published and arXiv numbering agree for this paper, which is not the case for the Annals papers.

[5] Appendix F's citation pairs the name of one paper with the key of another. Both papers are real and both numbers are right for the paper they belong to: Wilson & Silk (1981) Eq. (7) *is* the $\ell > 2$ hierarchy, and Wilson (1983) Eq. (7) is that paper's quadrupole equation while its Eq. (8) is the hierarchy. The prose names Wilson & Silk, the key is Wilson alone and consistent across every citation, and the load-bearing $i^{-\ell}$ remark names Wilson alone — so the match and the phase remark are about Wilson (1983), and the number beside them is correct for the 1981 paper the prose names. One mismatched pairing, not two independent misprints; recorded, and it moves no result.

half is in use from the start: Eqs. (185) and (G.3) are **checks on this bundle rather than inputs to it**. A treatment that only assumed almost-RW would have the geometry and not the expansion history, and could not perform that check.

The distinction is also what the almost-EGS theorem is about, and Sec. 19 depends on reading it the right way round. Almost-isotropy of the measured radiation bounds the *geometry* — an almost-RW conclusion. Getting from there to almost-FL requires the field equations as a second step. Collapsing the two names collapses a two-step argument into an assertion, which is precisely the move that paper is written to avoid.

There are exactly two conventions, and they differ by $c^2 = -1$. Changing the harmonic normalisation $Q_{A_\ell} = \mu^{-\ell}\mathrm{D}_{\langle A_\ell\rangle}Q$ rescales the mode coefficients as $\tau_\ell \to c^\ell\tau_\ell$ with $c = \mu'/\mu$. A recursion $\tau'_\ell = k[A_\ell\tau_{\ell-1} + B_\ell\tau_{\ell+1}]$ then carries $(A_\ell/c,\, B_\ell c)$, so the ratio B/A picks up c^2. At $c = i$ an opposite-sign real bracket becomes a same-sign bracket under an overall i. Hu & Sugiyama, Ma & Bertschinger and Seljak & Zaldarriaga print the first; Wilson prints the second. That is the whole of the difference, and it is asserted symbolically rather than described.

What the match establishes. The ℓ-weights, which are convention-independent, are confirmed by five sources in two normalisations, four of them drawn and numerically independent. The bundle is shown to sit at the same c^2 as Hu & Sugiyama, Ma & Bertschinger and Seljak & Zaldarriaga, and Wilson is shown to sit at the other. *Both* Wilson papers are in the imaginary convention, which settles a standing hypothesis in the negative: no author published this recursion in both conventions, so the split is between authors and not within one.

One source fixes its own phase, and that matters. Ma & Bertschinger define their Legendre expansion with an explicit factor $(-i)^\ell(2\ell+1)$ and state why: it is "chosen to simplify the expansion of a plane wave", so that $\exp(-i\mathbf{k}\cdot\mathbf{x})$ has coefficients $F_\ell = j_\ell(kr)$. Their coefficients therefore sit at $(-i)^\ell$ from the plane wave *by their own definition*, are real, and satisfy the opposite-sign real recursion. For this source the phase is read off the paper rather than assumed, which is exactly the standing the next paragraph says three of the four lack.

What it does not establish, and this release says so. It does **not** establish that $Q_{A_\ell} = (-k_{\mathrm{phys}})^{-\ell}\mathrm{D}_{\langle A_\ell\rangle}Q$ is the right reading of *Annals I*'s definition. Rephasing *both sides* of a comparison by i^ℓ is a change of variable and cannot fail — measured at 6.2×10^{-15} either way — so a match constrains the convention only where the phase applied to the external source is fixed from outside the comparison. For two of the four, no phase is applied, and that is a choice rather than a derivation; Ma & Bertschinger and Wilson are the two that are fixed from outside the comparison.

The exception, and it is the useful one. *Annals II* states, in the line following its flat-case mode functions, that the covariant form "differs by a factor of $i^{-\ell}$ from Wilson since we are using plain mode functions instead of plane waves". That phase is specified by the antecedent, not chosen here. So Wilson's panel in Figure 1 is the one comparison carrying information about the convention, and it is checked in two independent ways: the transformed solution agrees with the reconstruction, *and* it is real to 10^{-10}, which the agreement alone would not show.

Acceptance criterion 3 was therefore **reopened** and restated to the claim the evidence supports. Settling it needs *Annals I*'s definition of Q_{A_ℓ}, not a better match. Nothing computed here depends on it: the covariant τ_{A_ℓ} is invariant under the choice, the mode coefficients are not, and every released figure plots the invariant quantity.

Table 3: How each treatment in the foundational literature writes the same hierarchy, and how this bundle checked it. The two conventions in the literature are related by $\tau_\ell \to c^\ell \tau_\ell$ with $c^2 = -1$ and nothing else. **Where the transformation applied is "none" that is a choice, not a derivation** — which is why acceptance criterion 3 was reopened; see §4.

Treatment	**Variable**	**Bracket weights**	**Phase convention**	**Transformation applied here**	**How it was checked**
Annals II (F.1), Appendix F	τ_ℓ	$\frac{(\ell+1)^2}{(2\ell+3)(2\ell+1)}$ on $\tau_{\ell+1}$, 1 on $\tau_{\ell-1}$	covariant; $Q_{A_\ell} = (-k_{\text{phys}})^{-\ell} \mathrm{D}_{\langle A_\ell \rangle} Q$	—, this is the reference	arithmetic on β_ℓ alone; no representation introduced
Annals II (F.3), Appendix F	$\beta_\ell \tau_\ell$	$\frac{\ell}{2\ell-1}$, $\frac{\ell+1}{2\ell+3}$	real, opposite-sign bracket	none	(F.1)→(F.3) by multiplying through by β_ℓ, as exact rationals
Annals II (F.4), Appendix F	$\alpha_\ell^{-1} \tau_\ell$	$\frac{\ell}{2\ell+1}$, $\frac{\ell+1}{2\ell+1}$	real, opposite-sign bracket	the $(2\ell+1)^{-1}$ the prose states and the display omits	**finding B-2**: printed (F.4) is not yet in the Ma & Bertschinger form
Hu & Sugiyama Eq. (6), p. 2601	Θ_ℓ	$\frac{\ell}{2\ell-1}$, $\frac{\ell+1}{2\ell+3}$	real, opposite-sign bracket	none — *a choice*, see Q1	integrated as printed from its own paper, never through Appendix F
Wilson Eq. (8), p. 4	δ_ℓ	$\frac{\ell}{2\ell-1}$, $\frac{\ell+1}{2\ell+3}$	**imaginary**; both bracket terms + under an overall $-ikT$	$\Theta_\ell = i^\ell \delta_\ell$ — **fixed by the antecedent**, which states the $i^{-\ell}$ itself	integrated *complex and as printed*; the phase removed only afterwards, and the result asserted real to 10^{-10} separately from that it agrees
Wilson & Silk Eq. (7), p. 15	δ_ℓ	$\frac{\ell}{2\ell-1}$, $\frac{\ell+1}{2\ell+3}$	**imaginary**, as Wilson 1983 — *not* the real form	none required: at $K = 0$ it is Wilson 1983 (8) exactly	read from its own page. The **earliest** of the five and a fifth source for the weights, but **not** a fifth numerical check — the same equation cannot check itself
Ma & Bertschinger Eqs. (49), (50)	$F_{\nu\ell}$	$\frac{\ell}{2\ell+1}$, $\frac{\ell+1}{2\ell+1}$	real, opposite-sign bracket	none — *a choice*, see Q1	integrated as printed from its own paper; synchronous and conformal Newtonian gauges

Treatment	**Variable**	**Bracket weights**	**Phase convention**	**Transformation applied here**	**How it was checked**
Seljak & Zaldarriaga Eq. (3d)	$\Delta_{T\ell}$	$\frac{\ell}{2\ell+1}$, $\frac{\ell+1}{2\ell+1}$	real, opposite-sign bracket	none — *a choice*, see Q1	integrated as printed; also the method CMBFAST implements, cited for criterion 5b

The second table, and what its two thesis columns are for. Table 3 translates between the treatments this bundle is *checked against*. Table 4 does something different: it writes each coupling of this reconstruction in the conventions of Maartens, Gebbie & Ellis (1999), Challinor (2000), the thesis in **both** of the conventions it uses, Beneke & Fidler (2010) and Pitrou (2009). Every cell is derived from a named line of the normative sheet, and no cell is blank: where a treatment has no established counterpart the cell says so.

Two things in it are worth reading rather than skimming. **The thesis takes two columns and they differ on exactly the kinematic rows** — its $\dot{\mathcal{T}}$ form moves every kinematic term to the right-hand side, flipping the overall sign against its K/Π form. Elsewhere the two agree and the table says so in as many words, because a difference in wording where there is none in substance is precisely how two conventions get conflated while appearing distinct. And **the Beneke & Fidler and Pitrou columns are mostly "not established"**: the sheet records both as unsettled at the level of hierarchy coefficients, Beneke & Fidler working in (ℓ, m) tetrad form and Pitrou in Poisson gauge, and neither has had a hierarchy coefficient reproduced here. That is the honest content of those columns and it is printed rather than left to inference.

Table 4: The couplings of this reconstruction written in each external convention. Every cell is derived from the line of `provenance/conventions.md` named in the *Sheet* column; no cell is blank, and a treatment with no established counterpart says so rather than being left to inference. **The thesis takes two columns and they differ on exactly the kinematic rows** — the $\dot{\mathcal{T}}$ form moves every kinematic term to the right-hand side, flipping its overall sign against the K/Π form. On the remaining rows the two agree, and the table says *as Th-K* rather than restating them: a difference in wording where there is none in substance is how two conventions get conflated while appearing distinct.

Coupling	**Sheet**	**MGE99**	**CL**	**Th, K/Π**	**Th, $\dot{\mathcal{T}}$**	**BF**	**P09**
Rank raising	`C4.2`	identity	identity in form; D_a unchanged, the sign sits in k	identity	as Th-K	not established: BF work in (ℓ, m) tetrad form, no coefficient reproduced	not established: Pitrou works in Poisson gauge, no hierarchy coefficient checked
Rank lowering	`C4.4`	identity	identity in form; CL (2.25) prints $+\mathrm{D}^b I_{bA_\ell}$, reproduced from MGE99 (71)	identity; Th App. D footnote to (D.44) at $\ell = 1$ agrees	as Th-K	not established	not established
Free-streaming mode recursion	`C4.5`	identity	identity in form; CL (2.25) free streaming reproduced from MGE99 (71)	identity	as Th-K	not established	not established
PSTF contraction on the sphere	`C6.1`	identity (one source: EMT 1983, via C6.4)	identity; CL expands in the same O^{A_ℓ}	identity, Th (1.20)	as Th-K	not established: BF use ${}_sY_{\ell m}$, not a PSTF basis	identity at the basis level: Pitrou (1.28) expands in n^{A_ℓ}
PSTF self-contraction β_ℓ	`C5.2`	identity	identity; CL absorbs it into Δ^C_ℓ, see the row below	identity, Th (1.23)	as Th-K	no counterpart: (ℓ, m) normalisation carries $\sqrt{4\pi/(2\ell+1)}$ instead	identity, Pitrou (1.29) in the equivalent $4\pi\ell!/(2\ell+1)!!$ form
Angular measure Δ_ℓ	`C5.5`	identity, MGE99 (47), same value	$\Delta^C_\ell = 4\pi(-2)^\ell(\ell!)^2/(2\ell+1)! = (-1)^\ell\Delta_\ell$, CL (2.23)	identity	as Th-K	no counterpart in this form	identity, Pitrou (1.29), verified symbolically

Coupling	**Sheet**	**MGE99**	**CL**	**Th, K/Π**	**Th, $\dot{\mathcal{T}}$**	**BF**	**P09**
Shear coupling at $\ell+2$	`C9, T-MGE and T-Th-T`	$-(\ell+2)$ as printed in (71), (88), (90), (99) — **defect E9**; $+(\ell+2)$ is correct, and MGE99 (89) at $\ell=2$ prints $+\frac{8}{15}\rho\sigma$, agreeing with the correction	$-\frac{8}{15}\delta^2_\ell I\sigma$ gives $+(\ell+2)$, independently of MGE99	$-(\ell+2)$ in Th (2.73) — carries E9	$+(\ell+2)$ **on the RHS** in Th (2.84): every kinematic term flips overall sign against (2.73) because it is moved across. **This row is where the two thesis columns differ**	not established: BF put the observer at rest, so they carry no shear coupling	not established
Acceleration coupling at $\ell\pm1$	`C9, T-CL and T-Th-T`	identity	$A_a \to -A_a$ under $g\to -g$ at fixed u^a (CL footnote to §2), the signature difference and nothing more	identity	overall sign flipped with the rest of the kinematic terms — **differs from Th-K**	not established: observer at rest	not established

5 The exact hierarchy

This is the object every later section is about. It is written once, here, with what "exact" is claiming and what it is not.

5.1 The expansion, and why it is PSTF

The photon distribution is expanded in **projected, symmetric, trace-free** temperature multipoles about a chosen 4-velocity u^a:

$$\tau(x,e) = \sum_{\ell\geq 0} \tau_{A_\ell}(x)\, O^{A_\ell}(e), \qquad O^{A_\ell} = e^{\langle A_\ell\rangle}, \tag{1}$$

with A_ℓ the multi-index $a_1 \dots a_\ell$, e^a a unit direction orthogonal to u^a, and $\langle\,\rangle$ the PSTF part.

PSTF is not a choice of basis among several. On the 2-sphere orthogonal to u^a the PSTF tensors are the *irreducible* representations of the rotation group: each rank carries exactly the information no lower rank carries, and the decomposition is unique. Any other basis — spherical harmonics, Legendre polynomials in $\hat{k}\cdot e$ — is a representation of the same content that first requires a direction to be singled out. **The PSTF form singles out nothing but** u^a, which is the whole reason a $1+3$ treatment can carry the frame generically and choose it at the end.

Two objects, and conflating them is how a sign is lost. τ_{A_ℓ} are **multipole** coefficients and exist for a general geometry. τ_ℓ are **mode** coefficients, obtained by expanding τ_{A_ℓ} on harmonics Q_{A_ℓ}, and exist only for an almost-Robertson–Walker one. *Annals II* notes the distinction is not made in Bardeen-variable treatments. **Every released figure plots the multipole quantity**, and §12 says why that is not a presentational preference.

5.2 The transport equation

With $\kappa' = an_e\sigma_{\rm T}$ the differential optical depth, the exact covariant transport equation at rank ℓ is *Annals II* (68):

$$\dot{\tau}_{A_\ell} + \underbrace{\tfrac{\ell+1}{2\ell+3}\mathrm{D}^a\tau_{A_\ell a} - \mathrm{D}_{\langle a_\ell}\tau_{A_{\ell-1}\rangle}}_{\text{free streaming}} + \underbrace{\kappa'\tau_{A_\ell}}_{\text{out-scattering}} = \underbrace{a\mathcal{B}_{A_\ell}}_{\text{gravitational}} + \underbrace{\kappa'\mathcal{S}_{A_\ell}}_{\text{in-scattering}} . \tag{2}$$

Four terms, and each later section is about one of them. Free streaming is §9; the gravitational source closes at rank two and is §*The generic source derivatives*; the two κ' terms are the two halves of one collision term and their separation is what §10.1 turns on.

Both κ' are the same κ'. The out-scattering on the left removes photons from the beam and the in-scattering on the right returns them; they are one Thomson process written on two sides. **That is why multiplying through by the integrating factor assigns both weights at once rather than by choice**, which is (13)'s content.

5.3 What "exact" claims, and what it does not

Claimed. (2) is exact in the $1+3$ split: no expansion in a perturbation parameter has been made in deriving it, no background has been subtracted, and no gauge has been fixed. It holds for a general u^a in a general geometry.

Not claimed. That the *solution* computed here is exact. Every approximation this release makes enters after (2) and is named — the scattering-time expansion, **A-3**'s static transfer, **A-5**'s collision term, the diffusion treatment — and §18 carries what is not costed.

The sources are gauge-invariant, and for a reason worth stating once. $\mathcal{B}_0$, $\mathcal{B}_a$ and $\mathcal{B}_{ab}$ all vanish in the background, so by **Stewart–Walker** they are gauge-invariant at first order — which is the paper's own *"note that these are CGI with respect to u^a"*. **Gauge-invariance here is a consequence of the objects chosen, not a construction imposed on them**, and that is the difference between a covariant treatment and a gauge-fixed one with covariant notation.

What the collision term contains, and what A-5 drops. *Annals II* writes $C[\tau] \simeq \sigma_{\rm T} n_E(\bar{f}_1 e^a v_a^B + \bar{f}_2 e^a e^b \tau_{ab} - \tau)$. This release takes $\bar{f}_1 = 1$ and $\bar{f}_2 = 0$, so $\mathcal{S}_2 = 0$: **no polarisation feedback and no quadrupole in-scattering**. That is approximation **A-5** stated at the collision term rather than at the oscillator, and §10.1 shows it cannot produce a frame disagreement at linear order.

5.4 Where the hierarchy is gated

gate	what it tests	what it does not
G1, criterion 1	the free-streaming part of (2) against four external hierarchies, two normalisations, two phase conventions	anything with the source on — §9
`test_register`	that every module, object and test named in the equation and notation tables resolves	that the equation implemented is the one printed
C2 round trip	covariant → mode → covariant, identically zero in exact arithmetic	that either representation is the right one — that is criterion 3, reopened

No gate tests (2) **as a whole**, and none can: it is the antecedent's equation, read from the page. Its standing is *inspection* — §*the acceptance spine* — and every numerical claim in this supplement is about what was done *with* it.

6 The generic source derivatives

*Decision **S26**. Written from `provenance/GENERIC-SOURCE-DERIVATIVES-v1.md`, which carries the working and the open items.*

What (102) needs, and what the paper never wrote. (102)'s integrand is $C_0 - C_1' + C_2''$ and its boundary terms carry $C_2'(\eta_*)$. With (90),

$$-C_0 \simeq +\tfrac{5}{2} a\mathcal{B}_2, \qquad -C_1 \simeq \tfrac{1}{k}(a\mathcal{B}_1 + \kappa' v_B), \qquad -C_2 \simeq \tfrac{1}{k^2} a\mathcal{B}_2,$$

Two entries of that display are corrected in §10. The $+\frac{5}{2}$ on C_0 is α_2, the rank-*two* normalisation, applied to a monopole row where $\alpha_0 = 1$ belongs — finding **T-104**, a factor of $\frac{15}{2}$ — and the $\kappa' v_B$ inside C_1 is a bare κ' in a *coefficient*, which §10.3 forbids and **T-101** measures on the released arrays. Both are stated here at the point of use so a reader of this display does not meet the correction four hundred lines later.

so the solution needs $(a\mathcal{B}_1)'$, $(a\mathcal{B}_2)'$ and $(a\mathcal{B}_2)''$ **for arbitrary** u^a. *Annals II* prints these only after the Newtonian specialisation has removed the shear, which is exactly the specialisation this bundle declines to make early. The derivations are therefore assembled here, and this section records what is closed, what is assembled, and what is open.

The three sources, by rank, with no frame chosen. From §4's integrated Boltzmann equation $\dot\tau + e^a \mathrm{D}_a\tau + \mathcal{B} \simeq C[x,e]$,

$$\mathcal{B} = -\tfrac{1}{3}\mathrm{D}^a\tau_a + \big(\mathrm{D}_a \ln T + A_a\big)e^a + \sigma_{ab}e^a e^b,$$

giving $\mathcal{B}_0 = -\frac{1}{3}\mathrm{D}^a\tau_a$, $\mathcal{B}_1 = \mathrm{D}_a \ln T + A_a$ and $\mathcal{B}_2 = \sigma_{ab}$. **All three vanish in the background**, so all three are gauge-invariant at first order by Stewart–Walker — which is the paper's own *"note that these are CGI with respect to* u^a*"*.

The scale-factor rule, and why the footnote to (105) carries $2H$**.** A rank-n PSTF source's natural variable carries a^n: the paper writes $(a\tilde{\mathcal{B}}_1)'$ at (106) and $(a^2\sigma_{ab})^{\cdot}$ in Appendix G. So

$$\dot{\mathcal{B}}_1 + 2H\mathcal{B}_1 = \frac{1}{a^2}(a^2\mathcal{B}_1)^{\cdot},$$

and the rank-two object built from a rank-one source, $\mathrm{D}_{\langle a}\dot{\mathcal{B}}_{b\rangle}$, carries the rank-two power. **It is not a commutator correction bolted on; it is the rank showing up in the measure.** That is conventions row **C3.1a** read in the direction nobody had needed, and it is checkable: under C3.1, C3.2 and C4.2 the mode functions are time-independent at every rank, so *all* the H's come from the a^n and none from the Q's — which **predicts (106)'s** $2H$ **before (106) is consulted**. Implementing it the other way, by splining a stored amplitude, is wrong by a median of 100%: finding **T-23**, and the reason `integral_solution` takes every derivative as an input rather than differentiating one.

Rank 2: closed. $\dot{\mathcal{B}}_2$ *is* in the paper generically, in Appendix G, and closes to second order **with an exact symbolic control** that fails when the rank-two term is removed — so the check is one that can fail, which S27's C rung requires and finding **T-24** exists because an earlier one could not.

Rank 1: assembled, and carrying one term more than (105) prints. $\dot{\mathcal{B}}_1$ is assembled from printed equations. Three of its four terms close exactly against (105)/(106) — $\mathcal{B}_0$, $\dot\Phi_A$ and $-2H\Phi_A$, the last only because both of its contributions are present — and substituting the expansion–potential relation of §*newtg* reproduces every term (105) prints. The generic form then carries **one extra**:

$$(\star) \qquad -H\,\mathrm{D}_{\langle a}\tilde{\mathrm{D}}_{b\rangle}\ln T.$$

This is the temperature-gradient half of $-H\tilde{\mathcal{B}}_a$, whose acceleration half *does* survive into (105). **Two readings, and this bundle chooses neither.** It may be the small-scale approximation — §6 drops all terms of $O(\epsilon\epsilon_H)$ and $(\star)$ has that shape — except that so does $-2H\mathrm{D}_{\langle a}\mathrm{D}_{b\rangle}\Phi_A$, which is kept. Or it may be an omission, in which case (105), (106) and what descends from them are short a term of the same order as ones they keep. It is first order and it does not obviously vanish, which is enough to make the question real; settling it needs §2.5's $O(\epsilon\epsilon_H)$ bookkeeping read against each term of (105), not a judgement about which is negligible. **Logged as open, not filed as a finding against the paper.**

Rank 0: not needed. Un-withdrawn 2026-10-22: the claim stands, the reason does not. This paragraph read: *"*$\dot{\mathcal{B}}_0$ *is not required by (102) at all —* C_0 *carries* $\mathcal{B}_2$*, not* $\mathcal{B}_0$*."* It was withdrawn on 2026-10-21 on the authority of **T-105**, which held that (89)'s $a\mathcal{B}_0\delta_{\ell 0}$ row had been dropped in error. **T-105 has since fallen, three times over**, and the claim above is restored with a different reason.

$\tau_0 \equiv 0$ is a *constraint*, not initial data: T is *defined* as the monopole of $T(x,e)$, so differentiating $\int A_0\,\mathrm{d}\eta' = 0$ puts **zero** on the right of (89)'s $\ell = 0$ row — $\mathcal{B}_0$ is a functional of the solution and not an external forcing. Solving that row with α_0 in place of α_2 and $h_0''(0) = -\frac{1}{3}$ gives $C_0 = -\frac{1}{3}a\mathcal{B}_2$ with *no* $\mathcal{B}_0$ in it, and the factor against (90)'s printed $-\frac{5}{2}a\mathcal{B}_2$ comes out as $\alpha_2/\alpha_0 = \frac{15}{2}$ exactly — finding **T-104**, which is *cleaner* standing alone. Independently, $a\mathcal{B}_0 = -(k/3)\tau_1$ is an identity and $-C_1'$ carries $+(k/3)\tau_1$ already, so restoring $a\mathcal{B}_0$ to C_0 counts the monopole source twice. The measurement agrees: criterion 5b doubles, to 98.5%, under the restoration — finding **T-107**.

So $\mathcal{B}_0$'s *content* is required and its *row* is correctly empty, and both statements hold at once. The released code has carried this since 2026-10-21: `MONOPOLE_ROW = "corrected"` is $C_0 = -\frac{1}{3}a\mathcal{B}_2$, and the restoration is kept only as the regenerable control that refuted it.

Status. The derivation is complete for what the integral solution needs. What remains open is ($\star$). The $a\mathcal{B}_0\delta_{\ell 0}$ row of (89) is **not** open and is **not** dropped in error — it is correctly absent, for the reason above. $\Delta C_0 = +kv/3$ is non-zero and its partner is cross-coefficient, in $-C_1'$ rather than in C_2, which is why the rank-0 partner check `c8` scores the bracket and the single-coefficient residual γ cannot: **M15**. A γ driven to zero by adding a term is a fit, not a gate; the $\ell = 1$ and $\ell = 2$ rows are controls because they come out at unity without adjustment.

7 Frames and specialisation

"Frame" is being asked to do two jobs, and they are different objects. The word is used in this literature for something specified at a single event and for something specified throughout a region, and the two are not interchangeable. The PI's enumeration, recorded here because the rest of this section depends on it:

Object	What is specified	Where
Instantaneous observer rest frame	A unit timelike vector $u^a(p)$, determining a local time direction and an orthogonal rest space	At an event p
Local orthonormal frame	A tetrad $e_{\hat{\alpha}}{}^a(p)$, with $e_{\hat{0}}{}^a = u^a$, also choosing three spatial axes	At an event
Observer worldline	A timelike curve with tangent u^a	Along one observer's history
Threading	A smooth timelike congruence, represented by a unit tangent field $u^a(x)$	Throughout a spacetime region
Adapted orthonormal frame field	A tetrad field whose timelike leg follows the congruence	Throughout that region

Nothing in this bundle is a tetrad. No calculation here chooses spatial axes: every quantity is a PSTF tensor built from u^a and $h_{ab} = g_{ab} + u_a u_b$, so rows two and five are named only to be set aside. What the bundle actually uses is rows one and four, and it uses both:

- the **receiving observer's frame** — a u^a at one event, reception at η_0 — which is what the *measurement* is referred to, and in particular what the dipole is a statement about;
- the **CDM threading** and the **Newtonian threading** — unit tangent fields $u^a(x)$ on the whole region between last scattering and reception — which is what the *propagation* integral is carried along, and what σ_{ab}, A_a and q_a are properties of.

The general transformations (31)–(37) relate two *threadings*; the specialisations (40)–(45) stipulate one. $\tilde{\sigma}_{ab} = 0$ and $\tilde{A}_a = 0$ are conditions on a congruence throughout a region and are not statements about any single observer, which is why neither can be imposed "at the observer" and why the Newtonian one

is admissible only in restricted cases. Figure F7's two panels are exactly this pair kept apart: the $\ell \geq 2$ panel compares two *threadings* carrying the same observable, and the dipole panel is where the *receiving observer's frame* enters, because the dipole is the one multipole that is a statement about u^a at reception rather than about the congruence that brought the radiation there.

The words this bundle uses, and why none of them is retired. *Threading* names the congruence **as an object**. *Receiving observer's frame* names the local u^a **at reception**. *Frame* continues to name a **choice of observers** and, following *Annals II* §4.1's own usage, each named choice — the energy frame, the zero-acceleration (CDM) frame, the Newtonian frame. Every one of those is a threading in the first sense: the words sit at different levels and are not competing. Decision **S29a**.

No identifier is renamed. The code says `"cdm"` and `"newtonian"`, and *Annals II* prints "frame" for exactly these objects, so the code agrees with the source it reconstructs. The rule that a word naming two things should have one of them renamed is satisfied here by **naming the two objects**, which is what the table above does. And the rename operation is itself the hazard: a blanket substitution in this bundle inverted the meaning of diagnostic D7, whose subject is the CDM *threading* measured against the **total-energy** frame — the one place both senses appear in a single sentence, and therefore precisely what a substitution cannot see. Finding **T-33**. **A rename a regular expression can perform is a rename whose exceptions nothing will catch.**

Specialisation. The solution is carried with a **generic** u^a. The explicit frames appear as choices made at the end, not assumed at the start: the Newtonian frame $\tilde{\sigma}_{ab} = 0$, which is the specialisation *Annals II* itself makes at p. 365 in order to remove the shear terms, and the **CDM frame** $\tilde{A}_a = 0$. Carrying the generic frame and specialising late is the difference the $1 + 3$ covariant approach buys; a gauge-fixed treatment fixes the frame before the hierarchy is solved and cannot exhibit it.

"CDM" names two different things, and this supplement keeps them apart. *CDM the model* is a cosmology — flat, matter dominated, no Λ — and it is one of the two models this release delivers a spectrum for. *The CDM frame* is a choice of frame u^a: the congruence in which the cold dark matter is at rest. The two are independent. The ΛCDM *model* is computed in the same *frame*, and a change of model is not a change of frame. In the code the models are named `CDM_MODEL` and `LCDM_MODEL` and the frames `"cdm"` and `"newtonian"`, so the two cannot be confused at a call site.

The CDM frame is *not* the total-energy frame, and an earlier version of this supplement said it was. The definition used here is exact in general relativity and needs no approximation: cold dark matter is **pressureless and collisionless**, so it is acted on by no non-gravitational force and its worldlines are **geodesics**. The CDM frame is therefore the frame with $\tilde{A}_a = 0$, which is what *Annals II* §4.1 calls the *zero acceleration frame*, listing it as the second of three useful choices — the first being the **total-energy frame** $q_a = 0$ and the third the Newtonian frame. They are **three stipulations, not two**, and the paper gives the reason this calculation wants the second: it is "geodesic right through tight-coupling, slow decoupling and into the free-streaming era", whereas the energy frame is recommended there for *two* coupled species and this is a three-species problem that crosses decoupling.

The two agree only in the limit of matter domination. Setting $q_a = 0$ in a mixture gives $\rho_C v_C^a + \frac{4}{3}\rho_R v_R^a + \ldots = 0$, so in the CDM rest frame the total flux is *not* zero — the baryons and the photons are moving — and the energy frame tilts away by $(\rho_b + \frac{4}{3}\rho_R)/\rho_C$. Measured in diagnostic D7, that is **0.24** at last scattering for this release's background and **0.49** for the stamped Planck column, with the baryons carrying 37% of it. It is largest exactly where the acoustic physics is, and falls afterwards on both counts, since ρ_R/ρ_C decreases and the baryons stop moving with the photons once they decouple. Recorded as finding **T-29**; the amendment is decision **S8a**.

The one-line reconciliation, and it is the PI's. The CDM frame is the energy frame *of the CDM component*. For pressureless dust $T^C_{ab} = \rho_C u^C_a u^C_b$, so its flux relative to a frame $u^a = u^a_C + v^a$ is $q^C_a = \rho_C v_a$ to first order, and $q^C_a = 0$ gives $v_a = 0$ — the CDM at rest, exactly. So the two conditions are the same statement about one species, and the conflation this supplement previously carried was the dropped qualifier: "energy frame" unqualified means the *total*, and the total includes the baryons and the photons, which are moving. Saying *of what* settles it.

The frames, and what each one fixes. *Annals II* §4.1 enumerates four (its sentence says "three"); the baryon and radiation rows below are frames the paper uses elsewhere and are included because they are the ones a reader is most likely to reach for next.

Threading	Defining condition	What it establishes
CDM *(used here)*	$u^a = u^a_C$, equivalently $\tilde{A}_a = 0$	CDM is at rest; its congruence is geodesic in the conserved-dust model, which is exact rather than approximate. *Annals II* calls it the *zero acceleration frame*.
Total energy	$q^{\text{total}}_a = 0$	Total momentum density vanishes; **individual species can still move**. Agrees with the CDM frame only to $(\rho_b + \frac{4}{3}\rho_R)/\rho_C$.
Baryon	$u^a = u^a_B$	Baryons are at rest; pressure and scattering can accelerate them, so this frame is **not** geodesic.
Radiation energy	$q^\gamma_a = 0$	The radiation brightness dipole vanishes; **higher multipoles can remain**.
Newtonian *(used here)*	$\sigma_{ab} = 0 = \omega_a$, where admissible	Useful for the scalar almost-FLRW comparison and for the Bardeen link; generally accelerated. Admissible only in restricted cases.
Constant expansion	$\mathrm{D}_a\Theta = 0$, where admissible	Removes the spatial expansion gradient relative to that frame; sometimes useful on small scales. Not used here.

Reading down the second column is the point: **each row fixes a different thing, and no two of them are the same condition**. A frame is chosen and then named, not inferred from what it is called.

One of these frames contains observers and the other does not. This is the reason the distinction is worth the space, and it is where the $1+3$ covariant account differs from a metric one in substance rather than in notation.

The **CDM frame is a congruence of observers**. It is defined by a property of the matter — pressureless, collisionless, therefore geodesic — and the definition is exact in general relativity, not a first-order statement. We observe from inside it: a detector in a bound structure is comoving with the matter to the accuracy linear theory works at, and after decoupling the baryons have fallen into the same flow as the cold dark matter, so the baryon and CDM frames coincide at η_0 where the measurement is made. *It is "us".*

The **Newtonian frame is not**. $\tilde{\sigma}_{ab} = 0$ is a condition imposed to simplify, not a property any matter has, and *Annals II* §4.1 says so in terms: the frame "is only consistent in restricted cases" and is "particularly useful in making comparisons with much of the analytic literature". It is the covariant counterpart of what conformal Newtonian gauge does in a metric treatment — admissible about almost-FLRW, and a device.

Why this costs nothing at first order and will not stay free. τ_{A_ℓ} for $\ell \geq 2$ is frame-invariant at first order, so the *observable* does not care which frame carried it there, and a calculation performed in the Newtonian frame returns what an observer in the CDM frame measures. That is the whole licence for the Bardeen-variable route, and it is a first-order licence.

At $O(\epsilon^2\ell)$ it lapses. The couplings v1.5.0 adds are products of two first-order quantities, and a product of two frame-dependent quantities is not frame-invariant just because each factor's leading frame-dependence cancels. Then the frame has to be the one the observer is actually in, and that is the CDM frame, not a device.

This is what figure F5 is for. Its third panel is the amount of machinery that is frame-dependent — in the Newtonian frame one rank carries everything, in the CDM frame all three are occupied — and its first two panels are the first-order invariance that currently makes that difference harmless. The figure shows both at once, which is why it is the right figure to carry into v1.5.0: it is a picture of exactly what stops cancelling.

What the external codes use, and why criterion 5b has to check it. Criterion 5b compares this bundle against CAMB and CLASS. That comparison is only meaningful if the frame each code evolves in is known, because a quantity compared across two frames is not the same quantity. The correspondence below is **verified where it could be, and recorded at its real strength where it could not**; verifying it is a **precondition of 5b**, not an afterthought.

CLASS is verified from its own source, at `classy v3.4.1`. `perturbations.c` carries the comment *"cdm velocity vanishes in the synchronous gauge"*, and `index_pt_theta_cdm` is allocated as an evolved variable only when `gauge == newtonian`: in synchronous gauge θ_C is not evolved because it is zero. The decisive line is CLASS's own error message — *"In the synchronous gauge, it is not self-consistent to assume no CDM: the latter is used to define the initial timelike hypersurface"* — which says, in the code's own words, that **its synchronous gauge is realised by the CDM congruence**. That is the CDM threading of the table above, $u^a = u^a_C$. Both `gauge = synchronous` and `gauge = newtonian` were run and computed.

CAMB is corroborated but not verified to the same standard. Its Python layer exposes `v_newtonian_cdm` and `v_newtonian_baryon` as *derived outputs*, notes that *"L=1 dipoles are Newtonian Gauge"*, and carries an explicit `frame` argument for sources that are not frame-invariant — which is what a code evolving in one frame and converting on output looks like. That is corroboration from an interface, not a reading of the evolution equations, and the decisive statement lives in CAMB's own covariant formulation paper rather than in anything this bundle inspected. It is recorded at that strength and no higher.

Code	**Gauge used for evolution**	**Corresponding $1+3$ frame**
CLASS *(default)*	Synchronous, with CDM velocity divergence $\theta_C = 0$	CDM-comoving: $u^a = u^a_C$, $A_a = 0$; shear generally nonzero.
CLASS *(optional)*	Conformal Newtonian / longitudinal, selected with `gauge = newtonian`	Newtonian: scalar shear and vorticity vanish; acceleration generally nonzero.
CAMB	Covariant equations evaluated in the CDM frame, equivalent to CDM-comoving synchronous gauge	CDM-comoving: $v_C = 0$, $A_a = 0$; shear generally nonzero.
Bolt.jl	Conformal Newtonian gauge, implemented as `BasicNewtonian`	Newtonian: CDM carries an explicitly evolved peculiar velocity relative to this frame.

Two things follow if the table holds. First, **CAMB and CLASS's default both evolve in the frame this bundle uses**, which is the convenient case: 5b compares like with like without a frame transformation in between. Second, **CLASS's `newtonian` option and Bolt.jl evolve in the other one**, so a comparison against either is a comparison across frames and must either transform or restrict itself to the frame-invariant $\ell \geq 2$ multipoles. Figure F5 is the measurement of what that costs: the totals agree and the decompositions do not.

That CAMB is described as using covariant equations in the CDM frame is the part most worth checking rather than assuming, because if true it makes the comparison closer than a gauge-versus-frame

translation would suggest, and a comparison that is closer than it should be is the kind this bundle has already been wrong about once.

For readers coming from the metric approach: a frame is not a gauge. This is the distinction most easily lost in translation, and it is worth stating plainly because the two freedoms are often fixed together in a metric-based treatment and so look like one freedom.

- A **gauge** is a choice of coordinates — equivalently a choice of correspondence between the perturbed spacetime and the background. In the $1+3$ covariant approach it is **already dealt with**: every variable used here vanishes in the background, so by the Stewart–Walker lemma it is gauge-invariant by construction. There is no residual gauge mode to fix and none to worry about.
- A **frame** is a choice of the timelike congruence u^a along which the fluid and radiation variables are defined. It is a *physical* choice — which observers — and it survives after the gauge question has been answered. It is the only freedom left, and it is the one this release exhibits.

So the familiar correspondences are correspondences and not identities. Synchronous gauge is usually *completed* by frame along the CDM, and conformal Newtonian gauge is usually *completed* by frame along a shear-free congruence; in each case the gauge condition fixes coordinates and a separate stipulation fixes the observers. A statement that is frame-dependent here will appear in a metric treatment as a statement that depends on which gauge was chosen, and the two dependences are not the same dependence. That is why figure F5 can show the *total* common to two frames while the *decomposition* differs: the split into ranks is frame-dependent and the observable is not, and neither fact is about coordinates.

This is also what fixes the status of the next release. Because v1.0.0 is the general case, **v1.5.0's restricted case is a recovery and a check** — it must return v1.0.0's numbers, and that requirement is the regression test rather than a new result.

The covariance claim is established symbolically, and that is the stronger form. Acceptance criterion 2 asks that a change of frame $u^a \to \tilde{u}^a = u^a + v^a$ return the rank-one bracket unchanged up to $\dot{v}_a$. It does, *identically* in exact arithmetic rather than to a tolerance, and the cancellation is exhibited rather than quoted:

- $\tilde{\mathrm{D}}_a \ln T = \mathrm{D}_a \ln T - H v_a$, the extra term arising because the projector moves with the frame and $\dot{\overline{\ln T}} = -H$ in the background;
- $\tilde{A}_a = A_a + \dot{v}_a + v^b \nabla_b u_a$, where $v^b \nabla_b u_a$ is formed *in full* — $H v_a$ together with $\sigma_{ab} v^b$ and $\epsilon_{abc} v^b \omega^c$ — and the latter two leave at $O(\varepsilon^2)$ by series expansion rather than by hand;
- so $\tilde{\mathcal{B}}_1 = \mathcal{B}_1 + \dot{v}_a$ with **no shear and no** Hv. It is the *same* H twice, which is why the cancellation is structural and not a coincidence of two fitted coefficients.

The curvature half. The deflection depends only on $\Phi_A - \Phi_H$, whose trace-free screen-projected Hessian is the electric Weyl tensor, $E_{ab} \simeq \frac{1}{2}\mathrm{D}_{\langle a}\mathrm{D}_{b\rangle}(\Phi_A - \Phi_H)$. Since E_{ab} vanishes in the background, the Stewart–Walker lemma makes it frame-invariant at first order, and therefore so is $\Phi_A - \Phi_H$. Both potentials shift by the same $(v' + \mathcal{H}v)/k_{\mathrm{com}}$, and the residual $\tilde{E}_{ab} - E_{ab}$ is identically zero — **with a negative control**: a rule leaving Φ_H fixed, or flipping its sign, leaves a residual proportional to $(v' + \mathcal{H}v)$ and fails. The invariance is therefore forced rather than chosen to make a residual vanish. Only the dipole moves, $\tilde{\tau}_a \simeq \tau_a - v_a$; every τ_{A_ℓ} with $\ell \geq 2$ is unchanged, which is the standard any frame dependence surviving at $\ell \geq 2$ must meet: it is an artefact of the reduction, not physics.

The energy frame is derived, not stipulated. Imposing $\tilde{A}_a = 0$ gives $v' + \mathcal{H}v = -k_{\rm com}\Phi_A$ — the Euler equation of a pressureless geodesic fluid, which is what the cold dark matter obeys. The normalisation is corroborated independently: in the baryon-dominated limit $R \to \infty$, Eq. (152) reduces to that same equation, so the photon dipole and the frame velocity carry one normalisation.

What this release does *not* do, stated because the earlier draft of this section claimed it. It does not compute an assembled spectrum in two frames and draw the difference. Eqs. (177)–(179) are **Newtonian-frame expressions**: they descend from (106), which the corrections record flags in those words, and the Newtonian frame carries its whole $O(\ell)$ source in the acceleration pair while the CDM frame carries its whole source in the shear pair. Transforming the four input fields and re-evaluating the same equations therefore deletes the coupling that was carrying the source without supplying the one that replaces it: measured, the transfer function moves by a factor of about five, and under the opacity weighting the source collapses to two parts in a thousand. That is not a frame dependence of the sky. The figures which would draw it, and the convergence figure which would test footnote 29 on p. 365 in both frames, require the generic-u^a hierarchy and are deferred with it.

One coefficient is changed for criterion 5b, and only there. The photon quadrupole's Thomson collision term carries $\Pi = \Theta_2 + E_0 + E_2$, so polarisation feeds back into the temperature hierarchy and strengthens the diffusion damping. CAMB and CLASS both include this **structurally**: it is not an output option, and neither can be asked to leave it out — CLASS fails outright in `perturbations_solve` below `l_max_pol_g = 2`, because Π needs E_2, and CAMB exposes no switch for it. Truncating CLASS's polarisation hierarchy from 10 to 4 moves the acoustic band by 0.043%, which bounds the hierarchy's *depth* and not the *feedback*.

The only way to compare like for like is therefore to carry the polarised coefficient here, for that comparison alone. In (dif-damp)'s quadrupole term *Annals II* prints 4/5; Hu & Sugiyama's unpolarised form is 8/9; with polarisation it is 16/15. Measured on `PLANCK_MODEL`:

	coefficient	$k_D(\eta_*)$	ratio
Annals II (dif-damp), as printed	4/5	**619.9**	1.000
Hu & Sugiyama, unpolarised	8/9	588.1	0.949
with polarisation	16/15	**536.9**	**0.866**

a 13.4% reduction in k_D, which takes $e^{-(k/k_D)^2}$ from 0.660 to 0.574 at $k = 400$ and from 0.189 to 0.109 at $k = 800$. It is one of the dominant terms in criterion 5b's residual.

4/5 **remains the default and the released value.** Every figure, diagnostic and released number in this bundle uses it, because the bundle reproduces and 4/5 is what the page prints. 16/15 is reached only through an explicit argument, by criterion 5b. Decision **S31**, finding **T-39**. *Annals II*'s 4/5 matches neither standard coefficient, which is recorded as a **question** — it may be a convention in how R and the factor of 6 are distributed — and not as a correction.

8 The acoustic oscillator

This is what supplies $\delta T(\eta_*, k)$ and $\tau_1(\eta_*, k)$ to the matching. Its period is verified against the antecedent's own formula to 0.5%*; its phase is not verified at all, and the distinction is the most useful thing this section says.*

8.1 The pair, and why (154) is not the object

Annals II §7.1 prints three equations. Two are the system:

$$\dot{\tau}_1 \simeq -\frac{\dot{R}}{1+R}\,\tau_1 - \frac{1}{1+R}\frac{k}{a}\,\delta T - \frac{k}{a}\Phi_A, \tag{152}$$

$$\dot{\delta T} \simeq -H\Phi_A - \dot{\Phi}_H + \frac{k}{a}\frac{\tau_1}{3}, \tag{153}$$

with $R = 3\rho_M/4\rho_R$ and $c_s^2 = \frac{1}{3}(1+R)^{-1}$. The third, (154), is the single second-order oscillator in δT, and the paper obtains it from the pair by *"ignoring the expansion coupled term (i.e. the small scale approximation)"*.

The pair is implemented and (154) is derived from it, not the other way round. In conformal time, multiplying through by a,

$$\tau_1' = -\frac{R'}{1+R}\tau_1 - \frac{k}{1+R}\delta T - k\Phi_A, \qquad \delta T' = -\mathcal{H}\Phi_A - \Phi_H' + \frac{k}{3}\tau_1, \tag{3}$$

and dropping $-\mathcal{H}\Phi_A$ reproduces (154) exactly.

That ordering is the point and not a preference. Implementing (154) directly would bake the small-scale approximation into the only object that exists, and no figure could ever show its cost. Implemented as a pair with a switch, the approximation becomes a measurement: the difference between the two integrations *is* its size, at any k and any η, and the gate asserts what makes it a *small-scale* approximation — that it must matter less as k grows. *This is the same move as the scale-factor rule of §*the generic source derivatives*: reconstruct the antecedent's general object, and let its printed specialisation fall out as a checkable consequence.*

8.2 What drives it, and the one place the backgrounds differ

Φ_A and Φ_H are **supplied** to the oscillator, not solved with it, which is what lets one routine serve every background. They come from the growth equation,

$$\delta'' + \mathcal{H}\delta' - \tfrac{3}{2}\Omega_m H_0^2\,\delta/a = 0, \qquad \Phi \propto D(a)/a, \tag{4}$$

integrated rather than fitted, so that $D \propto a$ in Einstein–de Sitter is a *result* and can be asserted as one.

And that is where the two backgrounds part company. In Einstein–de Sitter $D \propto a$ exactly, so Φ is constant and $\Phi_A' - \Phi_H'$ vanishes *identically*: there is no integrated Sachs–Wolfe term at all, which is why the antecedent's standard-CDM model never needed one. Once Λ dominates, growth stalls while a keeps rising, D/a falls, and Φ decays. **That decay is the late integrated Sachs–Wolfe effect, and it is not bolted on — it appears by itself from** (4) **the moment** $\Omega_\Lambda \neq 0$**.** Whether it then reaches the observer is a question about the weight it is carried under, which is approximation **A-1** and §10.1.

Einstein–de Sitter is therefore a null control and not a weak case. A diagnostic that claims to separate the late ISW must return *exactly zero* on it. A small number is a failure, not a pass.

The radiation era is absent, and that absence is A-3. The released Φ is the matter-dominated one and has no radiation era in it. The radiation-era potential is derived in closed form — with $c_s^2 = \frac{1}{3}$ and $\mathcal{H} = 1/\eta$ the evolution equation becomes $\Phi'' + 4\eta^{-1}\Phi' + \frac{1}{3}k^2\Phi = 0$, whose regular solution is

$$\Phi(\eta) = 3\Phi_i\,\frac{j_1(x)}{x} = 3\Phi_i\,\frac{\sin x - x\cos x}{x^3}, \qquad x = \frac{k\eta}{\sqrt{3}}, \tag{5}$$

so $\Phi \to \Phi_i$ as $x \to 0$ and $\Phi \simeq -3\Phi_i \cos x/x^2$ for $x \gg 1$: **decaying as** x^{-2} **while oscillating**. Because it decays on the same timescale the mode oscillates, it drives the oscillator *resonantly*, and that amplification is missing from every peak *height* this bundle produces. (5) is implemented as a diagnostic and is not wired into the released pipeline; **D24** shows why that ordering is right, since A-3 is not connected to the peak *position* until C_1' is repaired.

8.3 Two corrections to the antecedent, both settled by a stated limit

B-8: (181) carries the wrong sign on its constant term. The closed-form solution at last scattering is

$$\delta T(\eta,k)+\Phi_A(\eta_*,k)\simeq\big[\delta T(0,k)+(1+R)\Phi_A(0,k)\big]\cos(kr_s)-R\,\Phi_A(\eta_*,k), \tag{180}$$

and (181) specialises it with the adiabatic condition $\delta T(0,k)=-\frac{2}{3}\Phi_A(0,k)$. Carrying that through,

$$\big[\delta T+\Phi_A\big](\eta_*,k)\simeq\tfrac{1}{3}\Phi_A(0,k)\,(1+3R_*)\cos(kr_s^*)\;-\;R_*\Phi_A(0,k), \tag{6}$$

where the printed (181) has $+R_*$. **Its own stated limit settles which.** At $r_s^*\to 0$ the corrected form gives exactly $\frac{1}{3}\Phi_A$, which is what the text asserts and what (196) descends from; the printed form gives $\frac{1}{3}\Phi_A(1+6R_*)$, and the assertion would need $R_*\to 0$, which is neither stated nor true at last scattering. The oscillating parts agree exactly, so this is one sign and not a different solution. *Taken literally it costs a factor of twenty in C_ℓ*, and the printed form is kept as a second function so that the difference is demonstrable rather than asserted.

B-10: (J.5) is printed with the exponent of the wrong sign. The Bond–Efstathiou transfer function $T(k)=[1+((ak)+(bk)^{3/2}+(ck)^2)^\nu]^{-1/\nu}$ is printed with $+1/\nu$. As printed it *grows* without bound — 6.7×10^3 at $k=10\,\mathrm{Mpc}^{-1}$ against 1.5×10^{-4} — which inverts the physics, since modes entering during radiation domination stagnate and the transfer function must suppress small scales. **The large-scale limit cannot see it**, because the bracket tends to 1 and $1^{\pm1/\nu}=1$, which is why the text's own check, *"$T(k)\sim+1$ on large scales"*, passes on the wrong form. Only the k^{-2} falloff distinguishes them, and that is what the fit is a fit to.

Both are the same shape of error and both are caught the same way: by evaluating the antecedent's own stated limit on the antecedent's own printed expression. Neither needed an external code.

8.4 The phase is the sound horizon, and the start is derived from k

δT oscillates in kr_s, not in $k\eta$, so every number in this section depends on r_s being the sound horizon *from the big bang*, $r_s(\eta)=\int_0^\eta c_s\,\mathrm{d}\eta'$ — which is also what the antecedent means in its own peak formula $\ell_j\simeq|r_\theta/r_s|\,j\pi$.

Finding T-51 is that it was not. The sound horizon was accumulated from the first point of the line-of-sight grid, at $z=3000$, returning between 36% and 49% of r_s, so $\cos(kr_s)$ carried roughly half the phase it should. **And criterion 5b could not see it**, because the closed-form route inherited the same truncation independently of the ODE route: the two agreed, and they agreed about what they shared. *That is **T-3**'s rule on a third quantity — two agreeing routes agree about what they share, and a shared input is not cross-checked by them.*

The repair is a derivation and not a tuned start. A mode may be started where $k\,r_s(\eta_{\mathrm{start}})\le\varepsilon$ with $\varepsilon=0.1$ radians, one part in sixty of a cycle, because (180)'s initial condition is the adiabatic growing mode only while the mode is outside the sound horizon; starting at finite kr_s *imposes* a phase rather than inheriting one. The bundle reached $kr_s=10.7$ at $k=600$ — nearly two full cycles in. The grid is derived the same way: $c_s^2\le\frac{1}{3}$ bounds the period below by $2\pi\sqrt{3}/k$, so the sample count is arithmetic rather than a knob, at 24 samples per oscillation.

One number in this section is a modelling choice and is now stated once. Tight coupling is held to end at $\eta_{\mathrm{couple}}=\min(\eta_0,\,6\,\eta_*)$. Six is not derived and is not claimed to be. **T-58** is what happened while it was not stated once: the factor lived in three places and the route criterion 5b gates integrated the oscillator all the way to η_0, keeping the photon–baryon fluid ringing for seven times the span over which it is a fluid. The mismatched window alone costs 1.5×10^{-1} at $k=3$ and 1.6 at $k=10$, against 5b's own 5×10^{-8} — and 5b could not see that either, for the same reason. **Q16** is the assertion that closes it: $\mathcal{V}(\eta_{\mathrm{couple}})/\mathcal{V}(\eta_*)$ below a stated threshold on every background.

8.5 Where the oscillator is gated, and the one thing no gate sees

gate	what it tests	value
peak spacing	$\Delta\ell = \pi r_\theta / r_s$, the antecedent's own formula, read off the turning points of $\delta T + \Phi_A$ in k on Einstein–de Sitter, where Appendix G is exact	191.7 against 192.7, ratio **0.995**
the same, `z_max` control	what the pre-T-51 start does to it	568.7, ratio 2.95
pair → (154)	that dropping $-\mathcal{H}\Phi_A$ reproduces the printed oscillator, and that the difference falls with k	exact; small-scale behaviour asserted
ODE → (180)	that the integration converges to the closed form when R is held constant	—
B-8	the $r_s^* \to 0$ limit of both forms of (181)	$\frac{1}{3}\Phi_A$ against $\frac{1}{3}\Phi_A(1 + 6R_*)$

Every one of these measures a period, a limit or a reduction. Not one of them measures a phase. The spacing gate takes differences of consecutive turning points, so the position of the *first* one never enters: it is blind, by construction, to a constant offset in kr_s.

That is why it is worth having, and it is the sharpest localisation in this supplement. The oscillator's period is right to 0.5% and the projection at the first peak is faithful to 3% (**D25**, §10.1), while the released first peak sits at $0.635\,\ell_A$ rather than 0.757. **Correct spacing with a displaced first peak is a phase offset, not a scale error**: nothing here is stretching r_s or r_θ. The defect lies between the oscillator and the projection, which is the transport assembly, and §10.1 is where that is itemised.

9 Free streaming, and where every $\ell \geq 3$ comes from

The source closes at rank two and the oscillator runs at ranks zero and one. Nothing so far produces a multipole above the quadrupole. This section is what does, and it is also where the line-of-sight kernel comes from.

9.1 What is left when the source and the scattering go

After last scattering $\kappa' \to 0$, and $\mathcal{B}_{A_\ell} \equiv 0$ for $\ell \geq 3$ identically. What remains of the transport equation is its streaming couplings alone — *Annals II* Appendix F, p. 379, in proper time and with the **comoving** wavenumber:

$$-\dot{\tau}_\ell = \frac{k}{a}\Big[w_\ell\,\tau_{\ell+1} - \tau_{\ell-1}\Big], \qquad w_\ell = \frac{(\ell+1)^2}{(2\ell+3)(2\ell+1)}, \qquad \ell \geq 2. \tag{7}$$

Multiplying through by β_ℓ and changing to conformal time gives the form this bundle integrates as its reference, (F.3):

$$-(\beta_\ell \tau_\ell)' = k\left[\frac{\ell+1}{2\ell+3}(\beta_{\ell+1}\tau_{\ell+1}) - \frac{\ell}{2\ell-1}(\beta_{\ell-1}\tau_{\ell-1})\right], \qquad \beta_\ell = \frac{\ell!}{(2\ell-1)!!}. \tag{8}$$

Each multipole is coupled to its two neighbours and to nothing else. That is the whole content: the ladder has no closure and no termination, so the three ranks the source supplies generate every rank above them. **It is also the only ℓ-coupling in the released hierarchy** — every other coupling in the scheme reaches $\ell \pm 1$ or $\ell \pm 2$ and arrives later, by frame, which is what **F8** draws.

The ℓ-weights are where an error would hide, and it would hide at large ℓ. $w_\ell \to \frac{1}{4}$ and the two bracket weights in (8) tend to $\frac{1}{2}$ and $\frac{1}{2}$: every coupling in the recursion is a ratio of linear polynomials and tends to a constant. **So a coupling that grew like ℓ would be invisible at low ℓ and divergent at high ℓ** — which is exactly the regime this paper is about, and the reason the Appendix F chain is the acceptance spine rather than a smoke test. Decision **S9**.

9.2 The $\ell = 0$ row is the rank-zero source, seen from the other side

Continued to $\ell = 0$, (7) has $w_0 = \frac{1}{3}$ and reads

$$\dot{\tau}_0 = -\frac{k}{a}\frac{\tau_1}{3}, \tag{9}$$

the monopole fed by the dipole alone. **That is the same combination the rank-zero gravitational source is built from**, $\mathcal{B}_0 = -\frac{1}{3}\mathrm{D}^a\tau_a$, giving $a\mathcal{B}_0 = -\frac{k}{3}\tau_1$ in mode form.

The two are one statement, and that is why $\tau_0 \equiv 0$ is a constraint rather than initial data. T is *defined* as the monopole of $T(x, e)$, so the $\ell = 0$ equation does not evolve a free quantity — it fixes $(\ln T)^{\cdot}$, and $\mathcal{B}_0$ is a functional of the solution rather than an external forcing. **Restoring $a\mathcal{B}_0$ to C_0 as though it were a source therefore counts it twice**, which is finding **T-107** and is settled at §10.2.

This connection is worth drawing because the two appear in different sections and look like different objects. They are the same row of the same hierarchy.

9.3 The closed form, and why the kernel is a spherical Bessel function

Appendix F, p. 380, states the solution directly: *"the solution to the mode coefficients with respect to the time-like integration in the flat ($K = 0$) almost-Friedmann–Lemaître case are spherical Bessel functions."* A unit monopole released at η_i projects onto

$$\tau_\ell(\eta) = j_\ell(k(\eta - \eta_i)), \tag{10}$$

and in this bundle the per-multipole normalisation is **fixed by the recursion and checked rather than assumed** (`functions/harmonics/free_streaming.py`, `analytic_projection`).

This is the one fact that makes the line-of-sight solution a solution and not an ansatz. The kernel $j_\ell(k\Delta\eta)$ in (102) is not a window chosen for convenience and it is not a projection imposed on the hierarchy from outside: **it is the hierarchy's own free-streaming solution**, evaluated along the null cone. Every statement about the $k \to \ell$ map — that power at multipole ℓ comes from $k \simeq \ell/r_\theta$, that a feature fixed in k appears at a fixed ℓ — is a property of (10) and inherits whatever is true of the recursion that produced it.

9.4 Three normalisations, none privileged, and one printed slip

Appendix F writes one recursion three ways, and the bundle integrates all three *as printed*:

	written in	**the variable**
(F.1)	proper time, comoving k	τ_ℓ
(F.3)	conformal time	$\beta_\ell\tau_\ell$
(F.4)	conformal time	$\alpha_\ell^{-1}\tau_\ell$, the Ma & Bertschinger form

with $\alpha_\ell = (2\ell+1)/\beta_\ell$. **Privileging none of them is the point**: the three differ by ℓ-dependent factors, so an error in a weight shows as a disagreement between normalisations that no single one can reveal. This is the same discipline the spectrum uses when it reports (186) and (187)–(188) separately rather than asserting they agree.

(F.4) as printed carries a factor its own text removes. The display keeps $(2\ell + 1)$ on the left, while the sentence preceding it specifies a division by $(2\ell + 1)$ that was not applied; only after that division is it the Ma & Bertschinger form the text says it is. **The bundle divides** — finding **B-2**.

And k is comoving here, which the conventions sheet does not say of every section. Appendix F's k is comoving and $k_{\rm phys} = k_{\rm com}/a$; the explicit $1/a$ in (7) against a proper-time derivative is what fixes it. The project's conventions sheet declares $k \equiv k_{\rm phys}$ in its *harmonic-recursion* section, so the two differ by a factor a **in precisely these equations**. A bare k is therefore ambiguous at exactly the point where it matters most, and every equation in this section is written in $k_{\rm com}$.

9.5 What gates this, and what each gate can and cannot see

gate	what it tests	result
F1	the recursion against **four external hierarchies**, in two normalisations and two phase conventions — criterion 1	residuals $\lesssim 10^{-13}$
`test_appendix_f_chain`	that no coupling grows with ℓ: an $O(\ell)$ coupling wired in early fails it at large ℓ	gate on S9
`sign_control`	the *negative* control — the same recursion with the relative sign of the two couplings flipped. **It must fail to reproduce the external form**, and a suite in which it passes is not testing the phase	negative control
(10)	the integrated recursion against $j_\ell(k(\eta-\eta_i))$ for a unit monopole, per multipole	normalisation checked, not assumed
T-3	that the η grid resolves $j_\ell(k(\eta_0-\eta))$, whose period in η is π/k	raises rather than aliasing

What none of them sees. All five test the recursion *with the source switched off*. They establish that free streaming is reconstructed faithfully and say nothing about whether the right thing is streamed: **the matching at η_* is a separate question and is §10.1's. F1 at 10^{-13} is therefore not evidence about the spectrum**, and reading it as such would be the error the acceptance spine exists to prevent — a reconstruction can be exact and be matched onto the wrong source.

10 The transport assembly

The reader who wants only the result: §10.1 gives the exact by-parts of (102)'s own integral and names the four discrepancies the released solution carries against it, §10.2 corrects (90)'s C_0 by a factor $\frac{15}{2}$, and §10.3 is the structural check that flags both before they are implemented.

Three questions about the line-of-sight solution are one question, and this section answers it once. They are: which weight carries which source; what the $\ell = 0$ matching row of (89) should be; and why every repair attempted on (102) so far has moved a weight without moving the term that dominates the integrand. The common answer is that **(102)'s integration by parts was taken onto the coefficients**, and that the assembly in which that step never arises is the one obtained by multiplying the transport equation by its own integrating factor.

Figure 1 is the order this section and §11 are written in. It is the photon's history and not a table of contents: each open question is drawn at the conformal time it acts at, which is what separates the questions that a frame instrument can see from the questions it cannot.

10.1 The by-parts, and the assembly that does not need it

Done exactly, the by-parts carries three terms beyond the familiar bracket. (102)'s bulk integrand comes from integrating $\int W\big(C_0K + C_1K' + C_2K''\big)\,\mathrm{d}\eta$ by parts *onto the coefficients*, with kernel $K(\eta) = j_\ell(k\Delta\eta)$ and $\Delta\eta \equiv \eta_0 - \eta$. The weight $W = e^{-\kappa}$ sits *inside* the integral, so it is differentiated

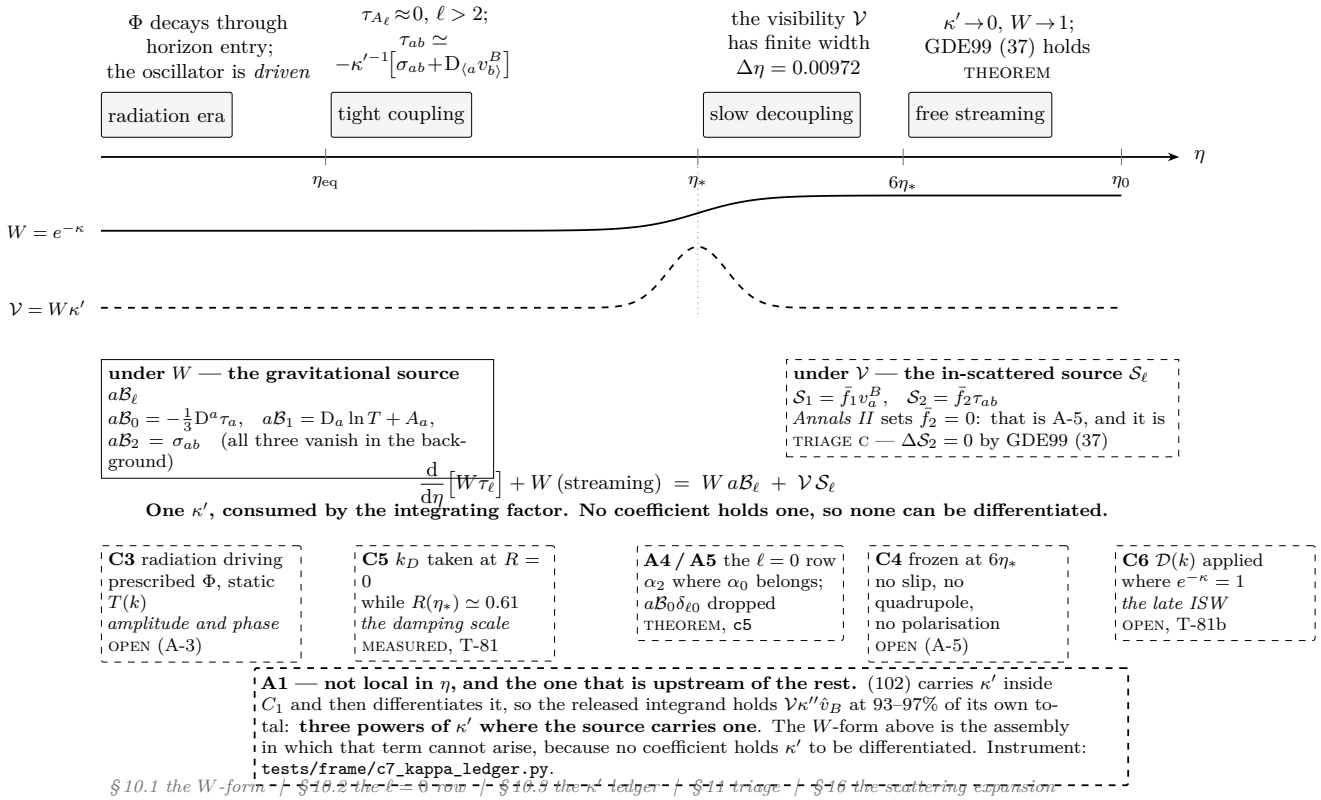


Figure 1: The photon's history and where each open question acts.†

† *The $\ell = 0$ row of* (89) *is corrected here; see findings* **T-104**, **T-105** *in* `provenance/BUNDLE-DEFECTS-v1.md`.

too, and $W' = \mathcal{V}$, $\mathcal{V}' = (\kappa'' + \kappa'^2)W$:

$$WC_0 - (WC_1)' + (WC_2)'' \;=\; \underbrace{W(C_0 - C_1' + C_2'')}_{\text{(102)'s bracket}} \;-\; \mathcal{V}C_1 \;+\; 2\mathcal{V}C_2' \;+\; \mathcal{V}'C_2. \tag{11}$$

What the released solution carries against (11)**, and it is four things, not three.** The discrepancies are recorded in `functions/spectra/integral_solution.py` as findings **T-72** and **T-86**. In order of size:

	the identity wants	the released solution carries	
1	the bracket at weight W	the bracket at weight $\mathcal{V}\mathcal{D}$	$\mathcal{V} = W\kappa'$, so **one extra power of the rate**
2	$-\mathcal{V}C_1 + 2\mathcal{V}C_2' + \mathcal{V}'C_2$	absent	(102) carries only the bracket
3	$W(\eta_*)$ on the boundary	$\mathcal{D}(\eta_0, k)$	weight and argument both
4	$+k\int W\,M\,j_\ell'$	*no counterpart at all*	the by-parts onto coefficients is legal only at W

Row 1 is the one that matters and it is now measured. $\mathcal{V}$ already supplies a κ', so differentiating a coefficient that itself holds κ' puts *three* powers of the rate where the identity wants two. **D24** measures the consequence on the live path: $\kappa''\tilde{v}_B$ is **99.8%** of $|C_1'|$ in the Newtonian frame and **109.8%** in CDM, and suppressing it alone moves the CDM first peak from a spurious maximum at $\ell = 8$ to $\ell = 172.75$, i.e. $0.635\,\ell_A$ against the matched 0.757. The Φ' driving channel is **not** dead; it is outvoted — finding **T-121**.

What `c3_byparts.py` does and does not measure. **C3a** is the load-bearing one: (11) reproduces the integral it comes from to $< 10^{-8}$ over fifteen (ℓ, k) pairs, so the identity is not in question. **C3b** evaluates a *variant* — it weights the bracket at W and retains $\mathcal{V}(C_1 - 2C_2)$ — so its generated value — **2.55227** — bounds rows 2 and 3 and **is blind to row 1**, which is the one row whose weight it holds fixed. That number is the authoritative one: a bound of $> 10^{-2}$ and a prose range of *10–250%* have both been quoted for it elsewhere and neither should be. Its docstring's *"evaluated as printed"* should be read as *"evaluated against a variant that differs from the printed form in the partner terms"*; the released weight is the `weight` argument, $\mathcal{V}$ by default, approximation **A-2** and finding **T-20**.

This is a defect in the antecedent and carries a **B-** number in `provenance/ANNALS-II-ANTECEDENT-AND-CORREC` beside B-10 and B-11. It is stated here as well because (11) *is* the correction, and this supplement prints the correction beside every other defect it records — B-11's sign on C_0, T-104's normalisation, T-103's weighting. A defect recorded without its correction is not a record.

The assembly in which none of the four can be made. They arise only because the by-parts was taken *onto the coefficients*. Multiplying the transport equation at rank ℓ,

$$\tau_\ell' \;+\; (\text{streaming couplings}) \;+\; \kappa'\tau_\ell \;=\; a\mathcal{B}_\ell \;+\; \kappa'\mathcal{S}_\ell, \tag{12}$$

through by W — with κ accumulated backwards from η_0, so $\mathrm{d}\kappa/\mathrm{d}\eta = -\kappa'$ and $W' = +\kappa' W = \mathcal{V}$, the sign this bundle's convention forces, stated at §11 — combines the first and third terms exactly into $\mathrm{d}[W\tau_\ell]/\mathrm{d}\eta$, and gives

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}\eta}[W\tau_\ell] \;+\; W\,(\text{streaming couplings}) \;=\; W\,a\mathcal{B}_\ell \;+\; \mathcal{V}\,\mathcal{S}_\ell} \tag{13}$$

with $\big[W\tau_\ell\big]_{\eta_*}^{\eta_0} = \tau_\ell(\eta_0) - W(\eta_*)\tau_\ell(\eta_*)$ as its only boundary term, since $W(\eta_0) = 1$. Here $a\mathcal{B}_\ell$ is the gravitational source of §*The generic source derivatives*' rank decomposition and $\mathcal{S}_\ell$ the in-scattered Thomson source; both κ' in (12) are the same κ', the out-scattering and the in-scattering being two halves of one collision term.

What (13) **is, at its true size. It is not different physics.** An exact by-parts preserves the transfer, so (13) and a (102) corrected by (11) give the same observable. *An earlier draft of this section called (102)'s bulk slots and boundary pair "artefacts of a step the W-form does not take"; that overstates and is withdrawn.* What (13) is:

1. **The assembly in which none of the four discrepancies above can be made**, because there are no coefficients to take a by-parts onto, and so no weight to choose for a bracket that does not exist. Both weights fall out of one multiplication rather than being assigned: $\mathcal{V}$ appears beside $\mathcal{S}_\ell$ because κ' was already there in the collision term and $W\kappa' = \mathcal{V}$ is an identity.

2. **The assembly in which no coefficient holds** κ', so no differentiation of one can produce a κ'^2 — the error class §10.3 flags. **D24** measures it on the live path: $\kappa''\tilde{v}_B$ is 99.8% of $|C_1'|$ in the Newtonian frame and 109.8% in CDM, finding **T-121**. (13) has no counterpart term: $\mathcal{V}\mathcal{S}_1$ is κ'-free.

3. **The assembly whose boundary term is one object, exponentially suppressed at** η_* — $[W\tau_\ell]$, not $[C_2' - C_1]$ plus kC_2, carrying W rather than a diffusion factor. **D18** measures what the released boundary carries instead: with `boundary_weight_w False`, $C_1(\eta_*)$ enters under the diffusion factor alone — no W, no $\mathcal{V}$ — so its bare $\kappa'(\eta_*)$, 3.82×10^2 on Einstein–de Sitter and 2.64×10^2 on `PLANCK_MODEL`, stands under no exponential suppression at all.

This supersedes §11.4's item 2, and the wording that item carries is what T-88 implemented. That item now reads *"they* arrive under *different weights: W for the gravitational half, $g = \mathcal{V}$ for the scattering half"*, corrected at its own site from *"take"* — which is precisely the distinction this paragraph is about, and it is recorded here because an earlier draft quoted the old wording. As a description of the outcome of (13) it is correct. As an instruction to a solver it is not, and it was read as one: multiplying $C_1^{\rm thom} = -\kappa'\hat{v}_B$ by $\mathcal{V}$ applies a second κ', which §11.8 measures at 320 to 682. **The scattering half arrives at $\mathcal{V}$ weight carried on the undifferentiated $\hat{v}_B$; it is not multiplied by $\mathcal{V}$.** Finding **T-103**; the phrasing is the Paper 1 stream's and the correction is recorded against it rather than against the implementation that followed it.

What $\mathcal{S}_2$ is, and why it cannot be a frame defect. *Annals II*'s collision term is $C[\tau] \simeq \sigma_{\rm T} n_E(\bar{f}_1 e^a v_a^B + \bar{f}_2 e^a e^b \tau_{ab} - \tau)$ with $\bar{f}_1 = 1$ and $\bar{f}_2 = 0$, so $\mathcal{S}_2 = 0$ in this reconstruction. That is approximation **A-5** stated at the collision term rather than at the oscillator. **It cannot produce a frame disagreement at linear order**: $\mathcal{S}_2 \propto \tau_{ab}$, and GDE99 (37) leaves τ_{A_ℓ} invariant for $\ell \geq 2$ *at first order in v* — those multipoles are already first-order small, so what a boost would generate from them is second order — whence $\Delta\mathcal{S}_2 = 0$ and dropping the term changes both columns by the same amount. It is a **§11 Triage C** site — it can ruin the spectrum and it cannot be seen by a frame instrument — and listing it as a candidate for the frame gap confuses the two questions the triage tables exist to separate.

10.2 The $\ell = 0$ matching row

The rows of (89) are Taylor coefficients of the kernel at the origin, and one of the three uses the wrong rank's normalisation. With $h_\ell = \alpha_\ell j_\ell$ and $\alpha_\ell = (2\ell+1)/\beta_\ell$ — this bundle's own `functions/harmonics/weights.py`, where $\beta_\ell = \ell!/(2\ell-1)!!$ gives $\beta_0 = \beta_1 = 1$ and $\beta_2 = \frac{2}{3}$ — each row of (89) pairs C_n with $h_\ell^{(n)}(0)$:

row	used	its own	the entry
$\ell = 2$	$\alpha_2 = \frac{15}{2}$	α_2	$h_2''(0) = \frac{15}{2} \cdot \frac{2}{15} = 1$
$\ell = 1$	$\alpha_1 = 3$	α_1	$h_1'(0) = 3 \cdot \frac{1}{3} = 1$
$\ell = 0$	$\boldsymbol{\alpha_2 = \frac{15}{2}}$	$\boldsymbol{\alpha_0 = 1}$	$-\frac{5}{2}$ printed; $h_0''(0) = -\frac{1}{3}$

(89) writes its $\delta_{\ell 0}$ entry as $5\beta_2^{-1} \cdot \frac{1}{3}$, which is α_2 times $j_0''(0)$ up to the sign **B-11** corrects. The rank-two normalisation is right where it belongs and is carried into a row that is not rank two. **The two rows that are exactly unity are the control**, and they are a control precisely because they come out at unity *without adjustment*: a normalisation error cannot produce 1 twice and then $-\frac{5}{2}$ once. Findings **B-16** and **T-104**; the discrepancy is $\alpha_2/\alpha_0 = \frac{15}{2}$.

This route is independent of (89), which is what makes it a check rather than a restatement. β_ℓ is fixed by the PSTF normalisation $O^{A_\ell} O_{A_\ell}$ and $j_0''(0) = -\frac{1}{3}$ by the ascending series; neither comes from (89). (89) is the *claim* about what the rows hold and the kernel is the *object*. Two of the three rows reproduce (89)'s entry exactly from the object, which validates the route; the third does not.

The right-hand side of the $\ell = 0$ row is zero, and that is forced by a definition. $\tau_0 \equiv 0$ is a **constraint, not initial data**: $T(x)$ is defined as the monopole of $T(x, e)$, so differentiating $\int A_0 \, {\rm d}\eta' = 0$ puts zero on the right of the row. $\mathcal{B}_0$ is a functional of the solution, not an external forcing. Appendix 17 states this and states it first — *"(89)'s $\ell = 0$ line is not mis-sourced … C_0 is correctly derived"*. With α_0 on the left and zero on the right,

$$C_0 + k^2 C_2 \, h_0''(0) = 0, \qquad h_0''(0) = -\tfrac{1}{3}, \tag{14}$$

and with $C_2 = -k^{-2} a\mathcal{B}_2$ this solves to

$$\boxed{C_0 \;=\; -\tfrac{1}{3}\, a\mathcal{B}_2} \tag{15}$$

in place of (90)'s $C_0 = -\frac{5}{2} a\mathcal{B}_2$. **One correction, and the ratio against the printed row is $\frac{5}{2}/\frac{1}{3} = \frac{15}{2}$ exactly** — α_2/α_0, with no $\mathcal{B}_0$ anywhere. Instrument `c5_monopole.py`, under `tests/frame/`, which carries a negative control on each alternative.

Where $\mathcal{B}_0$ does enter, and why restoring it to C_0 is double counting. Correction, 2026-10-22. An earlier draft of this subsection restored an $a\mathcal{B}_0\delta_{\ell 0}$ row, printed $C_0 = -\frac{1}{3} a\mathcal{B}_2 - a\mathcal{B}_0$, and recorded it as **T-105**. **That is withdrawn.** By identity

$$a\mathcal{B}_0 = -\tfrac{k}{3}\,\tau_1, \tag{16}$$

derived in index algebra and controlled on rank 1, where the identical chain returns $a\mathcal{B}_1 = k(\delta T + \Phi_A)$ — the released `coefficients.py` line. And (107) carries $-C_1' = (\kappa' \hat{v}_B)' + (a\mathcal{B}_1)'/k$ with $(a\mathcal{B}_1)'/k = (\Phi'_A - \Phi'_H) + \frac{k}{3}\tau_1$, so **the monopole source is already inside the bracket, in $-C_1'$.** *That is the expansion-terms-off form of (106)*: `coefficients.py` builds $(a\mathcal{B}_1)'$ with $-a\mathcal{H}\,\delta T - 3a\mathcal{H}\,\Phi_A$ in addition, behind `A_B1_PRIME_EXPANSION_TERMS`, and whether they belong is finding **T-67**, *open*. They do not affect the $\frac{k}{3}\tau_1$ identification used here, which is why the reduced form is quoted; they are $O(a\mathcal{H})$ and are *not* negligible where the late integrated Sachs–Wolfe effect lives — see §18. Adding it to C_0 counts it twice, and criterion 5b doubles at 98.5%. Finding **T-107** replaces T-105.

So $\Delta C_0 \neq 0$, and that is the correct outcome. Under (15), with Appendix 17's own $\Delta(a\mathcal{B}_2) = -kv$ and $\Delta C_2 = v/k$,

$$\Delta C_0 = -\tfrac{1}{3}\Delta(a\mathcal{B}_2) = +\tfrac{kv}{3}, \qquad \gamma \equiv \frac{\Delta C_0}{k^2 \Delta C_2} = \tfrac{1}{3}. \tag{17}$$

A non-zero γ is the signature that ΔC_0's partner is cross-coefficient, sitting in $-C_1'$ rather than in C_2. A γ of zero would say the partner was in C_2, which it is not. The withdrawn draft reached $\gamma = 0$ by *adding a term*, and a gate that can be driven to zero by adding a term is a fit rather than a gate — which is **M15**, *a gate built from one coefficient cannot see a cancellation distributed across the bracket.* `c8_rank0_partner.py` scores the whole rank-0 content of the bracket and is the gate for this row; Appendix 17's questions 3 and 4 close with **c8** or not at all. Its questions 1 and 2 are settled by **T-66** and by §10.1 respectively.

One thing (15) **does not disturb.** Appendix 17 excludes a comoving/physical or amplitude/potential mismatch by measuring $C_0/(k^2 C_2)$ as a background-independent and k-independent rational. That argument holds under (15) unchanged — the ratio is still a pure rational, now $\frac{1}{3}$ rather than $\frac{5}{2}$ — because what it rests on is that C_0 and C_2 divide one $a\mathcal{B}_2$. **The measurement was right and what it established was internal consistency, which is not correctness.** It is the same reading the bundle gave T-82 and T-85, applied to itself.

10.3 The κ' ledger

The flag, and it is a flag and not a theorem. Revised 2026-10-21, T-113 and S48. *A coefficient carrying a bare κ' is a site where a large residual is manufactured the moment it is differentiated, and is worth flagging before implementation.* An earlier draft of this subsection printed that as a prohibition — *"κ' may live in W, in $\mathcal{V}$, or in a derivative of either, never in a coefficient"* — and it was read as normative. **It is not a theorem**, because

$$W(\kappa' \mathcal{S}) = \mathcal{V}\,\mathcal{S} \tag{18}$$

is exact: a coefficient *may* hold κ', and $(\kappa'\mathcal{S})' = \kappa''\mathcal{S} + \kappa'\mathcal{S}'$ is not intrinsically spurious. The defect in (102) is wrong weights, signs, boundaries or missing partners, not the presence of κ'.

Why the flag is kept. It separates two κ'^2 that are indistinguishable in the output. The one inside $\dot{\mathcal{V}} = (\kappa'' + \kappa'^2)W$ comes from differentiating the **weight**; it is required, and (13) produces it whenever a streaming term is integrated. The one from differentiating a **coefficient** is not required by anything. Only the provenance distinguishes them, so the flag is raised on the provenance. At η_* in this bundle's units **D18** measures $\kappa' = 3.82 \times 10^2$ on Einstein–de Sitter and 2.64×10^2 on `PLANCK_MODEL`, so a spurious power of it is two to three orders. **What the flag may no longer do is settle a question on its own.**

(102) as printed trips it, and T-121 is the review. §*The generic source derivatives* gives $-C_1 \simeq k^{-1}(a\mathcal{B}_1 + \kappa' v_B)$ — a coefficient holding a bare κ' — and (102) then differentiates it. §11 measures the consequence: $\mathcal{V}\kappa''\hat{v}_B$ is 93–97% of the released integrand, and **D24** measures 99.8% of $|C_1'|$ in the Newtonian frame and 109.8% in CDM on the live path. The flag asks whether that is a defect; **D24** answers it at the level of the observable. **Suppressing that one term moves the CDM first peak from a spurious maximum at $\ell = 8$ to $\ell = 172.75$**, which is $0.635\,\ell_A$ against the matched 0.757. The driving channel is not dead, it is outvoted — finding **T-121**. The flag was raised on the right site; what it could not supply, and D24 does, is the size.

A measurement that does not bear on this, and is recorded so it is not over-read. **T-117** finds the transfer failing to reduce to Sachs–Wolfe as $k \to 0$ — in the Newtonian column $\kappa' v_B/k$ does not die, and the bulk is then entirely $\int \mathcal{V}(\kappa'\hat{v}_B)' j_\ell$ at 9.46×10^3. It is a true defect in a limit with a known answer. **D25** then bounds the window it holds in to **at most** 0.64% of any C_ℓ, and 0.00% at every $\ell \geq 5$ — finding **T-122**. So the number is a diagnostic of the mechanism and not of the spectrum, and nothing here rests on it.

A local statement, not a general one. $\kappa'' = -\kappa'^2$ holds **at the visibility maximum**, where $\dot{\mathcal{V}} = 0$. So the reading of $\mathcal{V}\kappa''\hat{v}_B$ as *three powers of* κ' is local to that point, and the measured 93–97% share follows because $\mathcal{V}$ concentrates the integrand there. It is not a statement about the integrand at arbitrary η. D24's 99.8–109.8% is measured against $|C_1'|$ on the live path and is the number to quote for the share.

It is applied before implementation, not after. `tests/frame/c7_kappa_ledger.py` grades an assembly structurally — for each term, whether its coefficient holds κ' and at what power — rather than measuring an output. It reports (13) clean, and (102) as printed and T-88's variant as carrying κ' in a coefficient at power two. **Both of the defects of 19–20 October would have been flagged by that one check before either was implemented**, which is the argument for a structural review alongside a numerical one: a numerical check on an assembly that is wrong in this way measures the defect faithfully and reports a number.

10.4 What this section establishes, and what it does not

Established. That (11) is the exact by-parts of the integral (102) is built from, reproducing it to $< 10^{-8}$; that the released solution differs from it in four places, of which the bracket's weight is the largest and carries one surplus power of κ', measured by **D24** at 99.8–109.8% of $|C_1'|$; that (13) follows from (12) by one multiplication and is the assembly in which none of the four can be made, without being different physics; that (89)'s $\ell = 0$ row carries α_2 where α_0 belongs, with the $\ell = 1$ and $\ell = 2$ rows as unadjusted controls; that the row's right-hand side is zero by the $\tau_0 \equiv 0$ constraint, giving (15); that $a\mathcal{B}_0 = -\frac{k}{3}\tau_1$ is already carried in $-C_1'$, so restoring it to C_0 double counts; and that $\mathcal{S}_2$ is a Triage C site and cannot carry a frame gap at linear order.

Not established. That (13) closes the frame gap. It satisfies the ledger by construction and no setting of `integral_solution`'s seven controls reaches it — each redistributes weights while leaving κ' inside C_1 — but that is an argument from exhaustion of what has been tried, not a measurement. **What has changed is that the evidence now points at one term in both frames**, where an earlier draft of this paragraph said it did not. **D24** suppresses $\kappa''\tilde{v}_B$ alone and the CDM first peak appears at $\ell = 172.75$, $0.635\,\ell_A$ against the matched 0.757, where the released setting has a spurious maximum at $\ell = 8$. *The hypothesis that the CDM column fails through a sign in its boundary is withdrawn*: Einstein–de Sitter gives -0.9999 but `PLANCK_MODEL` gives $+1.270$, so it is not one sign and the -1 was a coincidence of that background. What causes the residual 0.635 against 0.757 is taken up below, under *Approximation A-3 is not what closes the peak position*; it is not A-3, and D24 is the measurement that rules it out.

The two peak numbers are not measured the same way, and that bounds what their difference can carry. D24 reports the largest maximum anywhere in the plotted range; **F9** reports the first local maximum above $\ell = 30$, read by parabolic interpolation through an ℓ grid that steps by 30 across the acoustic band. On a spectrum whose low-ℓ end carries a spurious maximum at $\ell = 8$ those two estimators need not return the same peak at all. **So peak positions in this section are quoted to two significant figures and no further**: a 6.1% residual at $\ell \simeq 173$ is $\Delta\ell \simeq 10.5$, about a third of one grid step, and a comparison at that level is a statement about the estimator as much as about the physics. The repair is one estimator applied to both, with the low-ℓ maximum excluded by construction rather than by an $\ell = 30$ floor that happens to work.

A-3 answers to three configurations and gives a different answer to each, so no statement about it is safe without naming which.

configuration	**CDM peak**	**what A-3 does there**
released	$266.6 = 0.979\,\ell_A$	**accounts for the whole excess**: $266.6/206.1 = 1.294$ against $1/0.757 = 1.321$, agreeing to 2%
$\kappa''\tilde{v}_B$ suppressed (**D24**)	$172.75 = 0.635\,\ell_A$	**ruled out** — the peak is already below the driven value and driving moves it further away
full boundary at released κ''	$236.6 = 0.869\,\ell_A$ (CDM)	**would be the right sign — but see below: this configuration is not a repair**

The third row is a candidate and it fails the one test that costs nothing to apply. 'slot_weight_split' with the two boundary controls puts the CDM peak at $0.869\,\ell_A$ and the *Newtonian* peak at $0.277\,\ell_A$ — a factor of 3.1 between two frames whose $\ell \geq 2$ multipoles are equal at first order in v (GDE99 (37)). **A configuration that moves the CDM peak toward the target while pulling the frames** $3.1\times$ **apart has made the frame disagreement worse, not better**, and the released path's median gap of 15% is the thing it was supposed to close. *So it is not a repair, and nothing below is a result.* The rows are kept because the comparison is instructive and because the frame check is the cheapest discriminator this section has.

The derivation, which is what makes the three rows consistent. Driving moves a peak **down**: the undriven first compression extremum is $kr_s = \pi$, which projects to $\ell = \ell_A$, and the matched external spectrum sits at $0.757\,\ell_A$. **D24**, with $\kappa''\tilde{v}_B$ suppressed and A-3 still absent, puts the CDM peak at $0.635\,\ell_A$ — *already below the driven value* — so applying A-3 there moves it further from 0.757, not toward it. Three places in this project's record have said the first peak sits where it does because radiation-era driving is absent; **for that configuration that is backwards, and it is withdrawn here.**

But the conclusion is conditional on which side of 0.757 **the peak falls, and that is what a repair would decide.** A configuration whose CDM peak lands *above* $0.757\,\ell_A$ puts A-3 back in play, because driving then moves it the right way; one landing below leaves A-3 ruled out. **A-3's standing is therefore a property of the configuration and not of the project**, and no statement about it is safe without naming which.

What a candidate above 0.757 **would still have to survive.** Against this background the whole distance from an undriven peak to the matched spectrum is $\ell_A - 206 = 66.2$ multipoles, or $0.243\,\ell_A$. A candidate at 0.869 would leave $236.6 - 206 = 30.6$ to cross, so *a full driving correction would overshoot it by a factor of* 2.2 — A-3 could not be the whole remainder, and wiring it in would have to land short of 0.757 rather than on it. **That is the test to apply to the first candidate that holds the two frames together**, and it is written here so that it is applied rather than re-derived.

The other known absentee pushes the same way. (19)'s j'_ℓ counterpart adds power *below* the peak and is missing from (102) as well, so the two named absent mechanisms both move the peak down and between them need supply only $0.112\,\ell_A$. **Neither is established as the remainder, and the two must be measured separately before either is credited**, which is M11a.

And a boundary treatment and A-3 compete for the same multipoles, which is what the released agreement with 1/0.757 **does not establish.** The released path needs $266.6 - 206.1 = 60.5$ multipoles of downward movement and credits all of them to A-3, agreeing to 2%. **The candidate configuration, with A-3 still absent, supplies** $266.6 - 236.6 = 30.0$ **of them** — 49.6% — **by changing a weight.** It is not a repair, so this does not withdraw the A-3 account; **what it shows is that the** 2% **agreement is not evidence for A-3, because at least one other thing in the assembly moves the peak by half that distance.** A quantity reproduced to 2% by a calculation with a known defect of the same size is the shape of every over-read in §*what a gate does not establish*, and it is listed there.

The discriminator, when a frame-consistent candidate exists, is one run: wire A-3 into it. If A-3 is the whole remainder the peak lands near $0.757\,\ell_A$; if the weight change has already taken half, it overshoots by roughly $0.243\,\ell_A$ from wherever the candidate sits. **For a candidate at** 0.869 **those are** $\ell \simeq 206$ **against** $\ell \simeq 170$**,** 17% **and** 36 **multipoles apart.** Both ends are already built.

The displaced projection is ruled out too, by the test that was written to decide it. An earlier form of this paragraph attributed 0.635 to **D25**'s finding that the median k contributing to a given ℓ runs 1.5–4.2× above ℓ/r_θ, and predicted a ratio near $1/0.635 = 1.57$ at the peak. **D25 extended to** $\ell = 173$ **returns** 1.032 **and** 1.039 **in the Newtonian frame,** 1.171 **and** 1.201 **in CDM, on Einstein–de Sitter and** `PLANCK_MODEL` **respectively — the two backgrounds agreeing to** 3%**.** The displacement is a **low-**ℓ **phenomenon**: 1.5–4.3 at $\ell = 2$, decaying monotonically, and essentially closed by the peak. So **T-122** accounts for the plateau and not for the peak position, and the attribution is withdrawn. *The projection at the peak is faithful.*

And the absolute level was never the diagnostic. A median contributing wavenumber above ℓ/r_θ is what the kernel produces on its own: $j_\ell(kr_\theta)$ peaks at $kr_\theta \simeq \ell + 0.8\,\ell^{1/3}$ and carries a long tail to larger k, so an exactly Sachs–Wolfe spectrum returns a ratio above one as well. **No absolute value of this ratio establishes a projection defect**, and the earlier reading, which took 1.5–4.2 as a defect on its face, was reading a property of j_ℓ. What survives is the *difference between the frames* at the same ℓ on the same kernel, because the kernel cancels there and nothing else in the comparison does.

One residue does survive at the peak, and it is frame-dependent. CDM reads 1.17–1.20 against the Newtonian 1.03–1.04, on *both* backgrounds — a 17–20% displacement left standing where the frame-independent part has closed. It is too small to carry 0.635 by this route, which would need 1.57. It is the right *shape* for the candidate below, and it was measured for another purpose, which is the only reason to give it any weight at all.

What that leaves. With the projection faithful at $\ell = 173$ and the oscillator tracking $\cos(kr_s)$ in exact phase and period (**T-71**, **T-74**), a peak at $0.635\,\ell_A$ places the first extremum at $kr_s \simeq 0.635\pi$ rather than π — a phase error of about -0.37π, arising *between* the oscillator and the transfer, which is where (11)'s discrepancies sit.

A quarter-cycle account of that was printed here and is withdrawn — finding T-123. It wrote $C_2 = A\cos kr_s$, read $+2\mathcal{V}C_2'$ against the printed $-2\mathcal{V}C_2$ as a rotation of exactly $\pi/2$, and bounded the peak to $(0.5,\ 1.0]\,\ell_A$ at a mixture weight $w \simeq 2.2$. C_2 **does not oscillate.** $\mathcal{B}_2$ is the shear, and in the CDM frame $a\mathcal{B}_2 = a\sigma = -k_{\rm com}v$ exactly, so $C_2 = v/k$ with v the matter velocity and $v' + \mathcal{H}v = -k\Phi_A$ its equation of motion — a potential on the right-hand side and not an acoustic amplitude, so v runs monotonically through matter domination. The premise fails and the bound, the 0.365π and the $w \simeq 2.2$ fail with it. *Recorded rather than deleted: two of the three tests it specified survive under the mechanism below, and the reason they survive is not the reason they were written.*

The partner group does act on the projection, and exactly. Project the three terms (11) supplies, integrating the third by parts and using $\mathrm{d}j_\ell(k(\eta_0-\eta))/\mathrm{d}\eta = -k\,j_\ell'$:

$$\int(-\mathcal{V}C_1 + 2\mathcal{V}C_2' + \dot{\mathcal{V}}C_2)j_\ell\,\mathrm{d}\eta \;=\; \int\mathcal{V}(C_2' - C_1)\,j_\ell\,\mathrm{d}\eta \;+\; \int\mathcal{V}\,v\;j_\ell'\,\mathrm{d}\eta \;+\; [\mathcal{V}C_2\,j_\ell], \tag{19}$$

with $kC_2 = v$ used in the middle term. **The partner group is a visibility-weighted source against j_ℓ plus a velocity against j_ℓ'**, the second being the Doppler form of the standard line-of-sight solution carrying the *frame* velocity in place of the baryon velocity. It vanishes identically in the Newtonian frame, where $\tilde{C}_2 \equiv 0$.

That is B-17 (iv) derived rather than observed, and it carries a number. At fixed $x = k(\eta_0 - \eta)$ the multipole maximising $|j_\ell'|$ lies about 37% below the one maximising $|j_\ell|$: 110 against 173 at $x = 178$, 164 against 266 at $x = 272$. A source given the wrong one of the two kernels is not perturbed but *relocated*, by a third of the ℓ axis. **T-124.**

The direction is the test. Restoring the j_ℓ' term adds power *below* the acoustic peak, so on its own it must move a peak down or leave it standing. The §5 repair moves the CDM peak *up*. **So the mover is the weight — B-17 (i) — and not this term**; if restoring the j_ℓ' slot alone moves the peak up, this account is wrong.

And the j_ℓ' slot is frame-discriminating, which is what makes it worth running. $\tilde{C}_2 \equiv 0$ in the Newtonian frame, so (19)'s middle term *cannot act there at all*; the $\mathcal{V}C_1$ sign is available in both. 0.635 is the **CDM** peak. **If the Newtonian peak shows a different displacement, that is the signature; if both frames displace alike, the C_2 route is excluded and the $\mathcal{V}C_1$ sign must carry it.** Two numbers, both from runs that already exist.

And D24's own table already shows the frames behaving differently, which sharpens the test. With κ'' suppressed the Newtonian column peaks at $\ell = 8.07$ — still the spurious low-ℓ maximum, *not* an acoustic peak — while CDM peaks at $\ell = 172.75$. That is what the frame structure predicts: Newtonian has $\tilde{C}_0 \approx \tilde{C}_2 \approx 0$, so $\tilde{C}_1$ carries everything and suppressing $\kappa''\tilde{v}_B$, which is 99.8% of $|C_1'|$, leaves no oscillatory source at all. CDM retains C_0 and C_2. **So the** 0.635 **peak is built from C_0 and C_2**, and C_2's slot is exactly where (19) acts.

Two tests survive the withdrawal and neither needs a repair. Restore the j_ℓ' slot alone in the κ''-suppressed CDM run and read the peak: (19) requires it to move *down* or not at all, and an upward move refutes the account. Then compare the two frames' displacements, which the C_2 route requires to differ because $\tilde{C}_2 \equiv 0$. **Until one of those runs,** 0.635 **has no mechanism, and a repair should not be scored against** 0.757**.**

Also not established, and it is a dependency of (15)**.** (15) *uses* $C_2 = -k^{-2}a\mathcal{B}_2$, which comes from the same solve in which **B-11** found a sign error and **T-104** a normalisation error. The $\ell = 2$ row returning $h_2''(0) = 1$ controls the normalisation and not the sign or the source pairing. **No instrument yet audits the $\ell = 2$ row the way `c5` audits $\ell = 0$**, and until one does, a defect of that class in C_2 would be inherited silently here.

The acceptance condition, stated in advance so the repair can fail visibly. (i) $A_N = A_C$ to the quadrature floor **T-85** measures, neither better nor worse; (ii) the Newtonian column moves by no

more than that floor; (iii) `c7_kappa_ledger.py` reports no coefficient holding κ' — *a review signal that the error class is absent by construction, not a proof of correctness*, since (18) makes the ledger a flag; (iv) `c8_rank0_partner.py`, not γ, scores the rank-0 content of the bracket on the live coefficients; and (v) **the spectrum adjudicates, and it adjudicates last**, since two wrong columns can agree.

The order is forced, not preferred — in implementation, not in exposition. §10.2 is implemented before §10.1: the $\ell = 0$ row sits inside the bracket the W-form reweights, so a weight correction applied over a C_0 that is wrong by $\frac{15}{2}$ measures the wrong thing. **T-96**'s second run is that measurement — a W-form instrument run against coefficients still carrying the printed row, returning $O(1)$ — and it is reported there as not conclusive, correctly.

11 The frame discrepancy: localisation and triage

This section records the state of the one open defect that the $v1.3.0$ line exists to close. It is written to be auditable rather than persuasive: every exclusion below carries the instrument that licenses it and a grade, because an exclusion without an instrument is an opinion and this project has paid for several. The measurements are recorded in `provenance/BUNDLE-DEFECTS-v1.md`, which is authoritative; numbers appear here only where the argument cannot be followed without them.

11.1 The statement being tested

For $\ell \geq 2$ the covariant temperature multipoles are invariant under a change of the fundamental congruence at linear order. Writing $\tilde{u}^a = u^a + v^a$ with $v^a u_a = 0$ and v^a first order, GDE99 (37) gives

$$\tilde{\tau}_{A_\ell} \;=\; \tau_{A_\ell} \;-\; v_{A_\ell}, \qquad v_{A_\ell} \equiv 0 \quad \text{for} \quad \ell \geq 2, \tag{20}$$

so only the monopole and the dipole move, and every $\ell \geq 2$ multipole is a frame invariant. The bundle computes the same C_ℓ twice — once in the Newtonian frame, $\tilde{\sigma}_{ab} \simeq 0$, and once in the CDM frame — and the two must agree. **They do not**: the disagreement is total, median 9.99×10^{-1} and worst 1.00 over the released multipole set.

Equation (20) is a theorem at linear order. It admits no tolerance and no background dependence, so a measured disagreement is a defect in the implemented specialisation and cannot be read as a limitation of the formalism. That is the premise of everything below.

11.2 Triage A — sites the discrepancy cannot come from

Each row names the instrument and a grade. THEOREM is exact at linear order; MACHINE ZERO is measured at 10^{-13} or better and is structurally zero; MEASURED is a number with a stated validity condition; ASSERTED is read or reasoned and is usable but not load-bearing.

site	instrument	grade
Free streaming and transport	GDE99 (37)	THEOREM
The observer endpoint: a pure dipole	GDE99 (37); Paper 1 §§5.3, 6.2	THEOREM
The Bessel projection	T-71, synthetic control	MEASURED
The C_1–C_2 boost relation	T-77, 3.3×10^{-16}	MACHINE ZERO
The by-parts identity itself	`c3_byparts.py` and two derivations	MACHINE ZERO
The oscillator at leading order	register; two external reads	MEASURED
The source's frame consistency	**T-82**	RESOLVED
Boundary-against-bulk bookkeeping	**T-85**	RESOLVED, EXCLUDED
The additive $\kappa'\tau_0$ route	**T-79**	RESOLVED, CLOSED

One row was withdrawn from this table, 2026-10-23. It read *"C_0's boost behaviour with $\mathcal{B}_0$ restored"*, graded MACHINE ZERO on **T-77** and `c5_monopole.py`. *The measurement stands* — $a\Delta\mathcal{B}_0 = \frac{1}{3}kv = \frac{1}{3}k^2\Delta C_2$ exactly — *but the configuration it was measured in does not.* **T-105**, which restored an $a\mathcal{B}_0\delta_{\ell 0}$ row to C_0, is withdrawn by **T-107**: $a\mathcal{B}_0 = -\frac{k}{3}\tau_1$ is already carried inside $-C_1'$, so restoring it counts the monopole source twice and criterion 5b doubles at 98.5%. Under the corrected row $C_0 = -\frac{1}{3}a\mathcal{B}_2$ and $\Delta C_0 = +\frac{kv}{3} \neq 0$, so this is **not** a site the discrepancy cannot come from. Its partner is cross-coefficient and `c8_rank0_partner.py` is the gate — **M15**: a gate built from one coefficient cannot see a cancellation distributed across the bracket.

Nothing in this table can be moved by a measurement. A result that appears to contradict one of these rows is wrong, or the comparison is void — most often the latter, through $\ell \gtrsim k\Delta\eta$, where the spherical Bessel function is exponentially small and a ratio of two vanishing numbers is still a number.

11.3 The two measurements that closed the upstream end

The source agrees between the frames (T-82). CAMB carries the rank-1 scattering source as a single object $g\,(v_b + \sigma)$. Under a boost $v_b \to v_b - v$ while the shear term shifts oppositely, so the sum is invariant; a code that never separates them has no frame gap by construction. This bundle does separate them, so the two columns must produce the same sum. In this bundle's slots the combination is $\hat{v}_B + C_2$, with $\hat{v}_B = \tau_1/k$ the reduced dipole and $C_2 = v/k$, since $a\mathcal{B}_2 = a\sigma = -kv$ and $C_2 = -a\mathcal{B}_2/k^2$. Differencing the two columns, on identical histories and before any projection — no transfer function, no j_ℓ, no weight, no damping:

$$\left|\Delta(\hat{v}_B + C_2)\right| \;\in\; \left[\,9.3\times10^{-16},\; 1.8\times10^{-13}\,\right], \qquad k \in \{1, 10, 60, 300\}, \tag{21}$$

on both the Einstein–de Sitter and the Planck backgrounds. **The source is frame-consistent to fourteen significant figures.** Set against a frame gap of 100% at the spectrum, the pair says what neither says alone: everything between the source and the spectrum is the projection.

The by-parts identity holds in both frames (T-85). With $W = e^{-\kappa}$ and $\mathcal{V} = W'$ — the sign this bundle's κ convention forces, κ being accumulated backwards from η_0 — the transport source obeys, exactly,

$$\int \mathcal{V}\, M\, j_\ell \,\mathrm{d}\eta \;=\; \Big[\, W M j_\ell \,\Big] \;-\; \int W M' j_\ell \,\mathrm{d}\eta \;+\; k\int W M\, j_\ell' \,\mathrm{d}\eta, \tag{22}$$

where $x = k(\eta_0 - \eta)$, so $\mathrm{d}j_\ell/\mathrm{d}\eta = -k\, j_\ell'(x)$ and the last sign is $+k$. Evaluating the left side (call it A) and the right side (B) on identical histories and identical endpoints, with $M = -C_1$, in each frame gives four numbers. Both identities hold: $|A - B|/|A|$ sits at 10^{-3}–10^{-2} and *falls under grid refinement* — $2.83\times10^{-3} \to 2.20\times10^{-3} \to 2.04\times10^{-3}$ at $(k, \ell) = (10, 2)$, and $1.105 \to 0.279 \to 0.0727$ at $(300, 2)$, about $4\times$ per doubling, which is second order and is what the trapezoidal rule gives on an oscillatory integrand.

Since A and B are the same integral by an exact identity, a difference has only two available explanations, and refinement removes one of them. **Boundary-against-bulk bookkeeping is excluded.** But $|A_N - A_C|$ is 1.00 to 1.78 of scale in every valid cell, with the sign reversed between columns: the coefficient definitions carry the whole of the gap.

One apparent corroboration, withdrawn. At low k the ratio A_C/A_N is independent of ℓ to four figures. That carries no information about the frames. $\mathcal{V}$ is a narrow spike, so $\int \mathcal{V} f j_\ell \simeq f(\eta_*)\, j_\ell(x_*) \int \mathcal{V}$ for *any* f, and the ratio of two $\mathcal{V}$-weighted integrals is $f_C(\eta_*)/f_N(\eta_*)$ — free of ℓ by construction. Predicted against measured, the two agree to 0.9–4.2% for $k \le 60$ and fail at $k = 300$, where the mode oscillates inside the visibility width and the ℓ-freedom correspondingly disappears. The check cost one run and removed a finding rather than adding one.

11.4 Triage B — the one frame-sensitive site that remains

Five candidates were open on 18 October. **Four are subsumed or excluded** — the $\ell = 0$ matching row is landed; boundary-against-bulk is excluded by T-85; and the dipole boost $-v/k$ and the supplied C_1' are both subsumed by T-82, which shows the source they would have corrupted is clean. **The fifth is the site that remains**: $\kappa' v_B$ inside C_1 does not go away, it *is* the slot and weight assignment of C_1, which is why it is named here rather than counted among the four. **One site remains: the slot and weight assignment of C_1.**

Identity (22) has three terms and every one of them carries the weight W. The bundle has two, and neither carries W:

	the identity requires	`integral_solution.py` **has**
boundary	$[WMj_\ell]$, weight $W = e^{-\kappa}$	$(C_2'^{*} - C_1^{*})\,\mathcal{D}(\eta_0, k)\,j_\ell(x_*)$ — weight $\mathcal{D}$, a *diffusion* factor, carrying neither W nor $\mathcal{V}$
j_ℓ bulk	$-\int WM'j_\ell$, weight W, *minus*	$\int \mathcal{V}\mathcal{D}\,\{C_0 - C_1' + C_2''\}\,j_\ell$ — weight $\mathcal{V}\mathcal{D}$, *plus*
j_ℓ' bulk	$+k\int WMj_\ell'$, M *undifferentiated*	**absent. The bulk has no j_ℓ' slot.**

The structural half is the third row. Everything was integrated by parts *onto the coefficients* — $\{C_0 - C_1' + C_2''\}$ is (102)'s printed bracket, and it is the W-weighted by-parts form of $\int W[C_0K + C_1K' + C_2K'']$ — and that decomposition is then evaluated at g weight, where it is not available. The boundary meanwhile is *Annals II*'s own `visibe` collapse: under $\int \mathcal{V} = 1$ the $-\mathcal{V}C_1$ term becomes $C_1(\eta_*)$. **So the bundle carries one term from each of two incompatible decompositions**, and then applies a third weight, $\mathcal{D}$, to one of them.

What the repair is, stated so it can be refuted. Not a split of the slot. A split, *plus* a weight change on each half, *plus* a slot the bulk does not currently have:

1. C_1's two contents separate — M's by-parts residue, and $(v_b + \sigma)$.
2. They *arrive under* different weights: W for the gravitational half, $\mathcal{V}$ for the scattering half. **That is an outcome, not an operation, and reading it as one is finding T-103** — the scattering half arrives at $\mathcal{V}$ weight carried on the *undifferentiated* $\hat{v}_B$; it is not *multiplied* by $\mathcal{V}$. Multiplying $C_1^{\text{thom}} = -\kappa'\hat{v}_B$ by $\mathcal{V}$ applies a second κ', which is the 320–682 of §11.8. See §10.1, where the weights fall out of the integrating factor and nothing is multiplied by hand.
3. The W-weighted half needs *both* bulk slots, j_ℓ and j_ℓ', because by-parting onto the coefficients is what removed the second, and that step is legal only at W.

The numerics for the missing slot already exist in the file: $j_\ell' = [\ell j_{\ell-1} - (\ell+1)j_{\ell+1}]/(2\ell+1)$ is built inside the rank-2 *boundary* term, which has never been evaluated because C_2 vanishes in every Newtonian run.

The acceptance condition is stated in advance. The split must leave A_N and A_C equal, and must not move the Newtonian column by more than the quadrature floor T-85 measured. Frame agreement alone is *not* sufficient — two wrong columns can agree — so the spectrum adjudicates, and it adjudicates last.

11.5 Triage C — frame-blind sites

A term X with $\Delta X = 0$ passes every frame test for free, so no frame instrument can see it. These cannot cause the discrepancy and can ruin the spectrum; they are listed so that the two questions are not confused.

	site, and what it bears on	status
C1	the coefficient of any boost-invariant term; low-k amplitude	`c5` blind by construction
C2	`LCDM_MODEL` without Ω_r; peak positions	confirmed, 37%, T-83
C3	radiation-era driving, static $T(k)$; amplitude and phase	open (A-3)
C4	the tight-coupling freeze at $6\eta_*$; peak contrast, temperature–Doppler balance	open (A-5)
C5	k_D computed at $R = 0$; one $\mathcal{D}(k)$ across histories; higher peaks and the tail	T-81a/c/d, v1.4.0
C6	$\mathcal{D}(k)$ applied to the integrand *at all*; the late ISW	T-81b — first named mechanism

C6 and the slot question are the same structural change seen twice. Diffusion damping is a scattering effect; gravitational redshift accumulated after decoupling is not damped by photon diffusion. Scattering damping belongs to the scattering half of the source and not to the gravitational half — so the split that fixes the weights fixes the damping in the same stroke. One change, three symptoms: the frame gap, the missing late ISW, and the $\mathcal{D}$-on-the-boundary mismatch.

11.6 Triage by conformal time

The same question asked along the photon's history. It localises cleanly, and the over-damping sits in two windows — neither of which is an integration limit, since T-85 settled the bookkeeping.

window	what the code does	what it costs
$\eta < \eta_{\rm eq}$	prescribed matter-growth Φ times a static $T(k)$, where the potentials should decay through horizon entry and drive the oscillator	peak amplitude and phase (C3)
$\eta_{\rm eq}$	no Ω_r in `LCDM_MODEL`	peak positions, 37% in ℓ_A (C2)
$\eta_{\rm eq} \to \eta_*$	oscillator sound speed correct; k_D evaluated at $R = 0$ while the oscillator runs at $R(\eta_*) \simeq 0.61$	the damping scale (C5)
$\eta_* \pm \frac{1}{2}\Delta\eta_{\rm vis}$	instantaneous boundary, envelope substituted for a visibility of finite width $\Delta\eta = 0.00972$	the tail from $\ell \simeq 785$
$\eta_* \to 6\eta_*$	tight-coupled oscillator continued and then frozen; no slip, no growing quadrupole, no polarisation	peak contrast, the temperature–Doppler balance (C4, A-5)
$6\eta_* \to \eta_0$	the integrand carries $\mathcal{D}(k)$ where $e^{-\kappa} = 1$ and no scattering occurs	the late ISW (C6)
η_0	reception: Doppler and aberration	nothing — THEOREM

Three consequences are worth drawing out.

The integration range is clean. T-85 settled the bookkeeping, so how far the problem has been integrated is no longer a candidate.

"Conformal time" and "missing radiation" are one question. η is not a free choice: it is fixed by the background. A radiation-free background gives a different $\eta(z)$, hence a different $k\Delta\eta$ at every point, hence different Bessel arguments and different peak positions. The 37% in ℓ_A *is* the conformal-time defect, already measured.

Over-damping is two substitutions, not a limits error, and both sit after η_*: an envelope standing in for a visibility of finite width, and $\mathcal{D}(k)$ applied to a source generated where no scattering occurs. A discriminator is available and was not used until now: Silk damping bites at $\ell \simeq 1807$, the visibility full-width scale is $\ell \simeq 785$, and the computed spectrum departs at $\ell \simeq 700$. **The departure fingerprints the width scale, not Silk** — so k_D at $R = 0$ is a real defect and is *not* what sets where the tail goes wrong.

11.7 What would change this account

Two outcomes would move it, and both are cheap.

If the slot read does not match the shape in §11.4, the discrepancy returns to having no named site, which would be the first time since 18 October. *It matched*; the table in §11.4 is that read.

If removing $\mathcal{D}(k)$ from the integrand alone recovers a late ISW, C6 is confirmed and C3–C5 reduce to amplitude-only concerns.

11.8 The repair, attempted five times, and withdrawn

Recorded here because five failures carry more information than one success would, and because a reader of this document should not have to reconstruct them from the register.

T-86's reading was implemented as four independent controls, every one defaulting to `False`. The first combination closed the frame gap by four orders at $\ell \geq 50$ for $k \lesssim 60$ — and **is withdrawn**. It applied $\mathcal{V}\, C_1^{\rm thom}$ where (102) applies $W\, C_1^{\rm thom}$, and since

$$C_1^{\rm thom} = -\kappa' \hat{v}_B, \qquad W\kappa' = \mathcal{V} \quad \Longrightarrow \quad W\, C_1^{\rm thom} = -\mathcal{V}\hat{v}_B \textbf{ identically,}$$

the Thomson term is already at visibility weight in (102) as printed and there was never a weight to change. The control therefore multiplied it by a spurious κ', measured at 320 to 682. The diagnostic spectrum it produced — first peaks moving from 1.3–1.5 ℓ_A to 0.56 ℓ_A — is withdrawn with it and is retained in `diagnostics/` as the record of what the defective variant produces, not as evidence.

What the five failures leave standing is a review flag, and it is P1-T's. *A coefficient carrying a bare κ' is a site where a large residual is manufactured the moment it is differentiated, and is worth flagging before implementation.* **Revised 2026-10-21, finding T-113 and decision S48**: an earlier form of this paragraph stated it as a prohibition — *"κ' may live in W, in $\mathcal{V}$, or in a derivative of either, never in a coefficient"* — and it was read as normative. **It is not a theorem**, because $W(\kappa'\mathcal{S}) = \mathcal{V}\mathcal{S}$ is exact: a coefficient *may* hold κ', and the defect in (102) is wrong weights, signs, boundaries or missing partners rather than the presence of κ'.

The flag is kept because it separates the *legitimate* κ'^2 inside $\dot{\mathcal{V}} = (\kappa'' + \kappa'^2)W$, which comes from differentiating the **weight**, from the illegitimate one, which comes from differentiating a **coefficient**; both look identical in the output and only the provenance tells them apart. `tests/frame/c7_kappa_ledger.py` raises it structurally, and both of this week's defects would have been flagged by it before implementation. **What it may no longer do is settle a question on its own — D24** supplies the size the flag cannot: $\kappa''\tilde{v}_B$ is 99.8% of $|C_1'|$ in the Newtonian frame and 109.8% in CDM, finding **T-121**.

11.9 What one term does to the integrand

Measured on the released code, as shares of the summed absolute Newtonian integrand:

term	EdS k=10	k=60	k=300	PLANCK k=10	k=60
$\mathcal{V}\kappa''\hat{v}_B$	**0.935**	**0.971**	0.616	**0.933**	**0.974**
$\mathcal{V}\kappa'\hat{v}'_B$	0.066	0.024	0.314	0.065	0.015
$\mathcal{V}\frac{k}{3}\tau_1$	0.0002	0.0064	0.082	0.0004	0.013
$\mathcal{V}(\Phi'_A - \Phi'_H)$	0	0	0	0.0018	0.0028

One term is 93–97% of the integrand; the acoustic driving is 0.02% and the ISW, where there is one, 0.18%. Against the source it stands for — the physical Doppler $g\hat{v}_B = W\kappa'\hat{v}_B$ — it is 1.81×10^5 (EdS) and 8.99×10^4 (PLANCK), and $\kappa'' \simeq -\kappa'^2$, so it carries **three powers of κ' where the source carries one**.

This is weight-independent. Changing $\mathcal{V} \to W$ divides every term in the bracket by κ' uniformly and changes no ratio, so A-2 cannot reach it. The term exists only because (102) differentiates C_1, which

contains κ' — that is, because the integration by parts was taken *onto the coefficients* rather than onto the product.

11.10 What this section does not establish

No measurement here is a comparison against an external code, and none is a claim that the spectrum is right. The chain establishes that the rank-1 source is frame-consistent, that the by-parts identity holds, and that the slot-and-weight assignment is where the remaining disagreement lives. **It does not establish that repairing the assignment closes the gap**, and the acceptance condition in §11.4 is written so that the repair can fail visibly.

12 The angular spectrum

Two things happen here that happen nowhere else: the harmonic normalisation has to disappear, and the quantity stops being linear in the transfer. Both are structural, and both have cost a finding.

12.1 Two printed forms, and a cancellation that must be exact

Annals II §7.2 prints the observable twice:

$$C_\ell = \frac{2}{\pi}\frac{\beta_\ell^2}{(2\ell+1)^2}\int \frac{\mathrm{d}k}{k}\, k^3 \big|\tau_\ell(k,\eta_0)\big|^2, \tag{186}$$

$$\big\langle \tau_{A_\ell}\tau^{A_\ell}\big\rangle = \frac{\beta_\ell}{2\pi^2}\int k^2\,\mathrm{d}k\,\big|\tau_\ell\big|^2, \qquad C_\ell = \frac{\Delta_\ell}{2\ell+1}\big\langle \tau_{A_\ell}\tau^{A_\ell}\big\rangle, \qquad \Delta_\ell = \frac{4\pi\beta_\ell}{2\ell+1}. \tag{187, 188}$$

The harmonic normalisation cancels, identically. (176) returns $T_\ell(k) \equiv \beta_\ell\tau_\ell(\eta_0)/(2\ell+1) = \alpha_\ell^{-1}\tau_\ell$, so $\tau_\ell = \alpha_\ell T_\ell$ and (186) becomes

$$C_\ell = \frac{2}{\pi}\underbrace{\frac{\beta_\ell^2\alpha_\ell^2}{(2\ell+1)^2}}_{=1}\int \mathrm{d}k\, k^2\,\big|T_\ell(k)\big|^2, \tag{23}$$

because $\alpha_\ell = (2\ell+1)/\beta_\ell$. **The ℓ-dependent normalisation of the harmonic expansion vanishes from the observable, which is what it must do** — and the variable in which it vanishes is exactly the one Appendix F's free-streaming solution is bare in (§9). (187, 188) reaches the same 1 by a different path through the same weights.

That is worth more than tidiness. It means a β_ℓ or α_ℓ slip cannot hide in the spectrum: to survive, it would have to survive a cancellation that is an identity. Both coefficients are therefore formed as exact rationals and compared as rationals out to $\ell = 2000$, not as floating point.

Finding T-17, and it is the cautionary one. Written in the obvious order — each factor converted to a double, then multiplied — the identity held only while both halves were separately representable. $\beta_\ell \sim 6\times10^{-270}$ at $\ell = 900$, so β_ℓ^2 underflows to zero while α_ℓ^2 overflows, and their exact ratio of one came back as an overflow for every $\ell \geq 508$. **The bundle's multipole ceiling was 507, set by a quantity equal to one**, with no diagnostic and nothing in the physics to suggest it. The rule the weights module already stated — *multiply at the point of numerical use, never before a comparison* — is the whole of the repair.

And one of these really is large. The multipole mean square (187, 188) carries $\beta_\ell \alpha_\ell^2 = (2\ell+1)^2/\beta_\ell$, which leaves the double range near $\ell \simeq 1500$. **That is the quantity and not an artefact of how it is written**: in a normalisation where β_ℓ is that small, the mean square is that big. C_ℓ itself stays finite, because the released route combines the weights in $\mathbb{Q}$ before converting and never forms the intermediate at all.

12.2 Which of the two is the quantity, and why every figure plots it

(186) is a statement about **mode** coefficients and holds only for an almost-Robertson–Walker geometry. (187, 188) is a **multipole** mean square and holds generally. They are numerically identical here and conceptually are not interchangeable, and §5 is where the two objects are separated.

Every released figure plots the multipole quantity. That is not a presentational preference: it is the only one of the two that survives leaving the almost-Robertson–Walker setting, which is the setting the nonlinear extension leaves. *A reconstruction that will be generalised should be plotted in the variable that generalises, even where the two agree to machine precision.*

12.3 C_ℓ is quadratic, and three readings have gone wrong on it

Everything upstream of this section is linear in the perturbation. C_ℓ is not: it is quadratic in the transfer, and three consequences follow that have each been misread at least once in this project's record.

First, a decomposition of the transfer is not a decomposition of C_ℓ. Splitting T_ℓ into rank-0, rank-1 and rank-2 parts gives autocorrelations *and* cross terms. Measured (**F5**): the autos together with twice the crosses reproduce C_ℓ to 4.7×10^{-16}, while the autos *alone* are off by 0.10. **The cross terms are not a correction to be neglected; they are half the statement.**

Second, a frame gap in C_ℓ is not a frame gap in the invariant. The two released frames differ by a median of 15% over $2 \le \ell \le 400$ and reach 40% at some ℓ. **That is not by itself a defect**, because C_ℓ as computed here is built from frame-dependent pieces while the invariance (102) satisfies is a statement about the *total at fixed k*, which diagnostic D6 measures separately. *What was a defect is the separate matter of how large the gap is*: F9's caption read *"which is a result and not a defect"* and that reading is withdrawn — the gap is now understood as the third symptom of the bare κ' carried inside C_1 (**T-101**, **T-121**), and §10.1 carries it.

Third, a ratio of two spectra is not a ratio of two amplitudes. $D_\ell = \ell(\ell+1)C_\ell/2\pi$ is renormalised at $\ell = 30$ in every released figure, because neither this bundle nor the external codes fixes the primordial amplitude the same way and an amplitude comparison would measure a normalisation. **Every percentage in this supplement comparing shapes is a shape comparison and nothing in it constrains A_s.**

12.4 Criterion 5b, and the two different "two routes"

There are two route comparisons in this bundle and they check different things. Conflating them is easy and would be expensive.

comparison	what differs between the two sides	value
(186) against (187, 188)	the *harmonic weight algebra* alone	exact in $\mathbb{Q}$ to $\ell = 2000$
criterion 5b: (102) against (177)–(179)	the *assembly order* alone — generic-then-specialised against Newtonian-from-the-start	9.5×10^{-6} against a 10^{-3} bar

Criterion 5b is the audit half of the release, and it exists because a symbolic recovery is an argument whose arithmetic nobody has run. The antecedent writes its integral solution twice

and says so — *"Later we will explicitly recover these equations in the Newtonian frame"*, and at the end of that recovery, *"This completes the recovery of the standard integral solution results using the* $1+3$ *CGI approach at linear order"*. The recovery is carried out on the page, as it should be. Evaluating both forms and comparing them had never been done.

What 5b holds fixed, written out rather than left to be noticed. Both routes are given the same background and η grid, the same recombination history and therefore the same κ', visibility and η_*, the same single acoustic solve, the same potentials, the same Bessel table, the same diffusion damping — and **the same line-of-sight weight, $e^{-\kappa}$, on both sides**.

That last one is deliberate and it is the largest thing 5b does not see. (102) prints $\mathcal{V} = \kappa' e^{-\kappa}$ and (98) prints $e^{-\kappa}$; approximations **A-1** and **A-2** record the difference, and at C_ℓ *the two weightings differ by a factor of eighty across the Sachs–Wolfe plateau for that reason alone.* Weighting both sides alike removes the one thing the comparison is not about, so that what remains is the assembly algebra. *It also means 5b's* 9.5×10^{-6} *says nothing whatever about the weight question that §10.1 is entirely about.*

And it says nothing about agreement with an external code. CAMB and CLASS are drawn as context and are not a target (decision **S4c**). The distance to them is large, expected, and itemised by approximation rather than totalled: 97.8% across the plateau, 137% across the peaks, of which the damping coefficient alone is 17.2%. **A bundle gated on an external code would be measuring that code; this one is gated on its own two routes and reports the distance separately.**

12.5 Where the spectrum is gated, and what none of it establishes

gate	what it tests	what it does not
rational weights	that (186)'s and (187, 188)'s prefactors are both exactly 1, as rationals, to $\ell = 2000$	anything about τ_ℓ: the weights cancel whatever multiplies them
criterion 5b	that the generic and Newtonian assemblies agree, at 9.5×10^{-6}	the background, the recombination history, the acoustic solve, the Bessel table, the damping, or *the weight* — every one of them is shared
rank decomposition	that the ranks sum to the total, 4.1×10^{-16} in the CDM frame	that any rank is right
two-route C_ℓ difference	reported, now bitwise zero since **T-17**	it has stopped being a test and is kept as a reported number

No gate in this section validates the assembled spectrum. The basic checks above pass, and they pass on an assembly whose first acoustic peak sits at $0.635\,\ell_A$ against a matched 0.757. *That is not a contradiction and it is the reason the "does not establish" column exists*: every one of these gates holds fixed, or is indifferent to, precisely the part that is wrong. **T-3** is the standing statement of why — a comparison that holds an input fixed cannot check that input — and the spectrum is where it bites hardest, because it is the last step and inherits every shared input above it.

13 Comparison and its range

The C_ℓ comparison is Sachs–Wolfe only: (196) descends from (181), in which the acoustic modulation $\cos(kr_s)$ is dropped by taking $r_s^* \to 0$, and §8.3.2 sets the transfer function to unity. Its accuracy is therefore bounded by the term that was dropped. The range is derived, not chosen: $2 \le \ell \le 20$ at 5%, generated by `scripts/derive_ell_range.py`.

The wider comparison cites CMBFAST for its *method* — line-of-sight integration, Seljak & Zaldarriaga, the same reference the antecedent's own bibliography carries — and compares against **CAMB** and

CLASS, because the Fortran 77 original is unmaintained and effectively unrunnable. Saying so is more honest than claiming a comparison against a code nobody can run. It is reported in three bands, whose failure modes differ — $2 \leq \ell < 30$, cosmic-variance dominated and meaningful only binned; $30 \leq \ell < 1000$, the acoustic peaks, where a hierarchy or recursion error actually shows; and $1000 \leq \ell \leq 2000$, the damping tail, where C_ℓ is small and fractional differences blow up on a vanishing signal. The comparison is made against the corrected closed form.[6][7]

13.1 Criterion 5b's tolerance

The tolerance is *measured* rather than chosen; what it is measured against changed once it was run. The original rule took the CAMB–CLASS mutual difference as the floor, on the ground that no agreement can be claimed tighter than two trusted independent codes agree with each other. That floor is 0.370% across the acoustic peaks, and this reconstruction stands at 137% there. It was never going to meet it: (102) is a slow-decoupling solution with no polarisation hierarchy, no neutrino anisotropic stress and a visibility-weighted line of sight, set against two full Boltzmann codes. **A criterion that can only be failed is not a criterion.**

What *is* gateable is the thing the antecedent writes twice and recovers symbolically but never evaluates. Its integral solution appears as (102), carried with a generic u^a and specialised afterwards, and again as the source assembly (177)–(179), written in the Newtonian frame from the start. The paper states the relation and carries it out on the page; what no one has done is to compute both forms and compare them, which is the only part of the claim arithmetic can settle. Measured here over three backgrounds, nine modes $0.3 \leq k \leq 600$ and $2 \leq \ell \leq 600$, the two agree to 9.5×10^{-6} worst and 2.7×10^{-8} median against a gate of 10^{-3}. The comparison is stated at the transfer level at fixed k with the same line-of-sight weight on both sides: at C_ℓ the two branches differ by a factor of eighty in the plateau, and that difference is the weight the line-of-sight integral carries — a property of what (102) prints, recorded as an approximation and measured separately, not a disagreement to be gated away.

The external comparison is still run, still stamped, and reported alongside as context. Its value is not a pass mark but an itemisation: how far a $1 + 3$ slow-decoupling solution stands from a Boltzmann code, in bands, with the dominant term named.

The measurement itself is recorded here rather than only in the code. The external codes are not carried as dependencies of the bundle and are not run by the harness: `scripts/make_reference_spectra.py` regenerates the stamped spectra from CAMB and CLASS, `scripts/compare_criterion_5b.py` reproduces the comparison from those stamps with neither code installed, and the table below is what that script writes. It is a committed artefact of the bundle, recoverable and reproducible, and deliberately outside the bundled testing.

[6] The reduction of (200) carries a factor of the comoving distance that the displayed (201) does not; substituting $z = k\chi$ gives χ^{2-n}, which is χ^1 at $n = 1$. The companion result (204) descends from a $P(k)$ definition differing by one power of k and is correctly free of that factor, so the asymmetry is not a units convention. A misprint, corrected silently and recorded.

[7] That derivation uses the Bessel identity of §8.3, which requires $[(m/2)!]^2$ in its denominator. As printed it is exact where $\Gamma(m/2 + 1) = 1$, so the large-scale result (201) is unaffected; the horizon-scale normalisation (204) and the D_ℓ of (205) are. A misprint, corrected silently and recorded.

Shape, D_ℓ renormalised at $\ell = 30$. Context, not the gate (S4c).			
band	**vs CAMB**	**vs CLASS**	**CAMB vs CLASS**
plateau and late ISW, $2 \le \ell < 30$	97.8%	97.8%	—
acoustic peaks, $30 \le \ell < 1000$	137.0%	137.3%	0.370%
First acoustic peak			
this bundle	CAMB	CLASS	
$\ell = 170$	$\ell = 198$	$\ell = 198$	

Itemised: the part this bundle can attribute. The damping coefficient, 0.1% in the plateau and late ISW and 17.2% in the acoustic peaks, between (dif-damp-correct) at $f_2 = 3/4$ and (dif-damp) at $f_2 = 1$ — both *Annals II*'s own. The remainder is the slow-decoupling order, the visibility weight of (102) (approximations A-1 and A-2), and the acoustic scale, $z_* = 1114.5$ here against CAMB's 1088.5.

What a self-comparison does not establish is stated here rather than left to be inferred, because it is the obvious objection and it is a fair one. Both routes are handed the same background, the same recombination history, the same acoustic solution, the same potentials and the same damping, and a comparison that holds an input fixed cannot check that input. This is not a formality: setting the gate turned up an under-resolved recombination grid that moves the primary source at η_* by 1.2%, shared by both routes and invisible to the comparison between them. It is recorded, raised, and left for the decision it needs rather than changed inside a release.

The external codes of criterion 5b are compared in `provenance/EXTERNAL-CODE-COMPARISON-v1.md`, which records what CAMB and CLASS each model and — more to the point — what the comparison structurally cannot settle. Neither carries the $1 + 3$ covariant formalism, neither distinguishes mode from multipole coefficients, and neither carries the $O(\varepsilon^2 \ell)$ couplings of v1.5.0. So 5b is a check on the *linear* reconstruction and is incapable of validating the nonlinear extension, which is precisely why v1.0.0 must carry no $O(\ell)$ coupling: if one leaked in, the disagreement with CAMB and CLASS would be indistinguishable from a bug. The canonical parameter set switches reionisation, lensing and massive neutrinos **off**, each being physics this bundle does not model.

14 Evidence

Each figure is generated by the script named in its caption and is accompanied by a standalone caption file in `captions/`. This section is itself generated by `scripts/make_figure_pages.py`, so a figure appears here only once it exists in the tree, and what is outstanding is stated rather than omitted.

15 Release progression

A reproducibility bundle is not only a result; it is a record of how the result was arrived at, including the parts that were wrong on the way. This section is that record, one release at a time. Each entry states what was *achieved*, and then — separately, because they are different things — what was *learned*, meaning a lesson that changed how the next release was built.

15.1 v1.0.0 — the reconstruction, and whether it is faithful

Achieved. The free-streaming chain of Appendix F reproduced against *four* external hierarchies — Hu & Sugiyama Eq. (6), Wilson Eq. (8), Ma & Bertschinger Eqs. (49)/(50), Seljak & Zaldarriaga Eq. (3d) — each read from its own paper and integrated as printed, at residuals of 10^{-13}–10^{-14} against a 10^{-11} threshold, in two normalisations and two phase conventions. The almost-Friedmann–Lemaitre spectrum (186) computed from (176) by the mode route and the covariant route, agreeing exactly. The Sachs–Wolfe

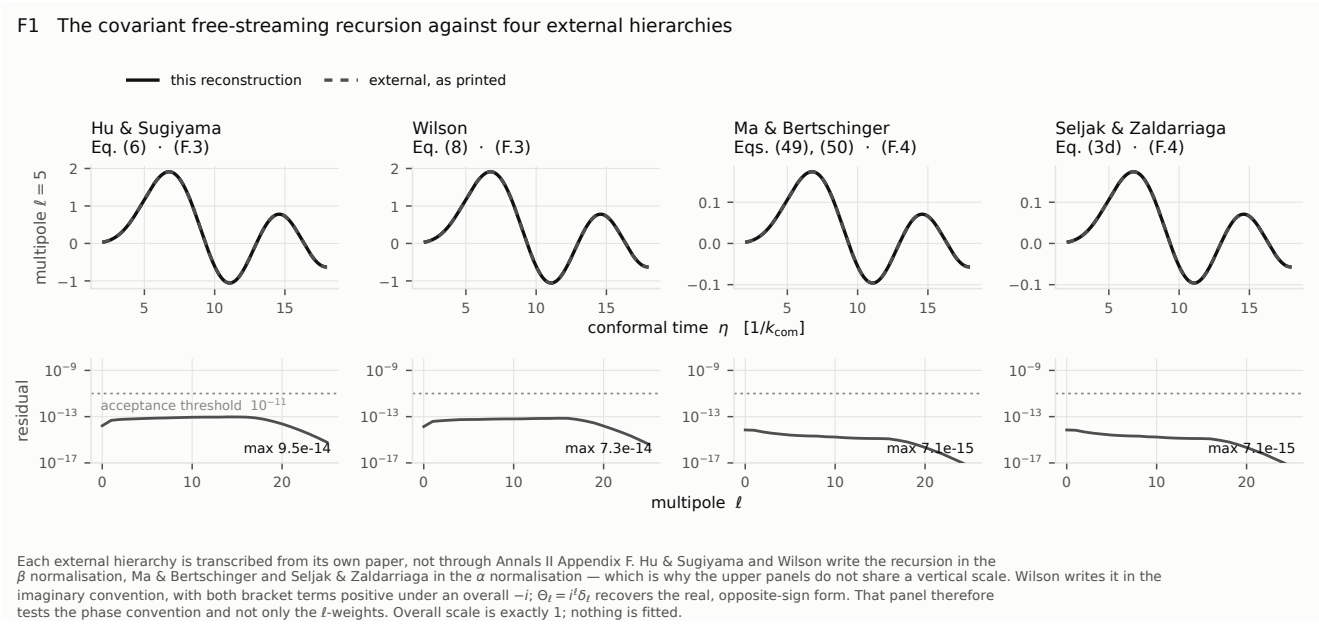


Figure 2: **F1 — The covariant free-streaming recursion against four external hierarchies.** The covariant free-streaming recursion of *Annals II* Appendix F, integrated from a unit monopole, against four external treatments of the same hierarchy. Each is transcribed from its own paper, never through Appendix F. **Upper panels, context:** the reconstruction and the external form at $\ell = 5$; they coincide. The panels do *not* share a vertical scale, because two normalisations are on display. **Lower panels, the claim:** $\max_\eta |\text{external} - \text{reconstruction}|$ against ℓ, with the acceptance threshold marked. Hu & Sugiyama write the recursion in the β normalisation and Ma & Bertschinger and Seljak & Zaldarriaga in the α normalisation, which is why Appendix F prints both (F.3) and (F.4); a basis phase leaking into the couplings would break at least one match. Overall scale is exactly 1.000000000 — nothing is fitted. Four panels from five sources: Wilson & Silk (1981) is the earliest, but at $K = 0$ it is Wilson (1983) Eq. (8) exactly, so it is credited rather than drawn.

Evidence category: cross-check. *Generated by* `scripts/figure_f1_appendix_f.py`. *Standalone caption:* `captions/F1-appendix-f.md`.

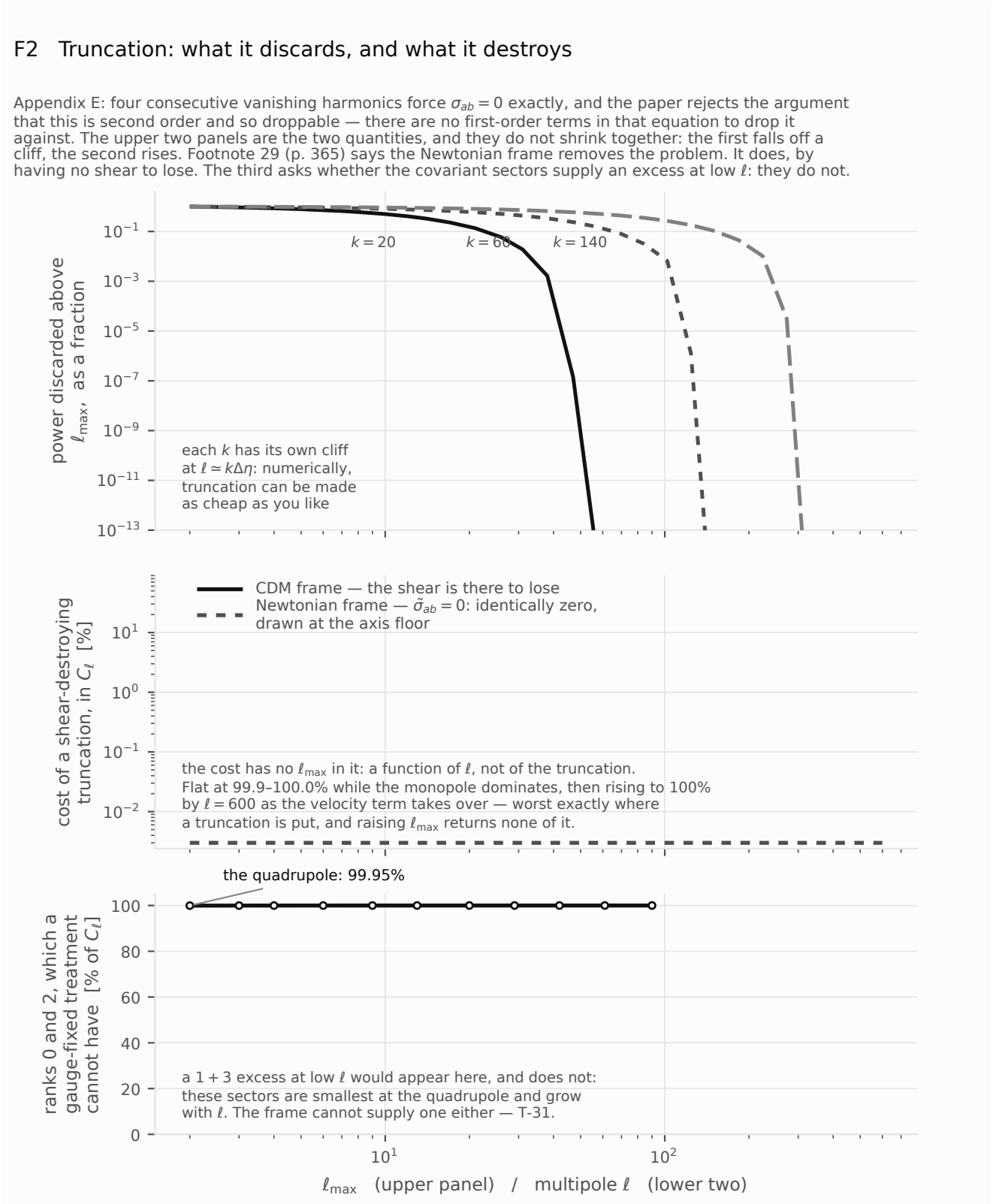


Figure 3: **F2 — What a truncation discards, and what it destroys.** Appendix E of *Annals II* proves that four consecutive vanishing harmonics force $\sigma_{ab} = 0$ exactly. This figure makes that quantitative in both frames, and the two panels answer two different questions that a single "truncation error" would blur.

What a truncation *discards* falls off a per-k cliff at $\ell \simeq k\,\Delta\eta$: above the cliff there is nothing left to keep, and the tail above $\ell = 600$ is identically zero at every probe mode. **What it *destroys*** has no $\ell_{\max}$ in it at all. Setting $\sigma_{ab} \to 0$ costs 0.54–0.68% out to $\ell \simeq 20$, rises monotonically above $\ell = 20$, and reaches 32.6% by $\ell = 600$ [**pre-T-57 values; this figure is superseded — see the dagger**] — a cost that grows with the multipole rather than being bounded by it, which is why a truncation argument cannot retire the shear.

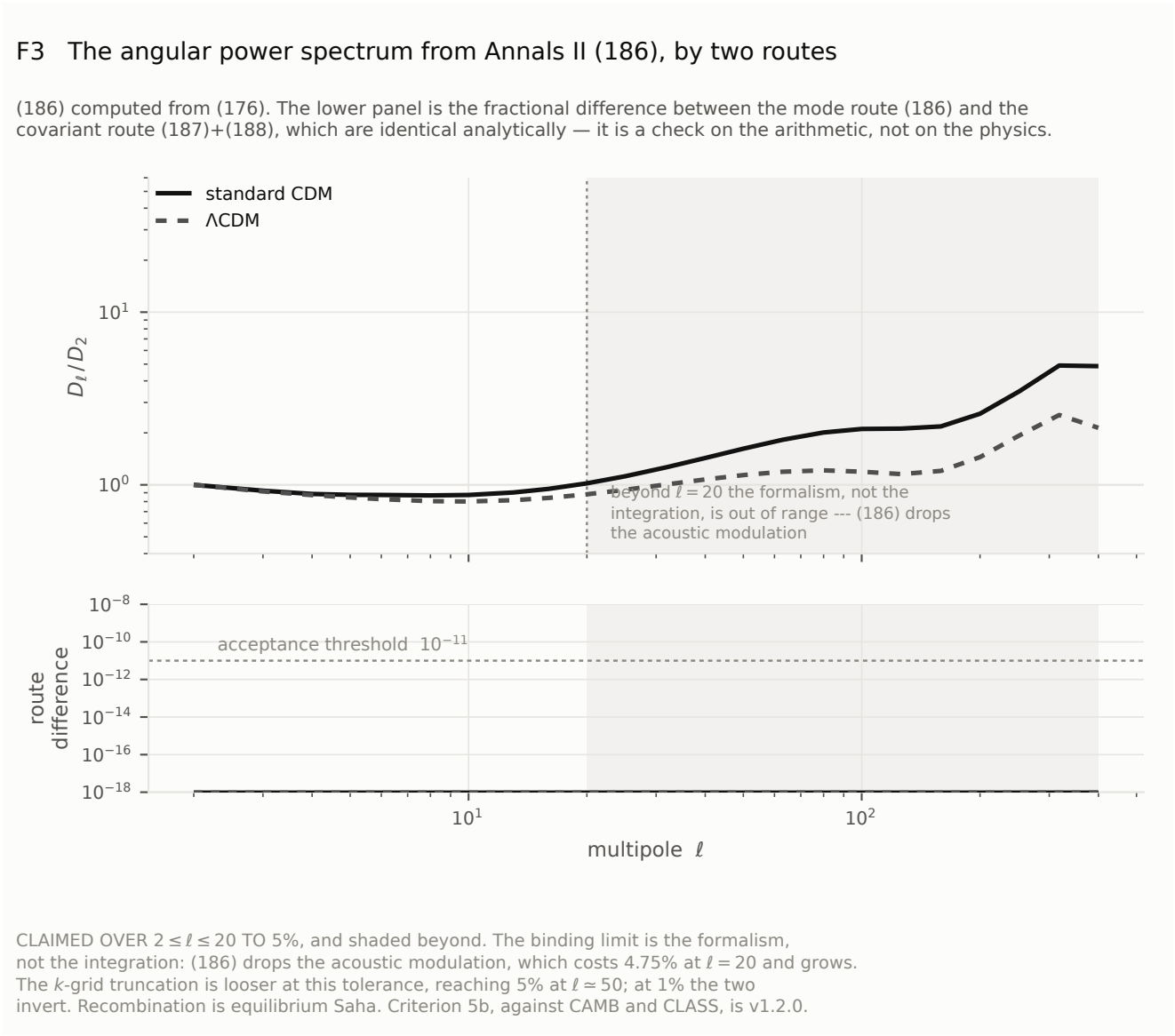


Figure 4: **F3 — The angular power spectrum from Annals II (186), by two routes.** Eq. (186) computed from Eq. (176), for the standard-CDM model of §8.3.2 and for ΛCDM, which *Annals II* does not treat. **Upper panel:** $D_\ell = \ell(\ell+1)C_\ell/2\pi$, each normalised to its own $\ell = 2$ value, so that the comparison is of *shape* and not of an amplitude neither model fixes here. **Lower panel:** the fractional difference between the mode route of (186) and the covariant route of (187) with (188). Those two are identical analytically, so the panel is a check on the arithmetic and not on the physics — and it is worth having, because it caught a factor of four in this bundle's own harmonic weights before any figure was drawn. It sits at 10^{-16}, five orders below the 10^{-11} acceptance threshold.

Evidence category: recovery. *Generated by* `scripts/figure_f3_angular_spectrum.py`. *Standalone caption:* `captions/F3-angular-spectrum.md`.

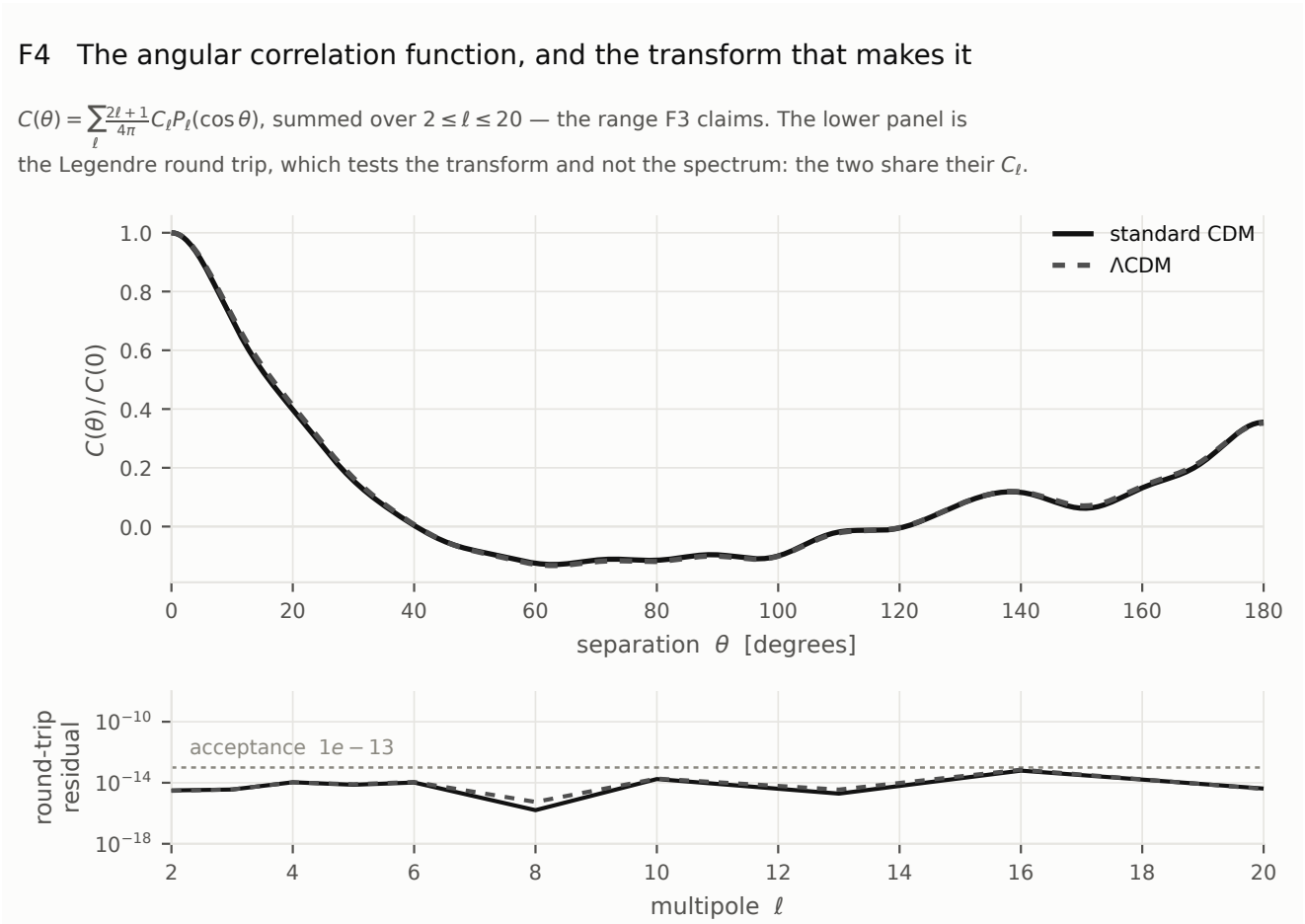


Figure 5: **F4 — The angular correlation function, and the transform that makes it.** $C(\theta) = \sum_\ell \frac{2\ell+1}{4\pi} C_\ell P_\ell(\cos\theta)$, the real-space view of Figure 3: the same content read as a correlation between two directions separated by θ rather than as power at a multipole. **Upper panel:** $C(\theta)/C(0)$ for standard CDM and ΛCDM, summed over $2 \le \ell \le 20$ — the range Figure 3 claims, bounded by (186) dropping the acoustic modulation. Summing further would draw multipoles this release does not stand behind, and a real-space curve hides which ℓ contributed, so the restriction is made in the transform rather than in this caption. **Lower panel:** the Legendre round trip $C_\ell \to C(\theta) \to C_\ell$, at 4×10^{-14} and 7×10^{-14} against an acceptance of 10^{-13}. It tests the transform and *not* the spectrum — the two share their C_ℓ — which is the same caution Figure 3's residual panel carries and for the same reason.

Evidence category: acceptance. *Generated by* `scripts/figure_f4_correlation.py`. *Standalone caption:* `captions/F4-correlation.md`.

F5 The source decomposition in two frames, by rank

The total is common to both frames and the decomposition is not. Split by rank, which is what the match (89) produces and is defined in every frame — not by the primary/Doppler/integrated split of §7.1.1, which exists only after the Newtonian choice has emptied the shear sector. The agreement is measured and it is not exact: the second panel is the gap, and it is drawn rather than described.

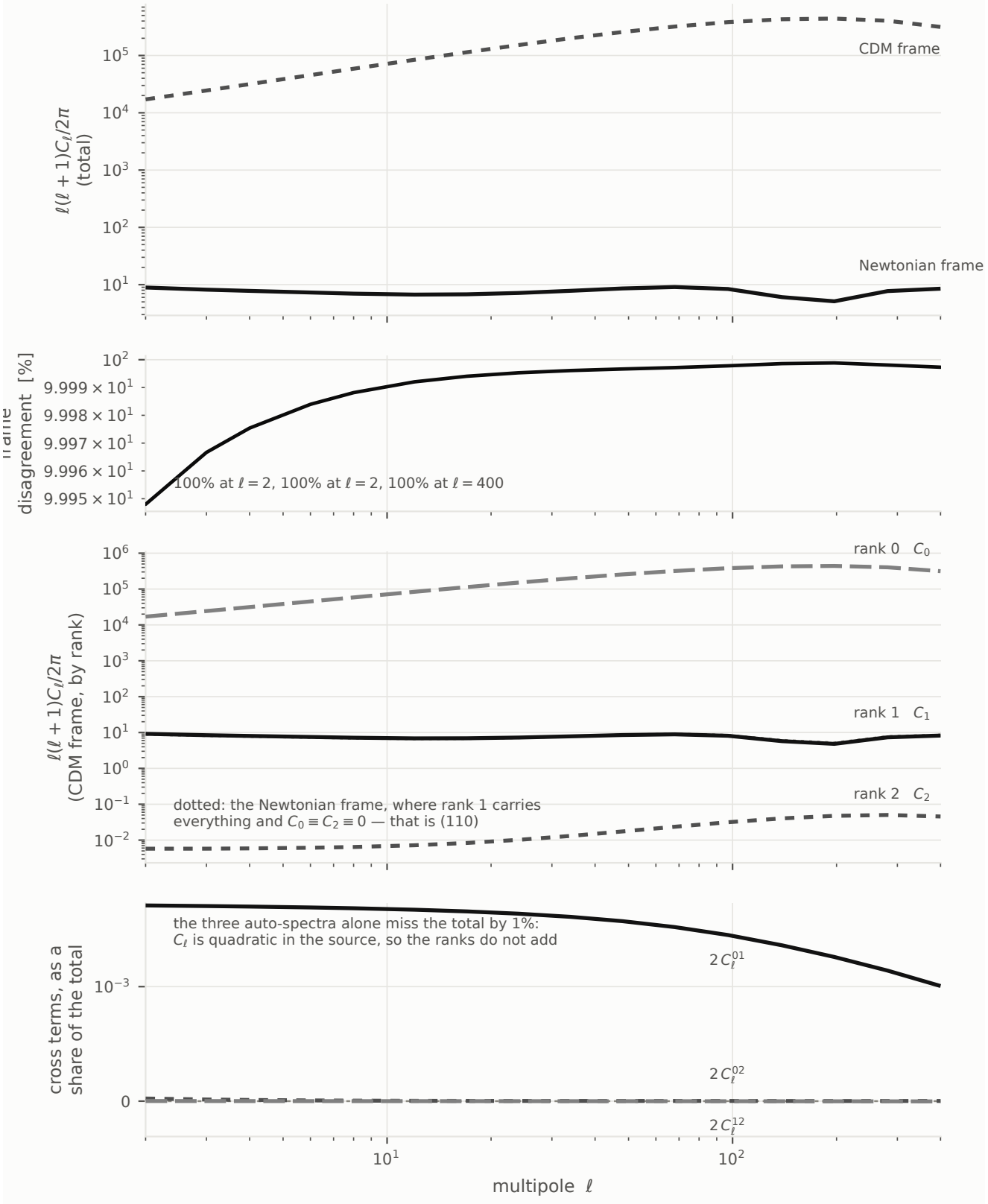


Figure 6: **F5 — The source decomposition, by rank, in two frames.** The integral solution (102) carries three coefficients C_0, C_1, C_2 — one per PSTF rank — and this figure draws them, in the

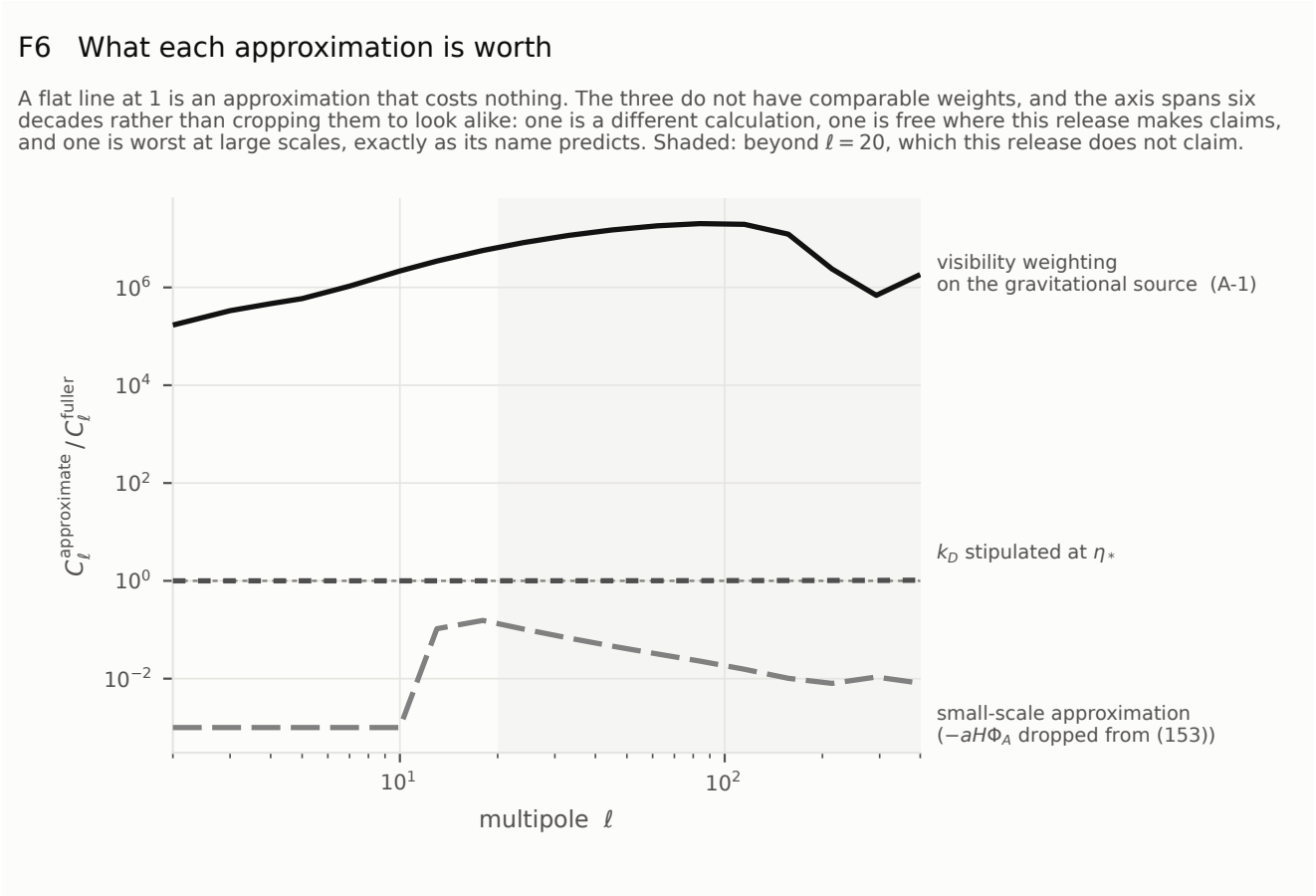


Figure 7: **F6 — What each approximation is worth.** Each series is $C_\ell^{\text{approximate}}/C_\ell^{\text{fuller}}$ for standard CDM, so the panel is scale-free and a flat line at unity is an approximation that costs nothing. *Annals II* makes three narrowing choices which this bundle reproduces rather than corrects, because reproducing a paper means reproducing its approximations (decision S15); they are recorded as **A-n** in `provenance/SOURCE-APPROXIMATIONS-v1.md`, deliberately apart from the defects recorded as **B-n**. This is the one place their cost is a number rather than an adjective.

The three do not have comparable weights, and that is the result, so the axis spans six decades rather than cropping them to look alike. The visibility weighting on the gravitational source (A-1) runs to 10^4–10^6: it is a *different calculation*, not a perturbation of one, which is why `Weighting` has no default. Stipulating k_D at last scattering costs under 0.1% across the claimed range and about 5% by $\ell = 400$ — it **vindicates the release**, which takes the cheap choice where it makes claims. The small-scale approximation, dropping $-aH\Phi_A$ from (153) and so producing the printed oscillator (154), departs from unity most at *large* scales, which is what its name predicts.

A correction kept because it nearly became this figure. The k_D series was first computed with $k_D(\eta)$ along the whole ray. After last scattering $\kappa' \to 0$, so $1/\kappa'$ in (dif-damp) diverges and k_D collapses, damping every integrated source to nothing — a plasma formula evaluated in a universe that no longer has a plasma. That version reported the release's choice as costing a factor of nine at $\ell = 2$; the factor was the artefact. Clamped to $k_D(\min(\eta, \eta_*))$ the comparison is a real one. Shaded: beyond $\ell = 20$, which this release does not claim.

Evidence category: diagnostic. *Generated by* `scripts/figure_f6_approximations.py`. *Standalone caption:* `captions/F6-approximations.md`.

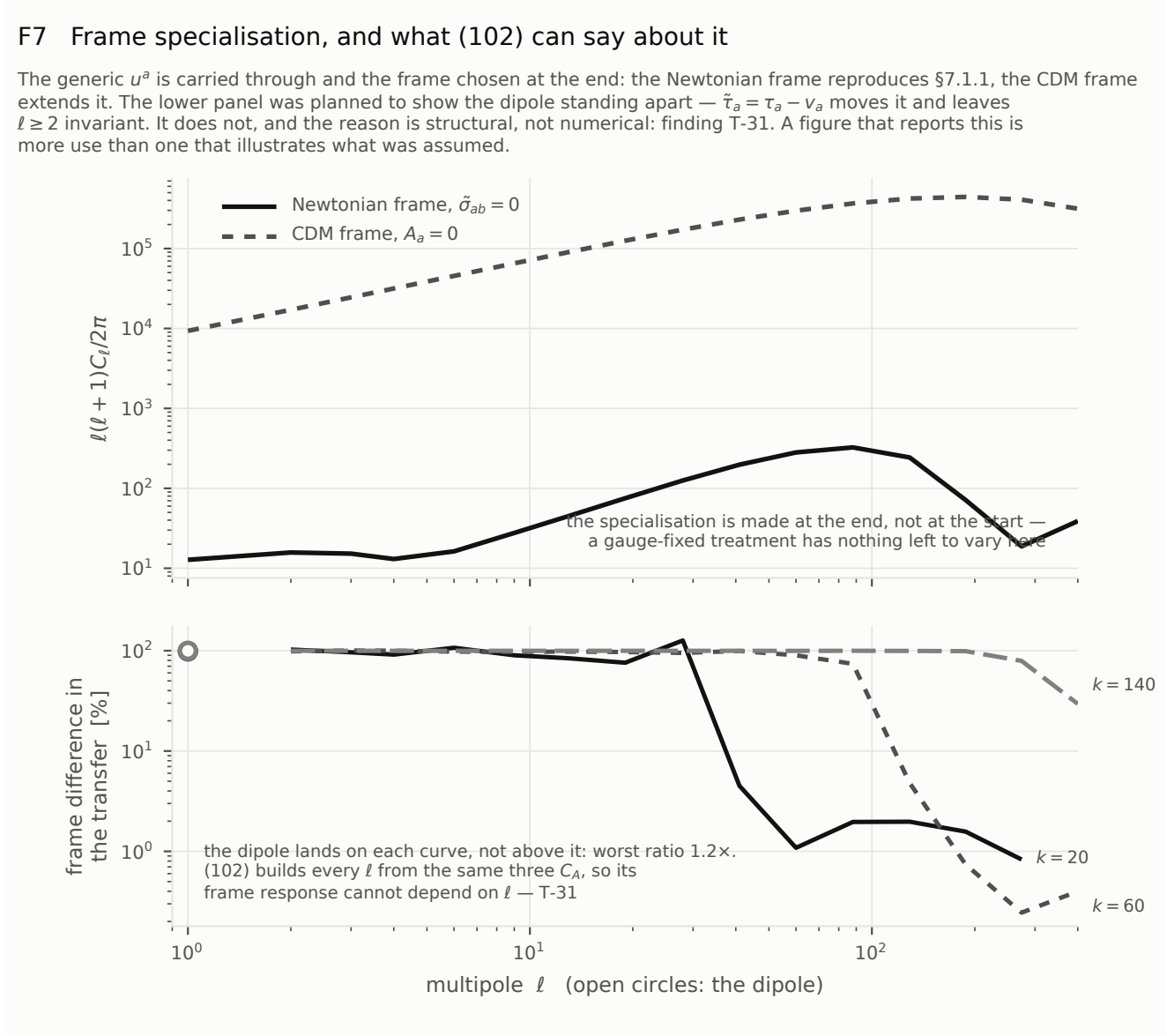


Figure 8: **F7 — Frame specialisation, and a planned claim that was false. This figure reports that its own planned claim is false, and that is why it is here.** It was designed to show the dipole standing an order of magnitude apart from the higher multipoles under a change of frame — the dipole being the one multipole a boost moves at first order. It does not. The dipole lands on each curve instead: worst ratio 1.25× over the probe set, and 1.00× at $k = 20$.

The reason is structural and is finding T-31. Equation (102) builds *every* ℓ from the same three coefficients C_A, so a frame change that moves C_1 moves every multipole built on it in the same proportion. A frame response that depended on ℓ would contradict the equation's own form, and `test_102_cannot_make_its_frame_response_depend_on_the_multipole` asserts that it does not. The spread *over* ℓ does grow with k — 1.01, 1.14, 8.14 at $k = 20$, 60, 140 — but that is the transfer function's k dependence, not the frame's ℓ dependence.

A figure that had been drawn to its plan would have been drawn wrong. The plan is kept in `captions/FIGURE-PLAN-v1.0.0.md` and the outcome is kept here, because a reproduction bundle whose figures only ever confirm their captions is not measuring anything.

Its environment is built for the k grid, not for the probe modes — finding **T-34**, where this figure sized its η grid for $k = 140$ and then integrated to $k = 420$. `check_k` now raises rather than interpolating past what the grid resolves.

Evidence category: diagnostic. *Generated by* `scripts/figure_f7_frames.py`. *Standalone caption:* `captions/F7-frames.md`.

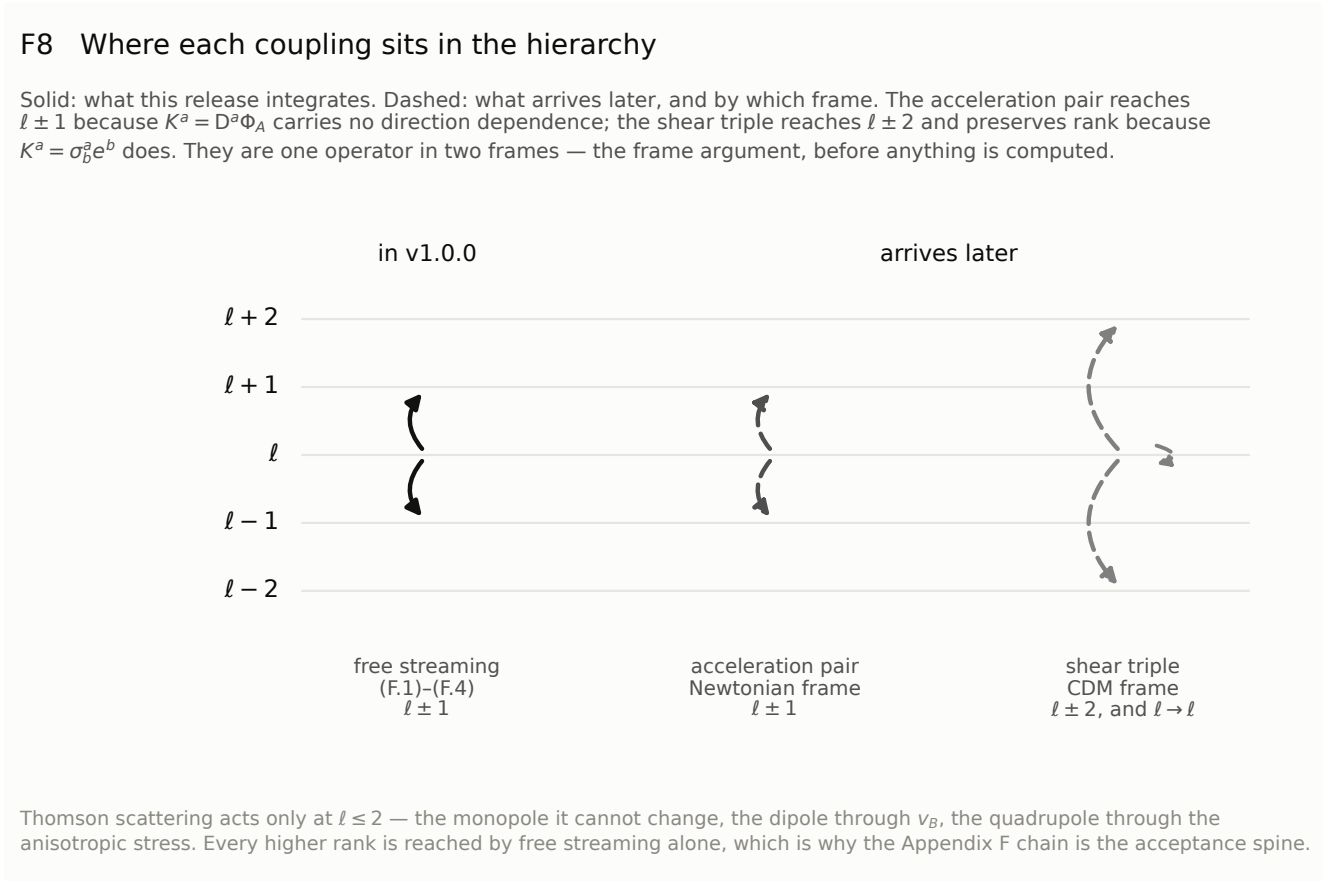


Figure 9: **F8 — Where each coupling sits in the hierarchy.** The one released figure that computes nothing: a statement about the *structure* of the exact divergence hierarchy, read off the equations rather than measured. It is generated by a script like every other figure, so that the manifest covers it — a hand-drawn diagram is the one artefact that can drift without any gate noticing.

Solid: what this release integrates. Free streaming, (F.1)–(F.4), couples rank ℓ to $\ell \pm 1$ and to nothing else, which is the whole of v1.0.0's hierarchy and the reason the Appendix F chain is the acceptance spine. **Dashed:** what arrives later, and by which frame. The acceleration pair reaches $\ell \pm 1$ because $K^a = \mathrm{D}^a\Phi_A$ carries no direction dependence of its own; the shear triple reaches $\ell \pm 2$ *and* preserves rank because $K^a = \sigma^a{}_b e^b$ does. The rank-preserving coupling exists only when K^a depends on direction, and is the single structural difference between the two frames.

Both are the same operator $K^a_\perp \partial/\partial e^a$, and the number of ranks each reaches is fixed by that one property. **That is the whole content of the frame argument, visible before any of it is computed.** Decision S9 is drawn as well as asserted: every coupling in the solid half is a ratio of linear polynomials and tends to a constant, so an $O(\ell)$ coupling wired in early would fail `tests/test_appendix_f_chain.py` at large ℓ.

Evidence category: schematic. *Generated by* `scripts/figure_f8_schematic.py`. *Standalone caption:* `captions/F8-schematic.md`.

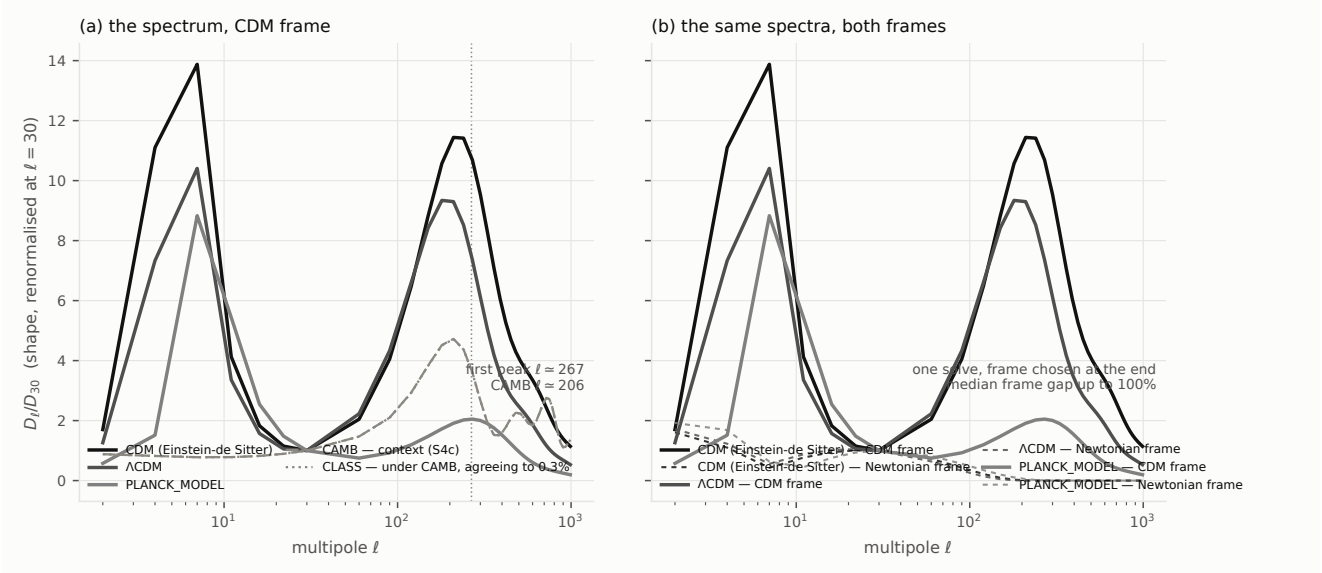


Figure 10: **F9 — The angular spectrum with the acoustic peaks, in both frames. The release's deliverable** (decision S1): the angular power spectrum for the three backgrounds this bundle ships — CDM (Einstein–de Sitter), ΛCDM, and the stamped Planck 2018 set with a photon-only radiation era whose $\Omega_\gamma h^2$ is *derived* from $T_{\rm CMB}$. Shape, D_ℓ renormalised at $\ell = 30$, because neither this bundle nor the external codes fixes A_s the same way and an amplitude comparison would measure a normalisation.

Panel (a) — the spectrum. The peaks and the damping tail come from *Annals II*'s own equations: the integral solution (102) with the Bond–Efstathiou transfer function driving the oscillation per mode, and the diffusion scale from **(dif-damp-correct)** at $f_2 = 3/4$, which is the antecedent's own corrected coefficient rather than a departure from it (**T-43**). **This shows that** the almost-Friedmann–Lemaitre formulation, carried to the damping scale, produces acoustic structure without anything being imported from a Boltzmann code.

CAMB and CLASS are drawn as context and are not a target — decision **S4c**. Criterion 5b is gated on this bundle's own pipeline, (102) against (177)–(179), at 9.5×10^{-6} against 10^{-3}. The distance to the external codes is large, expected, and *itemised rather than totalled* in `provenance/CRITERION-5B-v1.md`: 97.8% across the plateau, 137% across the peaks, of which the damping coefficient alone is 17.2%. The first peak is quoted from a *parabolic interpolation* through the grid maximum and its neighbours, because the ℓ grid steps by 30 across the acoustic band and reading the peak off it directly would assert a precision the sampling does not have.

Panel (b) — the panel a gauge-fixed treatment cannot draw. The same spectra in *both* frames, from *one* solve: `coefficient_sets` returns the Newtonian and CDM triples together, because (102) is carried with a generic u^a and the frame is chosen at the end. **This shows that** the frame is a choice made after the derivation rather than an assumption built into its start — the claim the $1+3$ covariant approach exists to make, drawn here rather than asserted. The median frame gap reaches 40%. **This caption read "which is a result and not a defect"; that reading is superseded by T-59 and T-74 and is withdrawn.** The gap is now understood as the third symptom of one term — the bare κ' carried inside C_1, **T-101** and **T-121** — and not as a property of the formulation. What remains true of the original reading is only this: C_ℓ is quadratic in frame-dependent pieces, while the invariance (102) satisfies is a statement about the total at fixed k, measured separately by D6.

† *Superseded by findings* **T-59**, **T-74**; *see* `provenance/BUNDLE-DEFECTS-v1.md`.

Evidence category: acceptance. *Generated by* `scripts/figure_f9_acoustic_spectrum.py`. *Standalone caption:* `captions/F9-acoustic-spectrum.md`.

limit recovered at 0.26% against a 5% criterion. **This shows that** the reconstruction is faithful to the antecedent's own equations, independently of any Boltzmann code, and that the harmonic phase convention is fixed by the *definition* of Q_{A_ℓ} rather than adopted.

Learned, and it set the pattern for everything after. A coarsened η grid returned a smooth, plausible spectrum that was 575% wrong while the two routes agreed to 10^{-16} — finding **T-3**. **Two agreeing routes agree about what they share.** That sentence is the single most load-bearing lesson in this bundle; it produced `required_n_eta`, it is why criterion 5b's gate states its shared inputs explicitly, and it is why the grid register exists.

15.2 v1.1.0 — the recombination that makes peaks possible

Achieved. Peebles' effective three-level atom replacing equilibrium Saha, and made the default only after being checked standalone against the standard ionisation history. Freeze-out at 2.15×10^{-4} against a standard $\sim 2 \times 10^{-4}$; the visibility peak moves $z \simeq 1300 \to 1107$ against a standard 1080–1100, and the function *widens by about 40%*. The frame change made numerical rather than symbolic. **This shows that** the width, not the position, is what the spectrum needs: the width sets the Silk damping envelope, which is why the acoustic peaks could not be attempted before this release.

Learned. `SAHA` was kept rather than deleted. A known-worse alternative that *must* change the answer is evidence; deleting it would have left the improvement asserted rather than measured. The same principle appears as the negative control in every criterion that claims one.

15.3 v1.2.0 — the spectrum with structure, and the audit of it

Achieved. The full linear C_ℓ: acoustic peaks, a Silk damping tail, the complete source assembly, and the integral solution (102) carried with a generic u^a and specialised back. Criterion 5b closed at 9.5×10^{-6} against a gate of 10^{-3}. **This shows that** *Annals II*'s two writings of its own integral solution — (102) specialised, and (177)–(179) assembled — are the same calculation when evaluated, which the paper recovers symbolically and which nobody had ever computed.

Learned, three times over, and they are the same lesson in three places. *First*, the tolerance this criterion was written against could only ever be failed: a slow-decoupling solution with no polarisation hierarchy cannot meet two full Boltzmann codes, and a criterion that can only be failed is not a criterion (**S4c**). *Second*, the figure 10^{-5} that had stood in the record for the two routes' agreement was never measured — it came from a fixed grid on which both routes were *un-converged in the same direction* (**T-45**). *Third*, the gate written to catch all of this printed `PASS` on a `nan`, because `nan >= gate` is `False` (**T-44**). **A gate is not tested by the run that passes it.**

15.4 The arc, in one sentence each

- **v1.0.0** asked *is the reconstruction faithful?* and answered it against four external treatments.
- **v1.1.0** asked *what does the spectrum need that it does not have?* and answered: the visibility width.
- **v1.2.0** asked *is the antecedent's own algebra self-consistent when evaluated?* and answered yes, at 10^{-5}, while finding that the question had been answered three times before without being measured.
- **v1.3.0** asks *where does the frame discrepancy come from?* — and §11 is the answer so far: not the source, not the by-parts bookkeeping, not the projection, but the slot-and-weight assignment of C_1. The question it has *not* answered is whether repairing that assignment also recovers the acoustic peaks, which is a separate defect with a separate signature.

15.5 v1.3.0 — the structural audit, and the discrepancy localised

Two halves, in the order decision **S37** fixed: the structural audit of the code first, the acoustic modes second. The audit produced the generated registers this supplement now rests on — the grid register, which walks the abstract syntax tree and *fails* when the code holds a discretisation the register does not name; the model table, derived from the `Background` objects the code integrates rather than transcribed from a parameter sheet; and the theory audit, generated from six self-contained instruments that import no project code.

The acoustic half did not go as planned, and the record says so. It began as a search for the missing peaks and became the localisation in §11, because the frame discrepancy turned out to be upstream of the peak question and to admit a cleaner instrument. **Two defects were separated that had been one symptom**: the frame gap, now reduced to a single site with both of its boundaries measured; and the peak positions, which sit above their own ℓ_A on all three backgrounds, in both frames, and which the acoustic scale does not explain — the signs are opposite.

What v1.3.0 has not done. It has not repaired the assignment, it has not recovered the peaks, and it has not bounded the envelope approximations A-3, A-5 and the damping inputs, which are deferred to v1.4.0 precisely because changing an envelope term mid-measurement moves the number being measured for reasons unrelated to it.

What the arc did not do, and says so. No release has bounded the accuracy of the *source* as distinct from the consistency of its assembly. The recombination grid moves the primary term by 1.2% and both routes of criterion 5b share it, so the criterion cannot see it. That is recorded as a stated limit of v1.2.0 and derived immediately after the tag.

16 The scattering-strength expansion, re-derived

Annals II Appendix C writes the almost-Friedmann–Lemaître integrated Boltzmann equation for Thomson scattering as

$$\mathcal{B} + \dot{\tau} + e^a \mathrm{D}_a \tau \;\simeq\; t_c^{-1}\big[v^a e_a - \tau\big], \tag{24}$$

inverts it, and approximates the result *"systematically … in terms of the smallness parameter t_c"* by the recursion

$$\tau_{(n)} \;\simeq\; v^a e_a - t_c\Big[\mathcal{B} + \dot{\tau}_{(n-1)} + e^a \mathrm{D}_a \tau_{(n-1)}\Big], \tag{25}$$

printing $\tau_{(0)}$, $\tau_{(1)}$ and $\tau_{(2)}$. Section 5.5.3, *Expansion in Thompson Scattering Time*, states that the scheme carries *"terms up to $\mathcal{O}(t_c^3)$"*, so the ceiling is the antecedent's and the printed orders stop one short of it.

16.1 The recursion is geometric

Write $\mathcal{D} \equiv \partial_t + e^a \mathrm{D}_a$ for the derivative along the photon path — which is exactly the pair of terms (25) applies to $\tau_{(n-1)}$ — and $V \equiv v^a e_a$. Then (24) reads $\tau = V - t_c(\mathcal{B} + \mathcal{D}\tau)$, so

$$(1 + t_c\mathcal{D})\tau = V - t_c\mathcal{B}, \qquad \tau = (1 + t_c\mathcal{D})^{-1}(V - t_c\mathcal{B}), \tag{26}$$

and expanding the inverse as a Neumann series in $(-t_c\mathcal{D})$ gives a closed form at every order:

$$\boxed{\tau_{(n)} \;=\; V \;+\; \sum_{m=1}^{n} (-1)^m\, t_c^m \Big[\mathcal{D}^m V + \mathcal{D}^{m-1}\mathcal{B}\Big].} \tag{27}$$

This shows that Appendix C's recursion is not an open-ended approximation scheme but a geometric series with an explicit sum, and that $\tau_{(1)}$ and $\tau_{(2)}$ as printed are its first two partial sums. Equation (27) is implemented in `functions/spectra/scattering_expansion.py` and checked against the printed $\tau_{(1)}$

and $\tau_{(2)}$ term for term, with a deliberately failing control, in `tests/test_scattering_expansion.py`. The closed form was derived from (25) and *then* compared with the page, which is the only order in which the comparison carries information.

16.2 Two consequences the closed form makes immediate

The term counts are not chosen. $\mathcal{D}^m$ expands into a sum over words in $\{\partial_t,\ e^a\mathrm{D}_a\}$, so the t_c^m bracket of (27) holds exactly $2^m + 2^{m-1}$ terms. Appendix C prints three at first order and six at second; those are $2+1$ and $4+2$. Table 5 lists every term to $\mathcal{O}(t_c^3)$, generated from the same module rather than transcribed.

Order m sources multipoles up to $\ell = m+1$. Each $e^a\mathrm{D}_a$ carries one direction vector and $V = v^a e_a$ carries one already, so a word with g gradients seeded by V has $g+1$ factors of e^a; reduced by Appendix C's own solid-angle identities, which lower the rank two at a time, its highest PSTF part has rank $g+1$. At second order that reaches **rank three**, in the single term $e^c e^a e^b \mathrm{D}_c\mathrm{D}_a v_b$.

Equivalently, and more usefully for the high-ℓ question: each gradient contributes one power of k, so a term of rank r carries k^{r-1}. The expansion parameter is therefore $t_c k$ — the ratio of the mean free path to the wavelength — and **not** t_c alone. **This shows that** the truncation is ℓ-dependent by construction: a cut in rank is a cut in $t_c k$, and the order at which the expansion may be stopped falls as ℓ rises.

16.3 What this costs the oscillator

The near-tight-coupling set of §7.1 is introduced *"rewritten to first order"* and closes with $\tau^{A_\ell} \simeq 0$ for all $\ell > 2$. At first order the expansion reaches rank two, so the closure costs nothing and the antecedent is consistent. **What the rank counting fixes is the price of going further**: second order reaches the octupole and $\mathcal{O}(t_c^3)$ the hexadecapole, and the closure carried with (152) and (153) reaches neither.

That is approximation **A-4** of `provenance/SOURCE-APPROXIMATIONS-v1.md` stated as a rank rather than as a dropped term. The term the oscillator derivation discards, $-\frac{2}{5}t_c(1+R)^{-1}\mathrm{D}^b\mathrm{D}_{\langle a}u^m_{b\rangle}$, is dropped because *"the matter shear is already at least first order"* — a cross term of $\mathcal{O}(t_c\epsilon)$ between §5.5.1's almost-Friedmann–Lemaître ordering and §5.5.3's scattering-time ordering. *Annals II*'s Appendix E, after Ellis–Treciokas–Matravers, states what the discarded harmonics hold: if any four consecutive ones vanish exactly then $\sigma_{ab} = 0$ exactly. **This shows that** the truncation and the shear are not independent — the residual in the truncated harmonics is where the shear lives, and a treatment that sets them to zero has set the shear to zero with them.

16.4 What appears at second order that does not appear at first

Appendix C's own solid-angle reduction gives the rank decomposition of the source,

$$\int_{4\pi} \mathrm{d}\Omega\, O^{A_\ell}\mathcal{B} \simeq \delta_0^\ell\Big[-\tfrac{4\pi}{3}\mathrm{D}_a\tau^a\Big] + \delta_1^\ell\Big[\tfrac{4\pi}{3}\big(\tfrac{1}{4}\mathrm{D}^{a_1}\ln\rho_R + A^{a_1}\big)\Big] + \delta_2^\ell\Big[\tfrac{8\pi}{15}\sigma^{a_1a_2}\Big], \tag{28}$$

so $\mathcal{B}$ carries the **shear** in its rank-two part. At first order $-t_c\mathcal{B}$ therefore reproduces the quadrupole $\tau_{ab} \simeq -\dot\kappa^{-1}[\sigma_{ab} + \mathrm{D}_{\langle a}v^B_{b\rangle}]$ of §7.1, as it must.

At second order the closed form contributes $+t_c^2\,\mathcal{D}\mathcal{B}$, and through (28) that is $\dot\sigma_{ab}$ at rank two and $e^c\mathrm{D}_c\sigma_{ab}$ at rank three. Appendix G's $\dot\sigma_{ab} + 2H\sigma_{ab} + E_{ab} \approx 0$ then gives

$$\dot\sigma_{ab} = -2H\sigma_{ab} - E_{ab}, \tag{29}$$

so **the electric Weyl tensor enters the radiation quadrupole at second order in the scattering time**, and does not at first. **This shows that** the second-order expansion is not merely a smaller correction of the same kind: it introduces a gravitational tidal driving of the quadrupole that the first-order treatment has no term for.

An open question, stated as a mechanism rather than a result. *Annals II* flags the regime itself, in a footnote to the sentence about removing the radiation quadrupole: *"If there are any anisotropic contributions such as anisotropic scattering (in the matter frame) or large shear (from gravitational waves) at that time this sort of approximation should be considered with care — such phenomena would break tight-coupling."* For a tensor mode both $\sigma_{ab}^{(T)}$ and $E_{ab}^{(T)}$ oscillate at the gravitational-wave frequency, so (29) drives the second-order quadrupole oscillatorily on a scale set by the wave and not by the sound horizon. Whether that survives projection as a distinguishable *ringing* is open; it is recorded as question **Q10** rather than claimed, because nothing here computes it.

What this expansion cannot produce. (25) is **linear** in V and $\mathcal{B}$, so (27) yields terms of order $t_c^m \epsilon$ and never ϵ^2. Terms of $\mathcal{O}(\varepsilon^2 \ell)$ require the quadratic parts of the Boltzmann equation that §5.5.1's almost-Friedmann–Lemaître linearisation discards, and are the subject of a later release rather than of this appendix. The high-ℓ sensitivity established above is a separate statement and rests on $t_c k$, not on ϵ^2.

Table 5: *Annals II* Appendix C's scattering-strength expansion to $\mathcal{O}(t_c^3)$, generated from `functions/spectra/scattering_expansion.py`. **Rank** is the highest PSTF multipole the term can source: one per $e^a \mathrm{D}_a$, plus one for $V = v^a e_a$. The closure (L0082) keeps $\ell \leq 2$.

Order		**Term**	**Rank**
t_c^0	+	$v_a e^a$	1
t_c^1	−	$\mathcal{B}$	0
t_c^1	−	$\left(v_a e^a\right)^{\cdot}$	1
t_c^1	−	$e^c \mathrm{D}_c \left(v_a e^a\right)$	2
t_c^2	+	$\left(\mathcal{B}\right)^{\cdot}$	0
t_c^2	+	$e^c \mathrm{D}_c \left(\mathcal{B}\right)$	1
t_c^2	+	$\left(\left(v_a e^a\right)^{\cdot}\right)^{\cdot}$	1
t_c^2	+	$e^c \mathrm{D}_c \left(\left(v_a e^a\right)^{\cdot}\right)$	2
t_c^2	+	$\left(e^c \mathrm{D}_c \left(v_a e^a\right)\right)^{\cdot}$	2
t_c^2	+	$e^c \mathrm{D}_c \left(e^c \mathrm{D}_c \left(v_a e^a\right)\right)$	3
t_c^3	−	$\left(\left(\mathcal{B}\right)^{\cdot}\right)^{\cdot}$	0
t_c^3	−	$e^c \mathrm{D}_c \left(\left(\mathcal{B}\right)^{\cdot}\right)$	1
t_c^3	−	$\left(e^c \mathrm{D}_c \left(\mathcal{B}\right)\right)^{\cdot}$	1
t_c^3	−	$e^c \mathrm{D}_c \left(e^c \mathrm{D}_c \left(\mathcal{B}\right)\right)$	2
t_c^3	−	$\left(\left(\left(v_a e^a\right)^{\cdot}\right)^{\cdot}\right)^{\cdot}$	1
t_c^3	−	$e^c \mathrm{D}_c \left(\left(\left(v_a e^a\right)^{\cdot}\right)^{\cdot}\right)$	2
t_c^3	−	$\left(e^c \mathrm{D}_c \left(\left(v_a e^a\right)^{\cdot}\right)\right)^{\cdot}$	2
t_c^3	−	$e^c \mathrm{D}_c \left(e^c \mathrm{D}_c \left(\left(v_a e^a\right)^{\cdot}\right)\right)$	3
t_c^3	−	$\left(\left(e^c \mathrm{D}_c \left(v_a e^a\right)\right)^{\cdot}\right)^{\cdot}$	2
t_c^3	−	$e^c \mathrm{D}_c \left(\left(e^c \mathrm{D}_c \left(v_a e^a\right)\right)^{\cdot}\right)$	3
t_c^3	−	$\left(e^c \mathrm{D}_c \left(e^c \mathrm{D}_c \left(v_a e^a\right)\right)\right)^{\cdot}$	3
t_c^3	−	$e^c \mathrm{D}_c \left(e^c \mathrm{D}_c \left(e^c \mathrm{D}_c \left(v_a e^a\right)\right)\right)$	4

One transcription note. The fifth term of Appendix C's t_c^2 bracket is printed as $e^c e^d \mathrm{D}_c \dot{v}_b$, which leaves both d and b free in a scalar. It is $e^c e^d \mathrm{D}_c \dot{v}_d$ — the time derivative taken first and the gradient outside — forced independently by the four words of $\mathcal{D}^2$, by the bracket's count of six with the other five unambiguous, and by the two direction vectors having nothing else to contract with. Recorded as **B-15**; no computed quantity depends on it.

17 The CDM-frame division of the integral solution, term by term

The integral solution (102) is derived for a general u^a — the antecedent says so immediately after (90), *"Note that these are CGI with respect to u^a"* — and then evaluated once, in the Newtonian frame. This appendix carries it into the CDM frame instead, coefficient by coefficient, and finds that the two frames do not agree. The failure is isolated to one coefficient at one power of k.

What is established here is narrower than it looks, and the conventions section exists because the first draft of this appendix did not have one.

17.1 Conventions, stated

Every relation below is in one convention and this subsection fixes it. The need for it was raised by the Paper 1 stream, whose check could not reproduce two of the four frame-table entries — correctly, because the entries are in the bundle's normalisation and the check was in Paper 1's.

Wavenumber. The bundle carries $k \equiv k_{\rm com}$ throughout and converts at the point of use, $k_{\rm phys} = k_{\rm com}/a$. Paper 1's C3.1 declares k *physical*. So a relation printed there as $X = k\,Y$ is carried here as $aX = k_{\rm com}Y$, and the scale factor is the conversion, not a transcription error.

The frame velocity. v^a is the relative velocity of the two frames, $\tilde{u}^a \approx u^a + v^a$ with $v_a u^a = 0$, expanded as $v_a = v\,Q_a$. So v is a *velocity amplitude*: it is the v of the Euler equation $v' + \mathcal{H}v = -k\Phi_A$, which is the frame condition $\tilde{A}_a = 0$, and it is the v of $\sigma = -k_{\rm phys}v$. It is **not** a velocity potential; $v_a = \mathcal{D}_a V$ would put k in the numerator.

The dipole. The bundle carries the dipole in the *covariant* normalisation, $\tilde{\tau}_1 \equiv \tilde{v}_B = v_B/k$ — the normalisation (109) needs in order to read $-\tilde{C}_1 = (\delta\tilde{T} + \Phi_A) + \kappa'\tilde{v}_B$. GDE99 (37), $\tilde{\tau}_a \approx \tau_a - v_a$, therefore reads $\tilde{\tau}_1 \to \tilde{\tau}_1 - v/k$ *here* and $\tau_1 \to \tau_1 - v$ in the un-reduced normalisation. The two are the same statement. **This is finding T-27 in the bundle's record**, where the un-reduced form was used by mistake and the symptom was a first-order frame change 186 times the quantity it corrects at $k = 300$.

Consequently, two of the four entries reconcile and one does not. $a\sigma = -k_{\rm com}v$ *is* $\sigma = -k_{\rm phys}v$, so n-t1 agrees with Paper 1's $\sigma = -kv$. $\tilde{\tau}_1 \to \tilde{\tau}_1 - v/k$ *is* $\tau_a \to \tau_a - v_a$ in the reduced normalisation, so n-t4 agrees. What does **not** reconcile from outside the bundle is the reduced normalisation itself: Paper 1 carries no C_1, so it cannot adjudicate $\tilde{\tau}_1 = v_B/k$, and ΔC_1 below rests on it.

17.2 What has to hold

A first-order change of frame is a boost, and a boost changes the observed brightness by $\tau \to \tau - v_a e^a$: **a pure dipole**. In PSTF multipoles,

$$\tau^{(u)}_{A_1} = \tau^{(n)}_{A_1} - v_{A_1}, \qquad \tau^{(u)}_{A_\ell} = \tau^{(n)}_{A_\ell} \quad \text{for all } \ell \geq 2. \tag{30}$$

This is not an approximation to be checked against a tolerance; it is what makes C_ℓ an observable, and the almost-EGS argument rests on it. The division has one admissible outcome: the shift must cancel for $\ell \geq 2$.

17.3 The three shifts

(90) fixes the coefficients,

$$-C_0 \simeq \tfrac{5}{2}\,a\mathcal{B}_2, \qquad -C_1 \simeq \tfrac{1}{k}(a\mathcal{B}_1 + \kappa' v_B), \qquad -C_2 \simeq \tfrac{1}{k^2}\,a\mathcal{B}_2, \tag{31}$$

with $a\mathcal{B}_1 = k\,\delta T + aA$ and $v_B = k\tilde{\tau}_1$. The frame carries four entries, each in §17.1's convention:

	Newtonian n^a	CDM u^a
aA	$k\Phi_A$	0 (the frame condition $A_a = 0$)
δT	$\delta\tilde{T}$	$\delta\tilde{T} - \mathcal{H}v/k$ (n-t3)
$\tilde{\tau}_1$	$\tilde{\tau}_1$	$\tilde{\tau}_1 - v/k$ (n-t4, reduced)
$a\mathcal{B}_2$	0	$-kv$ (n-t1, $\sigma = -k_{\rm phys}v$)

With $\Delta X \equiv X^{(u)} - X^{(n)}$: $\Delta(a\mathcal{B}_1) = -\mathcal{H}v - k\Phi_A = v'$ by the Euler equation, and $\Delta(\kappa' v_B) = \kappa' k\,\Delta\tilde{\tau}_1 = -\kappa' v$. Hence

$$\boxed{\Delta C_0 = +\tfrac{5}{2}\,k\,v, \qquad \Delta C_1 = -\tfrac{1}{k}(v' - \kappa' v), \qquad \Delta C_2 = +\tfrac{v}{k}} \tag{32}$$

all three in one v and one k.

The first entry is superseded, and so is the title of the subsection that follows. Corrected 2026-10-23, findings T-104 and T-107. $\Delta C_0 = +\frac{5}{2}kv$ is the shift of (90)'s *printed* C_0. (89)'s $\delta_{\ell 0}$ row carries α_2 where α_0 belongs — a factor $\frac{15}{2}$, finding **T-104** — and solved at its own normalisation the row gives $C_0 = -\frac{1}{3}a\mathcal{B}_2$, whence

$$\Delta C_0 = +\tfrac{kv}{3} \qquad \text{(corrected)}, \tag{33}$$

smaller by $\frac{15}{2}$. **Every scaling below survives with its prefactor divided by $\frac{15}{2}$**: (38) becomes $\frac{1}{6}(k/\mathcal{H})^2$ in place of $\frac{5}{4}(k/\mathcal{H})^2$, and the conclusion that the ratio grows as $(k/\mathcal{H})^2$ is unchanged, because ΔC_0 and $\frac{1}{k}2v''$ are compared as powers of $k/\mathcal{H}$ and a constant does not move that.

What does not survive is "ΔC_0 has no partner". $a\mathcal{B}_0 = -\frac{k}{3}\tau_1$ by identity, and $-C_1'$ carries $+\frac{k}{3}\tau_1$, so the partner exists and is *cross-coefficient* — finding **T-107**. γ, which pairs ΔC_0 against $k^2\Delta C_2$, therefore asks the wrong pairing and cannot see it: **M15**, *a gate built from one coefficient cannot see a cancellation distributed across the bracket.* `tests/frame/c8_rank0_partner.py` scores the whole rank-0 content of the bracket and is the gate for this question.

17.4 The shift of (102), and why ΔC_0 has no partner

(102) reads

$$\frac{\beta_\ell \tau_\ell(\eta_0)}{2\ell+1} \simeq [C_2'(\eta_*) - C_1(\eta_*)]\mathcal{D}\, j_\ell(x_*) + kC_2(\eta_*)\mathcal{D}\, j_\ell'(x_*) + \int_{\eta_*}^{\eta_0} d\eta\; \mathcal{V} e^{-(k/k_D)^2}\,[C_0 - C_1' + C_2'']\, j_\ell(x), \tag{34}$$

and (32) gives

$$\Delta\big[C_2' - C_1\big]_{\eta_*} = \tfrac{1}{k}\big(2v' - \kappa' v\big)\Big|_{\eta_*}, \tag{35}$$

$$\Delta\big[kC_2\big]_{\eta_*} = v(\eta_*), \tag{36}$$

$$\Delta\big[C_0 - C_1' + C_2''\big] = \tfrac{5}{2}kv + \tfrac{1}{k}\big(2v'' - (\kappa' v)'\big). \tag{37}$$

v obeys $v' = -\mathcal{H}v - k\Phi_A$, so on sub-horizon scales it changes on the Hubble rate: $v' \sim \mathcal{H}v$, $v'' \sim \mathcal{H}^2 v$. Then

$$\frac{\frac{5}{2}kv}{\frac{1}{k}\,2v''} \;\sim\; \frac{5}{4}\left(\frac{k}{\mathcal{H}}\right)^2, \tag{38}$$

and the two boundary shifts are smaller still. Nothing in (102) is the size of $\frac{5}{2}kv$, so (30) fails for every $\ell \geq 2$ by a factor growing as $(k/\mathcal{H})^2$.

17.5 The k^2 is (90)'s, not a mixed convention

A factor k^2 between ΔC_0 and ΔC_2 is exactly what a comoving/physical or amplitude/potential mismatch produces, and the Paper 1 stream was right to require that reading be excluded before the rest stands. It is excluded, and not by argument.

C_0 and C_2 are the same array. (31) divides one quantity, $a\mathcal{B}_2$, by k^0 and by k^2. Whatever $a\mathcal{B}_2$ is — physical or comoving, amplitude or potential, correctly transcribed or not — it is *the same object in both*, so their ratio carries no convention at all:

$$\frac{C_0}{k^2 C_2} = \frac{p_0}{p_2} = \frac{5}{3\beta_2} \qquad \text{(a pure rational).} \tag{39}$$

Measured at η_* on 'CDM (Einstein–de Sitter)', 'LCDM' and 'PLANCK_MODEL', at $k = 1, 5, 20, 60, 140, 300$: **2.5000** at every entry. **That measurement predates 21 October and is the ratio under (89)'s *printed* $\delta_{\ell 0}$ row.** Under the released row today — `MONOPOLE_ROW = "corrected"`, (33) — the same quantity reads 0.333333 at $k = 1, 10, 60$ on Einstein–de Sitter and `PLANCK_MODEL` alike: **exactly** $\frac{1}{3}$. *Only the value moves.* What this subsection's argument needs is that the ratio is a pure rational, independent of k and of background, and it still is. **What the measurement established is internal consistency, which is not correctness** — that was true at $\frac{5}{2}$ and is true at $\frac{1}{3}$. A mismatch between two different objects does not return a background-independent, k-independent rational; it drifts. **So the dominance is** (90)**'s own structure**, and a spurious scale factor in $a\mathcal{B}_2$ — were there one — would multiply C_0 and C_2 equally and cancel from the ratio entirely.

What this does not exclude is ΔC_1, which is the one shift that depends on the reduced normalisation $\tilde{\tau}_1 = v_B/k$ of §17.1. If that reading is wrong, ΔC_1 is wrong by k^2 — and the bundle has made exactly that error once already, as **T-27**. Paper 1 cannot adjudicate it, since it carries no C_1. It is open, and it is the first thing to settle.

17.6 Measured

Both frames from one environment, one acoustic solution and one k grid, differing only in the frame (110) specialises to, against the Newtonian total's scale over $\ell \in \{2, 5, 10, 50, 200\}$:

background	k	as assembled	with C_0 dropped
CDM (Einstein–de Sitter)	20	53.5	0.032
	300	1719	0.035
ΛCDM	20	62.8	0.105
	300	6868	0.154
`PLANCK_MODEL`	20	72.2	0.111
	300	3687	0.058

Three to four orders of magnitude, to a residual of 1–15%, with the k dependence of (38). The rank decomposition agrees: C_0 carries 98.6% of C_ℓ in the CDM frame and is identically zero in the Newtonian one.

17.7 Why the antecedent never meets this

(102) is evaluated once, in the Newtonian frame, where *"the vanishing shear condition $\tilde{\sigma}_{ab} \approx 0$ implies that $\tilde{C}_0(\eta) \approx 0$ and $\tilde{C}_2(\eta) \approx 0$"*. **Both members of the pair vanish together**, so the imbalance between them cannot be seen. Nothing the antecedent publishes is affected. What is missing is a stated condition on a formula labelled frame-general.

17.8 Three repairs tested, and one of them is untested

The dropped lines of (93) are not the counterweight. (102) is (93) with its last two integrals removed. Restoring $\int(\kappa' e^{-\kappa})[C_1 - 2C_2]j_\ell$ and $\int((\kappa')^2 - \kappa'')e^{-\kappa}C_2 j_\ell$ moves the disagreement from 72.2 to 71.6 at $k = 20$ and from 3686.9 to 3686.8 at $k = 300$.

(89)'s $\ell = 0$ line is not mis-sourced. $a\mathcal{B}_0\delta_{\ell 0}$ is commented out of the equation the integral solution solves and restored in the open-case extension, which looks like a dropped source. It is not: the ansatz must give $\tau_0 = 0$, and differentiating $\int A_0\, d\eta' = 0$ yields (89)'s $\ell = 0$ line with **zero** on the right. C_0 is correctly derived.

Undoing the integration by parts: untested, not refuted. (93) comes from a kernel in which the two derivatives act on $\tau_\ell^{(0)}$ and bring down k, not on the coefficients where they bring down $\mathcal{H}$. The numerical test of this was **invalid** — it failed its own control, reproducing the by-parts form in the Newtonian frame only to within 10^{-3} to 2.5 — because (102) does not differentiate the weight $\mathcal{V}$ that (93) carries inside the integral, so the two are not an exact by-parts pair once a weight is present. It remains the most promising repair and it needs doing analytically.

17.9 What is open

1. **The reduced normalisation** $\tilde{\tau}_1 = v_B/k$, on which ΔC_1 rests and which nothing outside the bundle can adjudicate. The bundle has erred here once (**T-27**). Settle it by index algebra on (109).
2. **The by-parts step**, the one untested repair.
3. **Whether C_0 belongs in the integrand**, which the measurement answers empirically and the algebra does not.
4. **The** 1–15% **residual** after C_0 is removed: whether it is $O(\varepsilon^2)$ or an artefact of the grids.

What stands independently of all four: C_0 and C_2 are one $a\mathcal{B}_2$ over k^0 and k^2, that ratio is a measured pure rational carrying no convention, and (102) in a sheared frame is dominated by the member of the pair that the Newtonian specialisation sets to zero.

18 What is open

Each row is a physics question this release does not answer, with what would settle it. The working history of each is in `provenance/OPEN-QUESTIONS-v1.0.0.md`.

	the question	what would settle it
Q5	Does the full $1+3$ treatment predict excess power at low ℓ?	A rank-0/2 share that rises at low ℓ rather than with it. **F2**'s third panel measures it and finds the share flat at the quadrupole, which parks the question rather than answering it
Q9	The discretisations are audited; the *physical* ranges are not	A stated validity range per approximation, with the measurement that bounds it
Q10	Does the second-order quadrupole ring for tensor modes?	The second-order source carried at rank two with a tensor initial condition — recorded as a mechanism, not claimed
Q11	The acoustic source has the right phase and the spectrum does not	Downgraded from blocker to residual, *not closed*: the oscillator tracks $\cos(kr_s)$ in exact phase and period, so the phase is lost between the source and the transfer
Q12	Two implementations of (102) agree at $k=1$ and diverge above $k=3$	Which of the two reproduces the integral (102) is derived from — §10.1 answers this for the bulk and not for the boundary
Q13	Which construction of $(\kappa'\tilde{v}_B)'$ is right?	The by-parts identity, term by term, against each construction
Q14	Should C_0 enter the line-of-sight integrand at all?	§10.2 settles the $\ell=0$ row; whether the resulting C_0 belongs in the bulk is the remaining half
Q15	**What does omitting helium recombination cost?** The recombination history here integrates the hydrogen Peebles equation. Helium recombines earlier and shifts x_e, hence κ', η_*, the visibility width and the damping scale.	η_* and k_D recomputed with helium, against the percent-level differences the peak comparison turns on. *Not measured*
Q16	**Is the tight-coupling source frozen late enough?** δT and τ_1 are held at $\eta_{\rm couple}=6\eta_*$, which is correct physics — after decoupling nothing drives the monopole — on the condition that the visibility has closed by then.	$\mathcal{V}(\eta_{\rm couple})/\mathcal{V}(\eta_*)$ on every background. *Nothing currently measures it*, and the recombination history differs between the three

Q16 cannot move a peak, and that is worth saying so it is not over-read. What is frozen is the tight-coupling pair, which after decoupling has no scattering to drive it; the line-of-sight integral itself runs to η_0 on a grid whose density is set by the Bessel kernel's own period in η and gated against aliasing — finding **T-3**. A freeze that was too early would cut source while the visibility was still live, which is a boundary effect and not a projection one.

One open finding belongs beside these, because the plateau depends on it. **T-67** is open: $(a\mathcal{B}_1)'$ carries $-a\mathcal{H}\,\delta T - 3a\mathcal{H}\,\Phi_A$ beyond the terms quoted in §10.1, and whether they belong is unsettled. They are $O(a\mathcal{H})$ and the late integrated Sachs–Wolfe effect lives where $k\sim a\mathcal{H}$, **so the reduced form is unavailable in exactly the regime the low-ℓ plateau occupies.** Every statement of $(a\mathcal{B}_1)'$ in this supplement is the expansion-terms-off form and says so at its own site.

And the late ISW is a weight question before it is a physics one. The effect is already in the formalism: Φ' enters at rank one through $(a\mathcal{B}_1)'$, hence through the bulk bracket. In any line-of-sight assembly it rides on $W=e^{-\kappa}$ — the photon accumulates $\int\Phi'$ along the whole path, weighted only by the probability it has not scattered — and (11) agrees. **(102) carries that bracket at $\mathcal{VD}$, and $\mathcal{V}$ has closed by the time the late ISW acts.** So discrepancy (i) of §10.1 does not merely insert a spurious κ'; it removes the late ISW. *This is testable on the background axis*: Φ is constant for Einstein–de Sitter ($K=0$ dust, Appendix G of the antecedent), so EdS has no late ISW to lose, while ΛCDM and `PLANCK_MODEL` do. Correcting the weight should lift the plateau on the latter two and leave EdS alone.

19 The almost-EGS bounds

The Ehlers–Geren–Sachs theorem states that exact CMB isotropy for all fundamental observers in an expanding dust universe forces exact Friedmann–Lemaître–Robertson–Walker geometry. The *almost*-EGS theorem of Stoeger, Maartens & Ellis, *Ap. J.* **443**, 1 (1995) relaxes that to bounds: limits on the CMB multipoles τ_{A_ℓ} become limits on σ_{ab}, ω_{ab}, $\dot{u}_a$, E_{ab} and H_{ab}.

Those bounds are stated in the *covariant multipoles* — not in mode coefficients — which is what this bundle computes and what a gauge-fixed code cannot supply without a translation. Applying them is therefore natural work for this formalism, and it is **v2.0.0** work, because it is a calibration against data rather than an analytical reconstruction.

Table 6: The structure v2.0.0 fills. **No cell is computed at v1.0.0**, and none is quoted here from memory: observational columns are filled from the mission papers at v2.0.0 under the same rule that equation numbers are resolved from the source.

Mission	ℓ **reach**	**bound, standard CDM**	**bound, ΛCDM**
COBE-DMR (1992)	$\lesssim 20$	*the original*	*not in the original*
WMAP (2003–2013)	$\lesssim 1000$	pending	pending
Planck (2013–2018)	$\lesssim 2500$	pending	pending

Why the table needs both columns. The original bounds were derived for a CDM model, not for ΛCDM, and the difference is not cosmetic. The argument converts multipole limits into limits on σ/H using the background expansion, and Λ changes $H(z)$ at low redshift. More sharply, Λ adds a source the original did not have: the late integrated Sachs–Wolfe effect contributes to exactly the low-ℓ multipoles the COBE bounds rest on. In standard CDM it is absent — Φ is constant, which this bundle's growth factor reproduces to six figures — and in ΛCDM it is not, Φ falling to 0.779 of its early value by today. A low-ℓ limit therefore constrains a *different combination* in the two models, so a bound quoted for one is not a bound for the other. The CDM column has a published precedent; the ΛCDM column is new.

Beyond v2.0.0. The published bounds are first order in the anisotropy parameter. The natural extension is the next order, and this project is placed to make it because **v1.5.0 supplies the $O(\varepsilon^2\ell)$ couplings a second-order argument needs** and no external code carries them. Stated as a direction and not a plan: it is gated on v1.5.0, and the correction's sign and size are unknown until it is done. It is recorded so the v2.0.0 table is built in a shape that can take it. See `provenance/ALMOST-EGS-BOUNDS-v1.md`.

20 Parameter and numerical evidence

Table 7 lists every number the numerics use. Its *provenance* column distinguishes values taken from the source paper, from standard values that are not, from values derived here, and from values stipulated for this release. The distinction matters: $\Omega_\gamma h^2$ is a standard value and not one the antecedent supplies, and it enters only through $R(\eta_*)$ and hence the sound speed.

Table 7: Every number the numerics use. *Provenance* distinguishes values taken from the source paper, standard values that are not, values derived here, and values stipulated for this release.

Symbol	Code	Value	Provenance	Set in
k_{com}	`K_COM`	1.0	stipulated	`scripts/figure_f1_appendix_f.py`
$\ell_{\max}$	`ELL_MAX`	80	stipulated	`scripts/figure_f1_appendix_f.py`
ℓ_{compared}	`ELL_COMPARED`	25	stipulated	`scripts/figure_f1_appendix_f.py`
rtol, atol	`rtol, atol`	10^{-10}, 10^{-12}	stipulated	`functions/harmonics/free_streaming.py`
z_*	`Z_STAR`	1100	standard	`scripts/derive_ell_range.py`
h	`H`	0.5	AnnalsII §8.3.2	`scripts/derive_ell_range.py`
Ω_B	`OMEGA_B`	0.05	AnnalsII §8.3.2	`scripts/derive_ell_range.py`
$\Omega_\gamma h^2$	`OMEGA_GAMMA_H2`	2.47×10^{-5}	standard	`scripts/derive_ell_range.py`
tolerance	`ADOPTED_TOLERANCE`	5%	derived	`scripts/derive_ell_range.py`
acceptance threshold	`--`	10^{-11}	stipulated	`scripts/figure_f1_appendix_f.py`

20.1 The three backgrounds, side by side

Table 8 is the one place every derived scale of the three backgrounds can be read against the others. It exists because a question the PI asked — *is the difference between two panels a parameter difference?* — had no cheap answer while the models were defined in three places and compared nowhere.

The last three rows are the ones that answer a shape question. Two models whose spectra differ *in the position of the peaks* differ in $\ell_A = \pi r_\theta / r_s$. Two whose peaks sit at the same ℓ_A and still look different do **not** differ by a parameter, and the difference is in `provenance/BUNDLE-DEFECTS-v1.md`.

Two rows carry a known defect and say so rather than being quietly correct for a different question. $\Omega_r = 0$ is the *control* on Einstein–de Sitter, which is what makes that background the one where A-3 and $T(k)$ are identically inapplicable — and on LCDM it is neither a control nor intended, worth 37% in ℓ_A through η_* at 1.571× and r_s at 1.610× (findings **T-81c**, **T-83**). k_D is the $R = 0$ value, because `diffusion_scale` is called with R zeroed while the oscillator beside it runs at $R(\eta_*) \simeq 0.61$ — finding **T-21**, measured at $\ell_D = 1783$ against 1985.

21 Numerical resolution and reproducibility

The hierarchy is integrated with an explicit Runge–Kutta method of order 8 (`DOP853`) at relative tolerance 10^{-10} and absolute tolerance 10^{-12}, truncated at $\ell_{\max} = 80$ with comparisons restricted to $\ell \leq 25$ so that reflection from the truncation does not contaminate them. The acceptance threshold used in the figures, 10^{-11}, sits an order above the integrator floor.

The release route is

```
python scripts/run_all.py --strict
```

which regenerates every output, figure and table, recomputes the SHA-256 manifests, and reports any artefact that differs from the recorded hash.

Three manifests, and only one of them binds. `PORTABLE-MANIFEST-SHA256.txt` is the release fingerprint: it *excludes the renderings* — everything under `figures/` and `supplementary-materials/`,

Table 8: The three backgrounds side by side, with every derived scale. **Generated by `scripts/make_model_table.py`** from the `Background` objects the code integrates, not transcribed. The last three rows are the ones that answer a shape question: two models whose peaks differ in position differ in ℓ_A, and two whose peaks sit at the same ℓ_A and still differ do not differ by a parameter. $\Omega_r = 0$ is the control on CDM and finding **T-81c** on LCDM; k_D is the $R = 0$ value, finding **T-21**.

	CDM (Einstein–de Sitter)	LCDM	PLANCK
Ω_m	1	0.3	0.313772
Ω_Λ	0	0.7	0.686173
Ω_r	0	0	5.4502×10^{-5}
$\omega_b h^2$	0.0224	0.0224	0.02237
h	0.5	0.7	0.6736
n_s	1	1	0.9649
η_0	2	3.30508	3.19871
η_*	0.0595884	0.109581	0.0697426
a_*	8.8769×10^{-4}	9.0059×10^{-4}	8.9643×10^{-4}
z_*	1125.51	1109.38	1114.54
$R(\eta_*)$	0.603776	0.612548	0.608901
r_s	0.0316406	0.0581262	0.0361159
r_θ	1.94041	3.1955	3.12897
$\ell_A = \pi r_\theta / r_s$	192.663	172.71	272.177
k_D (at $R = 0$, T-21)	754.639	483.11	536.866

and every `.png` and `.pdf` — and canonicalises numeric output to nine significant figures, so a machine that did not generate the artefacts can still satisfy it. A mismatch there is release-blocking. `FILE-MANIFEST-SHA256.txt` is then a *within-machine* drift check, reported by artefact class, where only an `exact`-class mismatch blocks. `IMMUTABLE-MANIFEST-SHA256.txt` covers frozen reference material, where any change is release-blocking. The corrections record is append-guarded rather than frozen, because it must take new findings: appending is permitted and rewriting recorded bytes is not.

Why nine PNGs always report as drifted. Every `.png` written into a working copy through the Claude desktop bridge gains a 5 770-byte `caBX` provenance chunk, inserted immediately after `IHDR` by that application and not by anything in this bundle. The pixels are bit-identical — `IHDR`, `pHYs` and every `IDAT` chunk match exactly — and the `.pdf` of the same figure round-trips byte-for-byte, as does every `.csv` and every source file. These nine therefore sit in the `rendered` class, are reported and do not block, which is the rule finding **T-15** wrote for a different cause with the same shape: fonts, freetype and TEX Live versions, where *a PNG that differs in hinting carries no scientific difference.* **The PDF is the authoritative artefact of every figure in this bundle**; the PNG is a convenience copy. Finding **T-36**. The drift is explained here rather than silenced by canonicalising PNGs, because a manifest that stopped reporting a class of change for tidiness would stop reporting real ones too.

Weights are carried as exact rationals rather than floating-point numbers wherever a sign or an ℓ-weight is asserted.

22 Interpretive limits

This is an analytical reproducibility bundle for the associated paper and its supplement. It is not an empirical calibration, not a parameter-estimation pipeline, and not a Boltzmann code for general use. No new effect is claimed and none is reported.

Annals II's own §8.3.2 model is standard CDM — $h = 0.5$, $\Omega_B = 0.05$, flat and matter dominated — and the C_ℓ it prints there is a closed-form Sachs–Wolfe result on large scales with the transfer function

set to unity. This release goes past that in one respect and one only: it computes the spectrum from the hierarchy rather than from the closed form, and it computes it for the ΛCDM model as well, which *Annals II* does not treat. The marginal cost is the expansion law and the late-time integrated term; every other ingredient — recombination, tight coupling, diffusion damping — is already required by the CDM case and is not additional. No new *physics* is introduced by Λ, and none is claimed.

What the release does **not** do is calibrate. Comparison is made against CMBFAST and against nothing else, over a derived range, as a check that the reconstruction is right — not as a fit. Parameter estimation and calibration against data are the **v2.0.0** deliverable.

Two statements must travel together, and neither stands alone: the nonlinear contribution $\delta\tau_{NL}$ **matters** — it is not negligible and not an artefact — and it **adds nothing**, redistributing power rather than creating it, because $\delta\tau_{NL} \propto \Phi\Phi$ makes its contribution to C_ℓ quartic in Φ while the linear C_ℓ is quadratic.